

**Search for high-mass leptoquark with
tau-lepton and missing transverse
momentum using 140 fb of pp collision
data at $\sqrt{s} = 13$ TeV in the LHC-ATLAS
experiment**

(LHC-ATLAS実験における重心系エネルギー13 TeVの140 fbの陽子陽子衝突データを用いたタウレプトンおよび消失横運動量を使った高質量レプトクォーク探索)

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Abstract

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Chapter 1

Introduction

1.1 Introduction

Modern high-energy physics seeks to understand the elementary constituents of matter and the fundamental interactions that govern their behaviour. At present, the most successful theoretical description of particle physics is the Standard Model (SM), a quantum field theory based on the gauge symmetry

$$SU(3)_C \times SU(2)_L \times U(1)_Y. \quad (1.1)$$

Within this framework, matter is described by three generations of quarks and leptons, while the electromagnetic, weak, and strong interactions are mediated by gauge bosons. The electroweak symmetry is spontaneously broken through the Higgs mechanism, giving masses to the weak gauge bosons and to the fermions through their interactions with the Higgs field. The discovery of the Higgs boson by the ATLAS and CMS experiments in 2012 completed the particle content predicted by the SM and marked a major milestone in modern particle physics.

The SM has been tested with remarkable precision over a wide range of energy scales and has successfully described a vast number of experimental observations. Nevertheless, it is widely regarded as an incomplete theory. It does not provide a particle candidate for dark matter, does not explain the observed matter–antimatter asymmetry of the universe, and does not account for the origin of non-zero neutrino masses. In addition, the pattern of fermion masses and flavour mixing remains unexplained, and the large separation between the electroweak and Planck scales suggests the possible existence of new physics beyond the SM. These open questions motivate direct searches for new particles and interactions at the highest experimentally accessible energies.

The Large Hadron Collider (LHC) at CERN provides a unique environment for probing such new phenomena. By colliding protons at tera-electronvolt centre-of-mass energies, the LHC allows the production of particles that may be associated with physics beyond the SM. During Run 2 of the LHC from year 2015 to 2018, proton–proton (pp) collisions were delivered at a centre-of-mass energy of $\sqrt{s} = 13$ TeV.

ATLAS is a detector located at one of the interaction points of the LHC. It is designed to observe and study a wide range of physics processes, mainly from the pp collisions, from precision measurements of SM interactions to searches for new particles. The detector consists of several sub-detectors surrounding the collision interaction point, including an inner tracking detector, electromagnetic and hadronic calorimeters, and a muon spectrometer. Together with a powerful magnet system, these components allow the reconstruction and identification of electrons, photons, muons, hadronically decaying τ -leptons, jets, and missing transverse momentum, denoted by $E_{\text{T}}^{\text{miss}}$.

Among the many possible extensions of the SM, leptoquarks (LQs) provide a well-motivated class of new particles connecting the quark and lepton sectors. They are hypothetical bosons that couple directly to a quark and a lepton, and they appear naturally in theories such as grand unified theories and models of quark–lepton unification. Leptoquarks are also of particular phenomenological interest because they can modify semileptonic processes and may therefore be connected to the observed tensions in flavour observables.

A search for the leptoquarks is presented in the thesis using 140 fb^{-1} of pp collision data recorded by the ATLAS detector at a centre-of-mass energy of $\sqrt{s} = 13$ TeV. This thesis focuses on the vector singlet leptoquark U_1 as the benchmark model. This model can contribute to processes involving b quarks, τ leptons, and neutrinos. Therefore, the analysis targets events with a hadronically decaying τ -lepton, large missing transverse momentum, and jets. The observed data are compared with SM background predictions in signal regions designed to enhance sensitivity to possible leptoquark contributions. In the absence of a significant excess over the SM expectation, constraints are placed on the allowed parameter space of the U_1 model, expressed in terms of the leptoquark mass and its couplings.

This dissertation is organised as follows. Chapter 2 provides an overview of the SM’s theoretical framework and the physics beyond it. Chapter 3 describes the leptoquark models and the searches performed at LHC. Chapter 4 introduces the LHC and the ATLAS detector. Chapter 5 provides the dataset and simulated samples. Chapter 6 details the reconstruction and the identification of the physics objects that are used for event selection. Chapter 7 discusses the optimisation for the event selection used for signal signature extraction. Chapter 8 presents the background estimation strategy. Chapter 9 discusses the systematic

281 uncertainties for the background and signals. Chapter 10 presents the statistical interpretation
282 of the results. Chapter 11 summarises this analysis.

Chapter 2

Theoretical Background

This chapter introduces the theoretical background which motivates this search performed in the thesis. The discussion begins with the Standard Model (SM), which provides the current theoretical framework for describing elementary particles and their interactions. The limitations of the SM are then summarised, followed by a discussion of several possible extensions beyond the SM. Particular emphasis is placed on leptoquark models, which provide a direct connection between the quark and lepton sectors and motivate the experimental search presented in this thesis.

2.1 The Standard Model

The Standard Model (SM) is a relativistic quantum field theory describing the known elementary particles and their strong, weak, and electromagnetic interactions. Its structure is determined by local gauge invariance under

$$SU(3)_C \times SU(2)_L \times U(1)_Y. \quad (2.1)$$

The group $SU(3)_C$ describes the colour symmetry of the strong interaction, while $SU(2)_L \times U(1)_Y$ describes the electroweak interaction [1–3]. In quantum field theory, particles are classified not only by their mass and spin, but also by their transformation properties under these gauge symmetries. Particles with half-integer spin are fermions and obey Fermi–Dirac statistics, while particles with integer spin are bosons and obey Bose–Einstein statistics [4]. The matter particles of the SM are spin-1/2 fermions, the interaction carriers are spin-1 gauge bosons, and the Higgs boson is a spin-0 scalar boson.

The minimal SM Lagrangian used here is constructed from the SM field content and is invariant under the local gauge group. Therefore, instead of introducing the Lagrangian as

an isolated expression, this section first summarises the field content and gauge quantum numbers of the SM. The principle of local gauge invariance is then introduced, followed by the gauge transformations of the strong sector, based on $SU(3)_C$, and of the electroweak sector, based on $SU(2)_L \times U(1)_Y$. The covariant derivative and field-strength tensors are then constructed from these symmetries, leading to the full SM Lagrangian. Finally, electroweak symmetry breaking and the generation of fermion masses are discussed.

2.1.1 Particle content and gauge group of the SM

The gauge quantum numbers of a field specify how it transforms under the SM gauge group. In this thesis, the electric charge is written as

$$Q = T^3 + Y, \quad (2.2)$$

where T^3 is the third component of weak isospin and Y is the weak hypercharge. This convention is used throughout the thesis.

For each generation, the SM fermions are arranged as

$$Q_L^i = \begin{pmatrix} u_L^i \\ d_L^i \end{pmatrix} \sim (3, 2, 1/6), \quad u_R^i \sim (3, 1, 2/3), \quad d_R^i \sim (3, 1, -1/3), \quad (2.3)$$

$$L_L^i = \begin{pmatrix} \nu_L^i \\ e_L^i \end{pmatrix} \sim (1, 2, -1/2), \quad e_R^i \sim (1, 1, -1), \quad (2.4)$$

where $i = 1, 2, 3$ is the generation index, and the notation denotes the representation under $SU(3)_C \times SU(2)_L \times U(1)_Y$, which are the QCD representation, the weak isospin representation and the weak hypercharge, respectively. In the minimal SM, no right-handed neutrino field is included. The scalar sector contains one complex Higgs doublet,

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \sim (1, 2, 1/2). \quad (2.5)$$

The gauge bosons are associated with the generators of the three gauge groups. The eight gluon fields G_μ^a , with $a = 1, \dots, 8$, are associated with $SU(3)_C$. The three weak gauge fields W_μ^I , with $I = 1, 2, 3$, are associated with $SU(2)_L$. The hypercharge gauge field B_μ is associated with $U(1)_Y$. Before electroweak symmetry breaking, W_μ^1 , W_μ^2 , W_μ^3 , and B_μ are gauge eigenstates. After electroweak symmetry breaking, they are reorganised into the physical electroweak bosons $W^\pm$, Z , and γ .

The main particle content of the SM is summarised in Table 2.1. For fermions and the Higgs boson, the table lists the corresponding electroweak quantum numbers using the convention in Eq. (2.2). For gauge bosons, the entries describe their gauge fields and the electric charges of the physical states where appropriate.

Table 2.1 Particle content of the SM and the corresponding spin and electroweak quantum numbers. For fermions and the Higgs doublet, T^3 and Y denote the third component of weak isospin and the weak hypercharge, respectively, with the electric charge defined as $Q = T^3 + Y$. For gauge bosons, only the electric charges of the physical states after electroweak symmetry breaking are shown.

Category	Type	Generation			Spin	Quantum numbers		
		1st	2nd	3rd		T^3	Y	Q
Fermion	Left-handed quark	u_L	c_L	t_L	1/2	+1/2	1/6	+2/3
		d_L	s_L	b_L		-1/2		-1/3
	Right-handed up-type quark	u_R	c_R	t_R	1/2	0	2/3	+2/3
	Right-handed down-type quark	d_R	s_R	b_R	1/2	0	-1/3	-1/3
	Left-handed lepton	ν_{eL}	$\nu_{\mu L}$	$\nu_{\tau L}$	1/2	+1/2	-1/2	0
		e_L	μ_L	τ_L		-1/2		-1
	Right-handed charged lepton	e_R	μ_R	τ_R	1/2	0	-1	-1
Gauge boson	Gluon	$g^a, a = 1, \dots, 8$			1	-	-	0
	Photon	γ			1	-	-	0
	Charged weak boson	$W^\pm$			1	-	-	± 1
	Neutral weak boson	Z			1	-	-	0
Scalar field	Higgs doublet	ϕ^+			0	+1/2	1/2	+1
		ϕ^0				-1/2		0

The quantum numbers listed above determine which interactions each field can have. They also determine which terms can be written in the SM Lagrangian. For example, explicit Dirac mass terms for charged fermions are not gauge invariant before electroweak symmetry breaking, because the left- and right-handed fermions transform differently under $SU(2)_L \times U(1)_Y$. Fermion masses instead arise from gauge-invariant Yukawa interactions with the Higgs field.

2.1.2 Local gauge invariance

The guiding principle in constructing the SM is local gauge invariance. A global transformation acts in the same way at every point in spacetime, while a local gauge transformation depends on the spacetime coordinate x . For a field $\psi(x)$ transforming under a gauge group with generators T^A , a generic local transformation is

$$\psi(x) \rightarrow \psi'(x) = U(x)\psi(x), \quad U(x) = \exp \left[i\alpha^A(x)T^A \right]. \quad (2.6)$$

Here $\alpha^A(x)$ are arbitrary real, spacetime-dependent gauge parameters. They specify how the field is transformed locally along each generator T^A of the gauge group, and the repeated index A is summed over all generators.

The ordinary derivative of the transformed field is

$$\partial_\mu \psi'(x) = U(x) \partial_\mu \psi(x) + (\partial_\mu U(x)) \psi(x). \quad (2.7)$$

The second term in Eq. (2.7) shows that $\partial_\mu \psi$ does not transform in the same way as ψ . Therefore, the kinetic term constructed using the ordinary derivative is not invariant under a local gauge transformation.

To restore local gauge invariance, the ordinary derivative is promoted to a covariant derivative D_μ satisfying

$$D_\mu \psi(x) \rightarrow D'_\mu \psi'(x) = U(x) D_\mu \psi(x). \quad (2.8)$$

This requirement introduces gauge fields. The gauge fields are therefore not added by hand as arbitrary new particles; they are required in order to compare fields at neighbouring spacetime points in a way that is independent of the local gauge convention. The following subsections describe how this principle is realised for the strong and electroweak gauge sectors of the SM.

2.1.3 The $SU(3)_C$ gauge sector

The strong interaction is described by quantum chromodynamics (QCD), a gauge theory based on the colour group $SU(3)_C$. Quark fields transform as colour triplets, while leptons and the Higgs field are colour singlets. For a colour triplet quark field $q(x)$, the local $SU(3)_C$ transformation is

$$q(x) \rightarrow q'(x) = U_C(x) q(x), \quad U_C(x) = \exp \left[i \alpha_C^a(x) \frac{\lambda^a}{2} \right], \quad (2.9)$$

where λ^a are the Gell-Mann matrices and $a = 1, \dots, 8$. The corresponding gauge fields are the eight gluon fields G_μ^a .

It is useful to introduce the matrix-valued gluon field

$$G_\mu(x) = G_\mu^a(x) \frac{\lambda^a}{2}. \quad (2.10)$$

With the convention

$$D_\mu = \partial_\mu - ig_s G_\mu^a \frac{\lambda^a}{2}, \quad (2.11)$$

where g_s is the strong coupling constant, the gluon field transforms as

$$G_\mu(x) \rightarrow G'_\mu(x) = U_C(x) G_\mu(x) U_C^\dagger(x) - \frac{i}{g_s} (\partial_\mu U_C(x)) U_C^\dagger(x). \quad (2.12)$$

This transformation law ensures that $D_\mu q$ transforms in the same way as q under local $SU(3)_C$ transformations.

The $SU(3)_C$ field-strength tensor is

$$G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc} G_\mu^b G_\nu^c, \quad (2.13)$$

where f^{abc} are the structure constants of $SU(3)$, defined by the commutation relation

$$\left[\frac{\lambda^a}{2}, \frac{\lambda^b}{2} \right] = i f^{abc} \frac{\lambda^c}{2}. \quad (2.14)$$

The last term in Eq. (2.13) arises from the non-Abelian nature of $SU(3)_C$. It gives rise to gluon self-interactions, which are a defining feature of QCD and are essential for asymptotic freedom at high energies and confinement at low energies.

2.1.4 The electroweak gauge sector

Before electroweak symmetry breaking, the weak and electromagnetic interactions are described together by the gauge group $SU(2)_L \times U(1)_Y$ [1–3]. The $SU(2)_L$ factor is the non-Abelian weak-isospin gauge symmetry, while the $U(1)_Y$ factor is the Abelian hypercharge gauge symmetry. They are introduced separately below, but the physical photon and Z boson arise only after electroweak symmetry breaking, when the neutral fields W_μ^3 and B_μ mix.

The $SU(2)_L$ gauge transformation

The $SU(2)_L$ gauge symmetry acts on left-handed fermion doublets and on the Higgs doublet:

$$Q_L^i, \quad L_L^i, \quad \Phi. \quad (2.15)$$

396 For an $SU(2)_L$ doublet $\chi(x)$, the local transformation is

$$397 \quad \chi(x) \rightarrow \chi'(x) = U_L(x)\chi(x), \quad U_L(x) = \exp \left[i\alpha^I(x) \frac{\tau^I}{2} \right], \quad (2.16)$$

398 where τ^I are the Pauli matrices and $I = 1, 2, 3$. The right-handed charged fermions u_R^i, d_R^i ,
399 and e_R^i are singlets under $SU(2)_L$ and therefore have $T_L^I = 0$.

400 The three gauge fields associated with $SU(2)_L$ are denoted by W_μ^I . They can be written
401 in matrix form as

$$402 \quad W_\mu(x) = W_\mu^I(x) \frac{\tau^I}{2}. \quad (2.17)$$

403 With the convention

$$404 \quad D_\mu = \partial_\mu - igW_\mu^I \frac{\tau^I}{2}, \quad (2.18)$$

405 where g is the $SU(2)_L$ coupling constant, the weak gauge field transforms as

$$406 \quad W_\mu(x) \rightarrow W'_\mu(x) = U_L(x)W_\mu(x)U_L^\dagger(x) - \frac{i}{g}(\partial_\mu U_L(x))U_L^\dagger(x). \quad (2.19)$$

407 This transformation law is the non-Abelian generalisation of the Abelian gauge-field trans-
408 formation. The matrix structure reflects the fact that the $SU(2)_L$ generators do not commute.

409 The $SU(2)_L$ field-strength tensor is

$$410 \quad W_{\mu\nu}^I = \partial_\mu W_\nu^I - \partial_\nu W_\mu^I + g\epsilon^{IJK}W_\mu^J W_\nu^K. \quad (2.20)$$

411 The last term arises from the non-Abelian nature of $SU(2)_L$ and leads to self-interactions
412 among weak gauge bosons.

413 The $U(1)_Y$ gauge transformation

414 The $U(1)_Y$ gauge symmetry is the Abelian part of the electroweak gauge group. Its cor-
415 responding gauge charge is the weak hypercharge Y , which determines how each field
416 transforms under a local $U(1)_Y$ phase rotation. For a field $\psi(x)$ with hypercharge Y , the local
417 $U(1)_Y$ transformation is

$$418 \quad \psi(x) \rightarrow \psi'(x) = \exp[iY\beta(x)]\psi(x), \quad (2.21)$$

where $\beta(x)$ is a spacetime-dependent transformation parameter. The corresponding gauge field is the hypercharge field B_μ . With the convention

$$D_\mu = \partial_\mu - ig'YB_\mu, \quad (2.22)$$

the gauge field transforms as

$$B_\mu(x) \rightarrow B'_\mu(x) = B_\mu(x) + \frac{1}{g'}\partial_\mu\beta(x). \quad (2.23)$$

This transformation ensures that $D_\mu\psi$ transforms in the same way as ψ .

The hypercharge assignments in Table 2.1 are chosen so that the electric charge is obtained from Eq. (2.2). For the left-handed lepton doublet, $Y = -1/2$. The upper component has $T^3 = +1/2$ and therefore $Q = 0$, corresponding to the neutrino. The lower component has $T^3 = -1/2$ and therefore $Q = -1$, corresponding to the charged lepton. Similarly, the left-handed quark doublet has $Y = 1/6$, giving charges $+2/3$ and $-1/3$ for the up- and down-type components.

Because $U(1)_Y$ is Abelian, the corresponding field-strength tensor is

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \quad (2.24)$$

There is no self-interaction term involving two B_μ fields in $B_{\mu\nu}$. Although $U(1)_Y$ is Abelian, it should not be identified with the electromagnetic gauge symmetry $U(1)_{\text{em}}$. Before electroweak symmetry breaking, the corresponding gauge field is the hypercharge field B_μ rather than the photon field. The electromagnetic gauge symmetry appears only after electroweak symmetry breaking, when B_μ mixes with the neutral weak gauge field W_μ^3 to form the photon and the Z boson.

2.1.5 Covariant derivative and gauge field strengths

Combining the three gauge symmetries, the covariant derivative acting on a generic SM field can be written as

$$D_\mu = \partial_\mu - ig_s G_\mu^a T_C^a - ig W_\mu^I T_L^I - ig' Y B_\mu. \quad (2.25)$$

Here g_s , g , and g' are the gauge couplings of $SU(3)_C$, $SU(2)_L$, and $U(1)_Y$, respectively. The matrices T_C^a and T_L^I are the generators acting on the particular field. For a colour triplet, $T_C^a = \lambda^a/2$, while for a colour singlet $T_C^a = 0$. For an $SU(2)_L$ doublet, $T_L^I = \tau^I/2$, while

for an $SU(2)_L$ singlet $T_L^I = 0$. The value of Y is the hypercharge of the field, as listed in Table 2.1.

The gauge kinetic terms are constructed from the field-strength tensors:

$$G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc} G_\mu^b G_\nu^c, \quad (2.26)$$

$$W_{\mu\nu}^I = \partial_\mu W_\nu^I - \partial_\nu W_\mu^I + g \varepsilon^{IJK} W_\mu^J W_\nu^K, \quad (2.27)$$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \quad (2.28)$$

The tensors $G_{\mu\nu}^a$ and $W_{\mu\nu}^I$ contain non-linear terms because $SU(3)_C$ and $SU(2)_L$ are non-Abelian groups. By contrast, $B_{\mu\nu}$ has the Abelian form because $U(1)_Y$ has only one generator.

2.1.6 The Standard Model Lagrangian

The SM Lagrangian is constructed from the field content described above and is invariant under the local gauge transformations of $SU(3)_C \times SU(2)_L \times U(1)_Y$. Using the covariant derivative and field-strength tensors introduced in the previous subsections, the minimal SM Lagrangian can be written as the sum of gauge-boson kinetic terms, fermion kinetic terms, the Higgs-sector terms, and Yukawa interactions. Ignoring gauge-fixing terms, ghost terms, and the QCD θ -term, it is written schematically as

$$\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}. \quad (2.29)$$

The four terms in Eq. (2.29) are fixed by Lorentz invariance, local gauge invariance, and renormalisability, once the SM field content is specified [5, 6].

The gauge-boson kinetic terms are

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4} G_{\mu\nu}^a G^{a\mu\nu} - \frac{1}{4} W_{\mu\nu}^I W^{I\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu}. \quad (2.30)$$

Here $G_{\mu\nu}^a$, $W_{\mu\nu}^I$, and $B_{\mu\nu}$ are the field-strength tensors of $SU(3)_C$, $SU(2)_L$, and $U(1)_Y$, respectively. The indices $a = 1, \dots, 8$ and $I = 1, 2, 3$ label the generators of $SU(3)_C$ and $SU(2)_L$, and repeated Lorentz and gauge indices are summed over. The common factor $-1/4$ gives the conventional canonical normalisation of the gauge-boson kinetic terms. With this normalisation, the quadratic parts of Eq. (2.30) describe the free propagation of the gluon, weak, and hypercharge gauge fields, while the non-Abelian parts of $G_{\mu\nu}^a$ and $W_{\mu\nu}^I$ generate self-interactions among the corresponding gauge bosons.

The fermion kinetic terms are

$$\mathcal{L}_{\text{fermion}} = \sum_{i=1}^3 \left[\bar{Q}_L^i i\gamma^\mu D_\mu Q_L^i + \bar{u}_R^i i\gamma^\mu D_\mu u_R^i + \bar{d}_R^i i\gamma^\mu D_\mu d_R^i + \bar{L}_L^i i\gamma^\mu D_\mu L_L^i + \bar{e}_R^i i\gamma^\mu D_\mu e_R^i \right]. \quad (2.31)$$

The sum runs over the three fermion generations. The fields Q_L^i and L_L^i are the left-handed quark and lepton doublets, while u_R^i , d_R^i , and e_R^i are right-handed singlets under $SU(2)_L$. The Dirac adjoint is defined as $\bar{\psi} = \psi^\dagger \gamma^0$, and γ^μ are the Dirac gamma matrices. The covariant derivative D_μ contains the gauge fields and couplings appropriate to the gauge quantum numbers of each fermion field. Therefore, Eq. (2.31) describes both the free propagation of fermions and their gauge interactions.

The Higgs-sector terms are

$$\mathcal{L}_{\text{Higgs}} = (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi), \quad V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2. \quad (2.32)$$

The first term is the kinetic term of the Higgs doublet and also contains the gauge interactions of the Higgs field through the covariant derivative. The function $V(\Phi)$ is the Higgs potential. With the convention used here, $\mu^2 > 0$ and $\lambda > 0$. The condition $\lambda > 0$ ensures that the potential is bounded from below, while the negative quadratic term causes the minimum of the potential to occur at a non-zero value of $\Phi^\dagger \Phi$. This leads to electroweak symmetry breaking through the Brout–Englert–Higgs mechanism [7–9].

The Yukawa interactions are

$$\mathcal{L}_{\text{Yukawa}} = -y_d^{ij} \bar{Q}_L^i \Phi d_R^j - y_u^{ij} \bar{Q}_L^i \tilde{\Phi} u_R^j - y_e^{ij} \bar{L}_L^i \Phi e_R^j + \text{h.c.}, \quad (2.33)$$

where $i, j = 1, 2, 3$ are generation indices, and y_u^{ij} , y_d^{ij} , and y_e^{ij} are complex Yukawa coupling matrices in flavour space. h.c. denotes the Hermitian conjugate. The conjugate Higgs doublet is defined as

$$\tilde{\Phi} = i\tau^2 \Phi^*, \quad (2.34)$$

where τ^2 is the second Pauli matrix. The field $\tilde{\Phi}$ transforms as an $SU(2)_L$ doublet with hypercharge opposite to that of Φ , allowing a gauge-invariant Yukawa interaction for up-type quarks. After electroweak symmetry breaking, the Yukawa interactions generate masses for the charged fermions. In the minimal SM, no right-handed neutrino field is included, and therefore no renormalisable neutrino Yukawa mass term appears in Eq. (2.33).

2.1.7 Electroweak symmetry breaking and the Higgs mechanism

The Higgs potential in Eq. (2.32) has a minimum at a non-zero value of $\Phi^\dagger \Phi$. For the convention used in Eq. (2.32), the minimum satisfies $\Phi^\dagger \Phi = v^2/2$, where $v = \sqrt{\mu^2/\lambda}$ is the Higgs vacuum expectation value and sets the electroweak scale. A convenient choice of vacuum expectation value is

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}. \quad (2.35)$$

The value of v is related to the Fermi constant by $v = (\sqrt{2}G_F)^{-1/2} \simeq 246 \text{ GeV}$ [10].

This vacuum breaks the electroweak symmetry according to

$$SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}. \quad (2.36)$$

The unbroken symmetry is electromagnetism. This mass-generation mechanism for gauge bosons is the Brout–Englert–Higgs mechanism [7–9].

In unitary gauge, the Higgs doublet can be written as

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \quad (2.37)$$

where $h(x)$ is the physical Higgs boson field. Expanding the Higgs potential around the vacuum gives

$$V(\Phi) = V(\langle \Phi \rangle) + \frac{1}{2}m_h^2 h^2 + \lambda v h^3 + \frac{\lambda}{4}h^4, \quad (2.38)$$

where the physical Higgs-boson mass is

$$m_h^2 = 2\lambda v^2 = 2\mu^2. \quad (2.39)$$

Unlike the W and Z boson masses, which are fixed at tree level by the electroweak gauge couplings and the vacuum expectation value, the Higgs-boson mass depends on the quartic coupling λ . Therefore, the SM does not predict the numerical value of m_h before λ is fixed experimentally. The Higgs boson was observed by the ATLAS and CMS experiments in 2012 [11, 12]. The observed Higgs-boson mass is approximately 125 GeV, corresponding to $\lambda \simeq m_h^2/(2v^2) \simeq 0.13$ [13].

The masses of the electroweak gauge bosons arise from the Higgs kinetic term,

$$(D_\mu \Phi)^\dagger (D^\mu \Phi), \quad (2.40)$$

after substituting the Higgs vacuum expectation value. It is useful to define the charged weak fields as

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2). \quad (2.41)$$

The terms quadratic in the electroweak gauge fields are then

$$(D_\mu \langle \Phi \rangle)^\dagger (D^\mu \langle \Phi \rangle) = \frac{g^2 v^2}{4} W_\mu^+ W^{-\mu} + \frac{v^2}{8} (g W_\mu^3 - g' B_\mu)^2. \quad (2.42)$$

The first term is the mass term of the charged weak bosons. The second term shows that the neutral gauge fields W_μ^3 and B_μ mix and must be diagonalised.

The weak mixing angle is defined by

$$\cos \theta_W = \frac{g}{\sqrt{g^2 + g'^2}}, \quad \sin \theta_W = \frac{g'}{\sqrt{g^2 + g'^2}}. \quad (2.43)$$

The corresponding neutral mass eigenstates are

$$A_\mu = B_\mu \cos \theta_W + W_\mu^3 \sin \theta_W, \quad (2.44)$$

$$Z_\mu = -B_\mu \sin \theta_W + W_\mu^3 \cos \theta_W. \quad (2.45)$$

The field A_μ is identified with the photon, while Z_μ is the neutral weak boson. The photon remains massless because it is associated with the unbroken gauge symmetry $U(1)_{\text{em}}$.

The resulting gauge-boson masses are

$$m_W = \frac{1}{2} g v, \quad m_Z = \frac{1}{2} \sqrt{g^2 + g'^2} v = \frac{m_W}{\cos \theta_W}, \quad m_\gamma = 0. \quad (2.46)$$

Thus, the SM predicts the relation $m_W = m_Z \cos \theta_W$. Inserting $v \simeq 246$ GeV and the electroweak couplings gives $m_W \simeq 80$ GeV and $m_Z \simeq 91$ GeV. Therefore, the Higgs mechanism gives masses to the W and Z bosons while leaving the photon massless.

After electroweak symmetry breaking, the unbroken gauge symmetry is $U(1)_{\text{em}}$, whose gauge field is the photon field A_μ . At energies much below the electroweak scale, the electromagnetic interactions of charged fermions are described by quantum electrodynamics (QED).

For a charged fermion field ψ with electric charge Q , the QED Lagrangian can be written as

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi, \quad (2.47)$$

where

$$D_\mu = \partial_\mu - ieQA_\mu, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (2.48)$$

Here e is the electromagnetic coupling constant, Q is the electric charge of the fermion in units of e , and $F_{\mu\nu}$ is the electromagnetic field-strength tensor. Thus, QED appears as the low-energy gauge theory associated with the unbroken $U(1)_{\text{em}}$ symmetry of the SM.

2.1.8 Fermion masses and flavour mixing

The Yukawa interactions generate fermion masses after electroweak symmetry breaking. When the Higgs field acquires its vacuum expectation value v , the Yukawa coupling matrices are converted into fermion mass matrices:

$$M_u^{ij} = \frac{v}{\sqrt{2}}y_u^{ij}, \quad M_d^{ij} = \frac{v}{\sqrt{2}}y_d^{ij}, \quad M_e^{ij} = \frac{v}{\sqrt{2}}y_e^{ij}. \quad (2.49)$$

Here M_u , M_d , and M_e are the mass matrices of the up-type quarks, down-type quarks, and charged leptons, respectively. The matrices y_u , y_d , and y_e are the corresponding dimensionless Yukawa coupling matrices, and the indices $i, j = 1, 2, 3$ label the fermion generations.

The physical fermion masses are obtained by diagonalising the mass matrices. In the quark sector, the up-type and down-type mass matrices are diagonalised by different rotations in flavour space. As a result, flavour mixing appears in the charged-current weak interaction. This mixing is described by the Cabibbo–Kobayashi–Maskawa (CKM) matrix [14, 15]. In the mass eigenstate basis, the charged-current interaction of quarks is

$$\mathcal{L}_{\text{CC}} = \frac{g}{\sqrt{2}}\bar{u}_L^i \gamma^\mu V_{ij}^{\text{CKM}} d_L^j W_\mu^+ + \text{h.c.} \quad (2.50)$$

Here g is the $SU(2)_L$ gauge coupling, u_L^i and d_L^j are the left-handed up-type and down-type quark mass eigenstates, and W_μ^+ is the charged weak gauge boson. The matrix element V_{ij}^{CKM} describes the strength of the charged-current transition from the down-type quark d^j to the up-type quark u^i . The matrices γ^μ are the Dirac gamma matrices, and the Hermitian-conjugate term describes the corresponding interaction involving W_μ^- . The CKM matrix is

essential for describing flavour-changing charged-current processes such as $b \rightarrow c\tau\nu$. Its complex phase also provides a source of CP violation in the quark sector of the SM.

In the minimal SM, neutrino masses are absent because no right-handed neutrino fields are included and no renormalisable neutrino Yukawa interaction can be written with the minimal field content. This provides one of the motivations for considering extensions of the SM.

2.2 Beyond the Standard Model

Although the SM has been tested with high precision and has successfully described a wide range of experimental observations, it is not a complete theory of nature. Several experimental and theoretical facts point to physics beyond the SM.

2.2.1 Gravity and the Planck scale

The SM describes the electromagnetic, weak, and strong interactions as quantum gauge field theories, but it does not include gravity. Gravity is instead described at macroscopic scales by general relativity. A complete theory of nature is therefore expected to provide a consistent quantum description of gravity and to explain its connection to the SM interactions.

The difficulty of connecting gravity with the SM appears in two related ways. First, quantum gravitational effects are expected to become important at the Planck scale, $M_{\text{Pl}} \sim 10^{19}$ GeV. This is much higher than the electroweak scale, $v \simeq 246$ GeV. This large separation of scales is closely related to the hierarchy problem discussed below.

Second, constructing a quantum theory of gravity is theoretically challenging. If general relativity is quantised perturbatively in the same way as ordinary quantum field theories, it is not renormalisable as a fundamental theory. This makes it difficult to combine gravity with the quantum-field-theoretic framework of the SM in a straightforward way. Several possible approaches to quantum gravity have been proposed, including string theory, loop quantum gravity, and asymptotic safety, although no complete theory has been experimentally confirmed.

2.2.2 The hierarchy problem

The electroweak scale is set by the Higgs vacuum expectation value, $v \simeq 246$ GeV, and is closely related to the masses of the weak gauge bosons and the Higgs boson. In the SM, the

Higgs-boson mass is determined at tree level by

$$m_h^2 = 2\lambda v^2, \quad (2.51)$$

where λ is the Higgs quartic coupling. The observed value $m_h \simeq 125$ GeV therefore corresponds to $\lambda \simeq 0.13$.

The hierarchy problem arises because the Higgs mass parameter is highly sensitive to physics at very high energy scales. If the SM is treated as an effective theory valid up to an ultraviolet cutoff scale Λ , loop corrections to the Higgs mass can be written schematically as

$$m_{h,\text{phys}}^2 = m_{h,0}^2 + \delta m_h^2, \quad (2.52)$$

where $m_{h,0}^2$ is the bare Higgs mass parameter and δm_h^2 denotes the quantum correction. At one-loop order, the correction contains terms quadratically sensitive to the cutoff scale,

$$\delta m_h^2 \sim \frac{\Lambda^2}{16\pi^2}, \quad (2.53)$$

If the cutoff scale is close to the Planck scale, $\Lambda \sim M_{\text{Pl}} \sim 10^{19}$ GeV, the magnitude of δm_h^2 is many orders of magnitude larger than the observed value $m_h^2 \sim (125 \text{ GeV})^2$. Therefore, the bare parameter $m_{h,0}^2$ and the quantum correction δm_h^2 must cancel each other with extremely high precision in order to leave the observed electroweak-scale Higgs mass. This required cancellation is known as fine-tuning.

The hierarchy problem is the question of why the electroweak scale remains so much smaller than a possible ultraviolet scale such as the Planck scale. It motivates many extensions of the SM, such as supersymmetry, composite Higgs models, and other scenarios in which new physics appears near the TeV scale and stabilises the Higgs mass against large ultraviolet corrections. More generally, the hierarchy problem provides one of the motivations for experimental searches for new particles at the TeV scale.

2.2.3 Neutrino masses

In the minimal SM, neutrinos are massless because no right-handed neutrino fields are included and no renormalisable neutrino mass term can be written using only the SM field content. However, neutrino oscillation experiments have shown that neutrinos change flavour as they propagate, which requires non-zero neutrino mass differences. Evidence for atmospheric neutrino oscillations was reported by the Super-Kamiokande experiment [16], and solar neutrino flavour transformation was established by the Sudbury Neutrino Obser-

vatory [17]. Therefore, neutrino masses provide direct experimental evidence for physics beyond the minimal SM.

2.2.4 Dark matter

Astrophysical and cosmological observations indicate the existence of non-luminous matter that cannot be explained by the known particles of the SM. Measurements of the cosmic microwave background, for example by the Planck satellite, are well described by a cosmological model containing a significant non-baryonic dark matter component [18]. The SM does not contain a suitable cold dark matter candidate. Although neutrinos are electrically neutral and weakly interacting, their small masses and relativistic behaviour in the early universe do not provide the required cold dark matter component.

2.2.5 Matter–antimatter asymmetry

The observable universe contains much more matter than antimatter. Sakharov identified three necessary conditions for dynamically generating such a baryon asymmetry: baryon-number violation, C and CP violation, and departure from thermal equilibrium [19]. The SM contains CP violation through the CKM matrix and baryon-plus-lepton number violation through non-perturbative electroweak effects. However, the amount of CP violation in the SM and the nature of the electroweak phase transition are not sufficient to explain the observed baryon asymmetry of the universe [20, 21]. This motivates new sources of CP violation and new dynamics in the early universe.

2.2.6 The strong CP problem

The most general QCD Lagrangian allows a CP-violating term of the form

$$\mathcal{L}_\theta = \theta \frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}, \quad \tilde{G}^{a\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} G_{\rho\sigma}^a. \quad (2.54)$$

Here g_s is the strong coupling constant, and $G_{\mu\nu}^a$ is the gluon field-strength tensor of the $SU(3)_C$ gauge group, with the colour index $a = 1, \dots, 8$ labelling the eight gluon fields. The tensor $\tilde{G}^{a\mu\nu}$ is the dual field-strength tensor, defined using the totally antisymmetric Levi-Civita tensor $\varepsilon^{\mu\nu\rho\sigma}$. The parameter θ is a dimensionless QCD vacuum angle that controls the size of this CP-violating interaction. The factor $g_s^2/(32\pi^2)$ is a conventional normalisation.

The operator $G_{\mu\nu}^a \tilde{G}^{a\mu\nu}$ is gauge invariant and has mass dimension four, so it is allowed by the gauge symmetry and renormalisability of QCD. However, it violates CP symmetry. Therefore, a non-zero value of the effective QCD vacuum angle would lead to CP violation in the strong interaction.

This term would contribute to the electric dipole moment of the neutron. Experimental limits on the neutron electric dipole moment strongly constrain the effective QCD vacuum angle $\bar{\theta}$ [22]. The effective parameter $\bar{\theta}$ includes both the QCD vacuum angle θ and phases from the quark mass matrix. The absence of observable strong CP violation, despite the fact that such a term is allowed by the SM gauge symmetries, is known as the strong CP problem.

2.2.7 Flavour anomalies and lepton flavour universality

The flavour structure of the SM is mainly determined by the Yukawa matrices. After electroweak symmetry breaking, these matrices generate the fermion mass matrices; their eigenvalues determine the fermion masses, while the mismatch between the diagonalisation of the up- and down-type quark mass matrices gives rise to quark flavour mixing through the CKM matrix.

Although the SM accommodates the observed masses and mixing angles, it does not explain their hierarchical pattern or their origin. Precision measurements in flavour physics therefore provide sensitive probes of possible physics beyond the SM.

A key property of the SM is lepton flavour universality (LFU), which means that the electroweak gauge interactions couple equally to the three charged-lepton generations, apart from effects due to their different masses. Semileptonic B -meson decays provide precise tests of LFU by comparing decay rates involving different charged leptons. In charged-current transitions of the form $b \rightarrow c\ell\nu$, the commonly studied ratios are

$$R(D^{(*)}) = \frac{\mathcal{B}(\bar{B} \rightarrow D^{(*)} \tau \bar{\nu}_\tau)}{\mathcal{B}(\bar{B} \rightarrow D^{(*)} \ell \bar{\nu}_\ell)}, \quad \ell = e, \mu. \quad (2.55)$$

These observables compare decays involving a τ lepton with the corresponding decays involving light charged leptons.

Neutral-current transitions of the form $b \rightarrow s\ell^+\ell^-$ can be tested through ratios such as

$$R(K^{(*)}) = \frac{\mathcal{B}(B \rightarrow K^{(*)} \mu^+ \mu^-)}{\mathcal{B}(B \rightarrow K^{(*)} e^+ e^-)}. \quad (2.56)$$

These ratios compare final states with muon pairs and electron pairs. Recent measurements of $R(K)$ and $R(K^*)$ are compatible with SM predictions [23, 24]. However, measurements of

$R(D)$ and $R(D^*)$ continue to show tensions with SM predictions. The HFLAV CKM 2025 average gives a combined discrepancy of about 3.8σ with respect to the SM prediction [25] as shown in Figure 2.1. These observables remain important motivations for models that couple differently to different lepton generations.

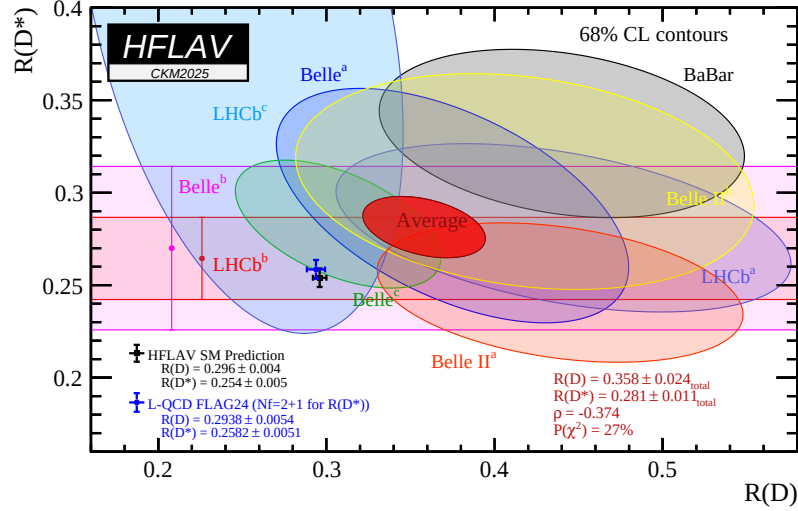


Fig. 2.1 Average of $R(D^*)$ and $R(D)$ [25] measured by the BaBar [26, 27], Belle [28–31], Belle-II [32, 33], and LHCb [34–36] experiments.

2.3 Possible Solutions to the Remaining Problems of the Standard Model

Many extensions of the SM have been proposed to address one or more of the problems described above. This section briefly introduces several representative ideas: supersymmetry, axions, and leptiquarks. These models are not mutually exclusive, and in many cases they address different aspects of the SM's limitations.

2.3.1 Supersymmetry

Supersymmetry (SUSY) is a spacetime symmetry relating bosons and fermions [37, 38]. In supersymmetric theories, each SM particle is accompanied by a superpartner whose spin differs by one half unit. The fermionic superpartners of bosons are conventionally named by adding the suffix '-ino', such as the gluino, wino, bino, and higgsino, while the bosonic superpartners of fermions are named by adding the prefix 's-', such as squarks and sleptons.

If supersymmetry were exact, particles and their superpartners would have equal masses. Since no such superpartners have been observed, SUSY must be broken if it is realised in nature.

One of the main motivations for SUSY is the hierarchy problem. Loop corrections to the Higgs mass from SM particles can be cancelled by corresponding contributions from their superpartners, reducing the sensitivity of the electroweak scale to very high energy scales [39]. In models with conserved R -parity, supersymmetric particles are produced in pairs and the lightest supersymmetric particle (LSP) is stable [40]. If the LSP is electrically neutral and weakly interacting, such as a neutralino in many SUSY scenarios, it can provide a dark matter candidate [41, 42]. SUSY can also improve the unification of gauge couplings at high energy, since the additional superpartners modify the renormalisation-group evolution of the gauge couplings [43, 44]. However, no direct evidence for supersymmetric particles has yet been observed, and the simplest low-scale SUSY scenarios are increasingly constrained by collider searches [45].

2.3.2 Axions and the strong CP problem

The Peccei–Quinn mechanism provides a dynamical solution to the strong CP problem [46, 47]. It introduces a new global chiral symmetry, usually denoted as the Peccei–Quinn symmetry $U(1)_{\text{PQ}}$, which is spontaneously broken. The associated pseudo-Nambu–Goldstone boson is the axion [48, 49]. The axion field dynamically relaxes the effective value of $\bar{\theta}$ to zero, thereby explaining the absence of observable strong CP violation. In many models, axions or axion-like particles can also contribute to the dark matter density of the universe [50].

2.3.3 Leptoquarks

Leptoquarks provide another well-motivated class of particles beyond the Standard Model. They are hypothetical bosons that couple directly to a quark and a lepton, thereby connecting the quark and lepton sectors. Such particles appear naturally in several extensions of the SM, including models based on quark–lepton unification, such as the Pati–Salam framework [51].

Leptoquarks are particularly interesting because they can affect semileptonic processes. This makes them relevant to the flavour anomalies observed in several B -meson decay measurements, where the experimental results have shown tensions with the SM predictions in observables testing lepton-flavour universality.

Among the possible leptoquark scenarios, the vector singlet leptoquark U_1 is especially well motivated for charged-current anomalies such as $b \rightarrow c \tau \nu$. This motivates collider

738 searches for final states containing hadronically decaying τ leptons. The following chapter
739 introduces the leptoquark models in more detail and the leptoquark searches at the LHC.

Chapter 3

Leptoquark Search at the LHC

3.1 Leptoquark Models and Phenomenology

3.1.1 General properties of leptoquarks

Leptoquarks (LQs) are hypothetical colour-charged bosons that couple directly to a quark and a lepton and therefore mediate interactions connecting the quark and lepton sectors. Depending on their spin, leptoquarks can be classified as scalar leptoquarks with spin zero or vector leptoquarks with spin one. They appear in several theoretical frameworks, including grand unified theories such as $SU(5)$ and $SO(10)$, quark–lepton unification models such as the Pati–Salam model, composite models, and models motivated by flavour anomalies [52, 51, 53].

Leptoquarks are particularly interesting in the context of semileptonic flavour physics. Their tree-level exchange can generate effective four-fermion interactions involving two quark fields and two lepton fields. Such interactions can modify flavour observables in processes such as $b \rightarrow c \tau \nu$ and $b \rightarrow s \ell^+ \ell^-$. In addition, the flavour structure of LQ couplings is model dependent and can be arranged to favour interactions with third-generation fermions. This feature is motivated by the distinctive role of the top quark, bottom quark, and τ -lepton in many flavour-motivated scenarios.

At the LHC, leptoquarks can be searched for directly. Since they carry colour charge, they can be pair-produced through QCD interactions. Additional production modes, such as single production or non-resonant exchange, are also possible, depending on the LQ coupling to SM quarks and leptons. Therefore, the LQ searches in ATLAS provide information complementary to low-energy flavour measurements: flavour observables probe virtual effects through precision measurements, while the search at pp collider probes direct LQ production or high-energy virtual exchange.

3.1.2 The simplified U_1 leptoquark model

Several leptoquark representations have been studied in connection with semileptonic B -meson anomalies, including the scalar singlet S_1 and the scalar doublet R_2 . In this thesis, however, the benchmark interpretation is based on the vector singlet leptoquark U_1 . The U_1 benchmark is well motivated because its left-handed coupling to quark and lepton doublets can generate the SM-like vector operator for $b \rightarrow c\tau\nu$. In addition, flavour structures with enhanced couplings to third-generation fermions naturally lead to final states involving τ -leptons, neutrinos, and jets.

The U_1 leptoquark transforms as $U_1^\mu \sim (3, 1, 2/3)$ under the SM gauge group. It is therefore a colour-triplet, weak-singlet vector boson with hypercharge $Y = 2/3$. Since it is an $SU(2)$ singlet, its electric charge is $Q = T^3 + Y = 2/3$.

A commonly used simplified-model Lagrangian [54] for U_1 is

$$\begin{aligned} \mathcal{L}_U = & -\frac{1}{2}U_{1\mu\nu}^\dagger U_1^{\mu\nu} + M_U^2 U_{1\mu}^\dagger U_1^\mu \\ & -ig_c(1 - \kappa_c)U_{1\mu}^\dagger T^a U_{1\nu} G^{a\mu\nu} - i\frac{2}{3}g_Y(1 - \kappa_Y)U_{1\mu}^\dagger U_{1\nu} B^{\mu\nu} \\ & + (U_1^\mu J_\mu + \text{h.c.}). \end{aligned} \quad (3.1)$$

Here U_1^μ is the vector leptoquark field and M_U is the leptoquark mass. The first term is the kinetic term, while the second term is the mass term. The field-strength tensor of U_1 is defined as

$$U_{1\mu\nu} = D_\mu U_{1\nu} - D_\nu U_{1\mu}, \quad (3.2)$$

with the covariant derivative

$$D_\mu = \partial_\mu - ig_c G_\mu^a T^a - i\frac{2}{3}g_Y B_\mu. \quad (3.3)$$

Here g_c and g_Y are the colour and hypercharge gauge couplings. The fields G_μ^a and B_μ are the gluon and hypercharge gauge fields, and $G^{a\mu\nu}$ and $B^{\mu\nu}$ are their corresponding field-strength tensors. The matrices T^a are the generators of $SU(3)$. For a colour-triplet field, they are given by $T^a = \lambda^a/2$, where λ^a are the Gell-Mann matrices and $a = 1, \dots, 8$.

The parameters κ_c and κ_Y control the interactions of the vector leptoquark with the SM gauge fields. $\kappa_c = \kappa_Y = 0$ corresponds to the Yang–Mills scenario, whereas $\kappa_c = \kappa_Y = 1$ corresponds to the minimal-coupling scenario [55]. The Yang–Mills scenario is adopted throughout this thesis.

The last term in Eq. (3.1) describes the interaction of U_1 with SM fermions through the current J_μ . It is given by

$$J_\mu = \frac{g_U}{\sqrt{2}} [\beta_L^{i\alpha} (\bar{q}_L^i \gamma_\mu \ell_L^\alpha) + \beta_R^{i\alpha} (\bar{d}_R^i \gamma_\mu e_R^\alpha)]. \quad (3.4)$$

Here $i = 1, 2, 3$ labels the quark generation and $\alpha = 1, 2, 3$ labels the lepton generation. q_L^i and ℓ_L^α are left-handed quark and lepton doublets, while d_R^i and e_R^α are right-handed down-type quark and charged-lepton singlets. The parameter g_U denotes the overall strength of the U_1 coupling to fermions. The matrices $\beta_L^{i\alpha}$ and $\beta_R^{i\alpha}$ are complex 3×3 flavour-coupling matrices for the left- and right-handed interactions, respectively. The strength of a given flavour coupling is therefore proportional to $g_U \beta_{L,R}^{i\alpha}$.

The down-quark and charged-lepton mass eigenstate basis is adopted for the $SU(2)$ multiplets:

$$q_L^i = \begin{pmatrix} V_{ji}^* u_L^j \\ d_L^i \end{pmatrix}, \quad \ell_L^\alpha = \begin{pmatrix} \nu_L^\alpha \\ e_L^\alpha \end{pmatrix}, \quad (3.5)$$

where V is the CKM matrix.

Expanding the left-handed current from Eq. 3.4 gives

$$\bar{q}_L^i \gamma_\mu \ell_L^\alpha = V_{ji} \bar{u}_L^j \gamma_\mu \nu_L^\alpha + \bar{d}_L^i \gamma_\mu e_L^\alpha. \quad (3.6)$$

This expression shows that a single left-handed flavour coupling $\beta_L^{i\alpha}$ gives two related interactions. One involves an up-type quark together with a neutrino, while the other involves a down-type quark together with a charged lepton.

The simplified U_1 model allows general flavour couplings to different quark and lepton generations. Meanwhile, we focus on the charged-current $b \rightarrow c \tau \nu$ transition and the signatures containing a τ -lepton and a neutrino. Therefore, the couplings to second-generation leptons in our benchmark are set to zero, which is $\beta_L^{22} = \beta_L^{32} = 0$, and only the couplings involving third-generation leptons are retained. Finally, the flavour-coupling matrices are taken to be

$$\beta_L = \begin{pmatrix} 0 & 0 & \beta_L^{13} \\ 0 & 0 & \beta_L^{23} \\ 0 & 0 & \beta_L^{33} \end{pmatrix}, \quad \beta_R = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \beta_R^{33} \end{pmatrix}. \quad (3.7)$$

The third-generation coupling is normalised to $\beta_L^{33} = 1$, while g_U remains a free parameter controlling the overall fermionic coupling strength.

Within this benchmark, β_L^{23} controls the $s\tau$ and $c\nu_\tau$ coupling through the CKM factor V_{cs} , while β_L^{33} controls the $b\tau$ coupling and contributes $c\nu_\tau$ coupling as $V_{cb}\beta_L^{33}$.

The heavy U_1 field can be integrated out at the matching scale $\mu \sim M_U$. At tree level, its exchange generates four-fermion operators involving left- and right-handed quark and lepton fields [56]. The corresponding effective Lagrangian can be written as

$$\mathcal{L}_{\text{eff}}^U = -\frac{2}{v^2} \left[C_{LL}^{ij\alpha\beta} \mathcal{O}_{LL}^{ij\alpha\beta} + C_{RR}^{ij\alpha\beta} \mathcal{O}_{RR}^{ij\alpha\beta} + \left(C_{LR}^{ij\alpha\beta} \mathcal{O}_{LR}^{ij\alpha\beta} + \text{h.c.} \right) \right]. \quad (3.8)$$

The four-fermion operators are defined as

$$\mathcal{O}_{LL}^{ij\alpha\beta} = (\bar{q}_L^i \gamma_\mu \ell_L^\alpha) (\bar{\ell}_L^\beta \gamma^\mu q_L^j), \quad (3.9)$$

$$\mathcal{O}_{RR}^{ij\alpha\beta} = (\bar{d}_R^i \gamma_\mu e_R^\alpha) (\bar{e}_R^\beta \gamma^\mu d_R^j), \quad (3.10)$$

$$\mathcal{O}_{LR}^{ij\alpha\beta} = (\bar{q}_L^i \gamma_\mu \ell_L^\alpha) (\bar{e}_R^\beta \gamma^\mu d_R^j). \quad (3.11)$$

The Wilson coefficients are

$$C_{LL}^{ij\alpha\beta} = C_U \beta_L^{i\alpha} (\beta_L^{j\beta})^*, \quad (3.12)$$

$$C_{RR}^{ij\alpha\beta} = C_U \beta_R^{i\alpha} (\beta_R^{j\beta})^*, \quad (3.13)$$

$$C_{LR}^{ij\alpha\beta} = C_U \beta_L^{i\alpha} (\beta_R^{j\beta})^*. \quad (3.14)$$

The overall coefficient is defined as

$$C_U = \frac{g_U^2 v^2}{4M_U^2}, \quad (3.15)$$

where $v \simeq 246$ GeV is the Higgs vacuum expectation value. The coefficient C_U sets the overall strength of the effective interaction and scales as g_U^2/M_U^2 , while the flavour dependence is encoded in the coupling matrices β_L and β_R .

The effective Lagrangian for the $b \rightarrow c\tau\nu$ transition can be written as:

$$\mathcal{L}_{\text{eff}}^{b \rightarrow c\tau\nu} = -\frac{4G_F}{\sqrt{2}} V_{cb} \left[(1 + C_{LL}^c) (\bar{c}_L \gamma_\mu b_L) (\bar{\tau}_L \gamma^\mu \nu_{\tau L}) - 2C_{LR}^c (\bar{c}_L b_R) (\bar{\tau}_R \nu_{\tau L}) \right], \quad (3.16)$$

where G_F is the Fermi constant. The coefficient C_{LL}^c modifies the SM-like left-handed vector interaction, while C_{LR}^c is associated with the scalar operator arising from the combination of left- and right-handed fermion couplings of the U_1 leptoquark.

For the flavour structure introduced above, the corresponding coefficients are

$$C_{LL}^c = C_U \frac{(\beta_L^{33})^*}{V_{cb}} (V_{cd}\beta_L^{13} + V_{cs}\beta_L^{23} + V_{cb}\beta_L^{33}), \quad (3.17)$$

$$C_{LR}^c = C_U \frac{(\beta_R^{33})^*}{V_{cb}} (V_{cd}\beta_L^{13} + V_{cs}\beta_L^{23} + V_{cb}\beta_L^{33}), \quad (3.18)$$

Using the normalisation $\beta_L^{33} = 1$ and neglecting the small first-generation coupling β_L^{13} , the coefficients reduce to¹

$$C_{LL}^c \simeq C_U \left(1 + \frac{V_{cs}}{V_{cb}} \beta_L^{23} \right), \quad (3.19)$$

$$C_{LR}^c \simeq C_U (\beta_R^{33})^* \left(1 + \frac{V_{cs}}{V_{cb}} \beta_L^{23} \right). \quad (3.20)$$

3.1.3 Low-energy fit and benchmark scenarios

The Wilson coefficients C_{LL}^c and C_{LR}^c introduced above can be constrained using measurements of semileptonic charged-current B -meson decays. Ref. [56] performed a combined fit to low-energy $b \rightarrow c \tau \nu$ observables and determines the preferred region in the (C_{LL}^c, C_{LR}^c) plane. The fit result is shown in Fig. 3.1. The Wilson coefficients in this figure are evolved to a common reference scale of 1 TeV to facilitate comparison with high-energy observables. The best-fit point is $(C_{LL}^c, C_{LR}^c) = (0.05, -0.02)$, and the preferred region is compatible with both pure-left-handed ($\beta_R^{33} = 0$) and sizeable right-handed ($\beta_R^{33} = -1$) scenarios. Although the sign of β_R^{33} affects the low-energy relation between C_{LL}^c and C_{LR}^c , the collider predictions used in this thesis are identical for $\beta_R^{33} = +1$ and -1 . The new-physics amplitudes with different fermion chiralities do not interfere, and the amplitude proportional to a product of left- and right-handed couplings does not interfere with the SM amplitude. The collider interpretation therefore depends only on $|\beta_R^{33}|$.

Pure-left-handed scenario. For $\beta_R^{33} = 0$, Eq. 3.20 gives $C_{LR}^c = 0$. Therefore, the U_1 leptoquark contribution to $b \rightarrow c \tau \nu$ is described only by the left-handed vector operator, as given by Eq. 3.19. This is the primary benchmark scenario considered in this thesis.

¹The small first-generation coupling $\beta_L^{13} = (V_{td}^*/V_{ts}^*)\beta_L^{23} \sim 0.2\beta_L^{23}$.

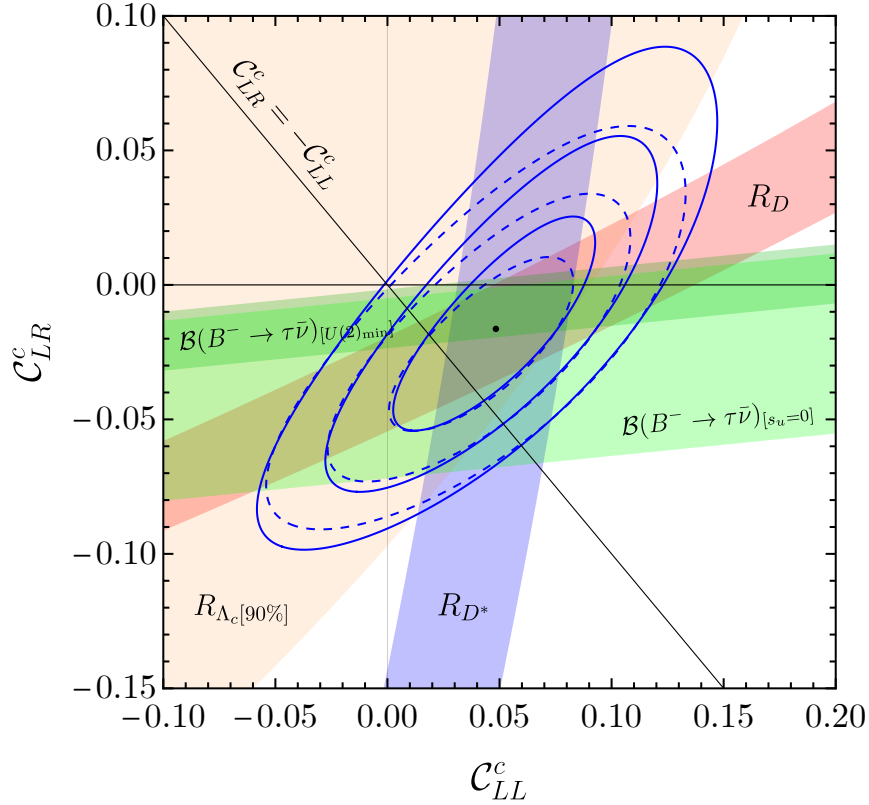


Fig. 3.1 The result in (C_{LL}^c, C_{LR}^c) plane obtained from the fit to low-energy $b \rightarrow c \tau \nu$ observables. The blue solid ellipses correspond to the 1, 2 and 3 σ contours. The blue dotted lines are obtained including also the constraint from $\mathcal{B}(B_u^- \rightarrow \tau \bar{\nu})$. The black point indicates the best-fit point of the fit, $(C_{LL}^c, C_{LR}^c) = (0.05, -0.02)$. The coloured bands show the $\Delta\chi^2 = 1$ regions preferred by the individual observables. The pure-left-handed scenario ($\beta_R^{33} = 0$) and the equal magnitude right-handed scenario ($\beta_R^{33} = -1$) are given by the black solid lines [56].

The corresponding B -anomaly-favoured region can be obtained from the intersection of the low-energy fit with the line $C_{LR}^c = 0$ in Fig. 3.1.

Right-handed scenario. The second benchmark is obtained by allowing a sizeable right-handed third-generation coupling. For $\beta_R^{33} = -1$, the relation between the Wilson coefficients becomes

$$C_{LR}^c = -C_{LL}^c. \quad (3.21)$$

This benchmark therefore corresponds to the diagonal line in Fig. 3.1, where the right-handed third-generation coupling has the same magnitude as the left-handed coupling.

The two scenarios are described by the parameter space (β_L^{23}, M_U, g_U) with different fixed values of β_R^{33} . Particular emphasis is placed on β_L^{23} , since charged-current low-energy data alone do not provide a stringent constraint on this coupling. For fixed M_U and g_U , β_L^{23} has a strong impact on C_{LL}^c because its contribution is enhanced by the CKM ratio V_{cs}/V_{cb} .

Indeed, one can constrain the size of β_L^{23} by considering possible ultraviolet (UV) completions of the simplified U_1 model. While the simplified model parametrises the relevant interactions directly in terms of the couplings β_L^{ij} , the UV completion provides a more fundamental description of their origin. One example is the “4321” model, which is based on the extended gauge group $\mathcal{G}_{4321} = SU(4) \times SU(3)' \times SU(2)_L \times U(1)'$. In the flavour structure considered in this model, the U_1 couplings are non-universal, with $|\beta_L^{23}| < |\beta_L^{33}|$ [54].

Additional constraints can be introduced when considering $\Delta F = 2$ observables. These are processes in which a quark-flavour quantum number changes by two units, including neutral-meson mixing such as $B_s - \bar{B}_s$, $K - \bar{K}$, and $D - \bar{D}$ mixing. In a given UV completion, the flavour-symmetry breaking that generates non-universal U_1 couplings may also be transmitted to the neutral-current sector, inducing flavour-changing neutral currents (FCNC) and consequently additional contributions to $\Delta F = 2$ processes. The strong experimental constraints on neutral-meson mixing therefore restrict the allowed flavour-violating structure of the UV model and indirectly constrain β_L^{23} .

Combining such considerations of the UV flavour structure with constraints from $\Delta F = 2$ observables, Ref. [56] identifies

$$\beta_L^{23} \in [0.06, 0.16] \quad (3.22)$$

as a theoretically motivated range for experimental searches. **However, this constraint should not be interpreted as a model-independent experimental bound on β_L^{23} .** Additional particles, flavour symmetries, or cancellation mechanisms may suppress flavour-

changing transitions and thereby allow larger values of β_L^{23} . Therefore, large values of β_L^{23} remain a possibility within the U_1 model, and it is important to test this complementary region directly.

3.1.4 Leptoquark searches at the LHC

Leptoquarks can be searched for at the LHC through several production modes, including pair production, resonant single production, and non-resonant t -channel exchange. Pair production is dominated by QCD interactions. For a fixed κ choice, its cross section depends strongly on M_U but weakly on the couplings g_U and $\beta^{i\alpha}$. Pair-production searches therefore provide relatively robust constraints on the leptoquark mass.

ATLAS and CMS have performed searches for pair-produced leptoquarks coupling to third-generation fermions in final states containing τ -leptons, b -jets, and, depending on the decay mode, missing transverse momentum [57, 58]. For the vector U_1 , the mass limit extends to approximately 1.5–1.8 TeV as shown in Fig. 3.3. The limit also depends on the choice of the U_1 -gluon coupling, such as the Yang–Mills or minimal-coupling scenario, as well as on the decay pattern determined by the assumed fermionic couplings.

For single production, the cross section also depends strongly on the leptoquark–fermion couplings. Searches for singly produced leptoquarks decaying into $b\tau$ have been performed in the $b\tau\tau$ final state, where the leptoquark is produced in association with a τ -lepton as shown in Fig. 3.2a [57, 59]. In addition, virtual t -channel U_1 exchange can contribute to final states such as $\tau\tau$ or $\tau\nu$, enhancing the event yields in the high-mass region. ATLAS and CMS have therefore searched for non-resonant leptoquark contributions in high-mass $\tau\tau$ production [60, 61]. These searches can remain sensitive to masses beyond the reach of the direct pair-production searches, particularly for large g_U , since the sensitivity is approximately governed by g_U/M_U .

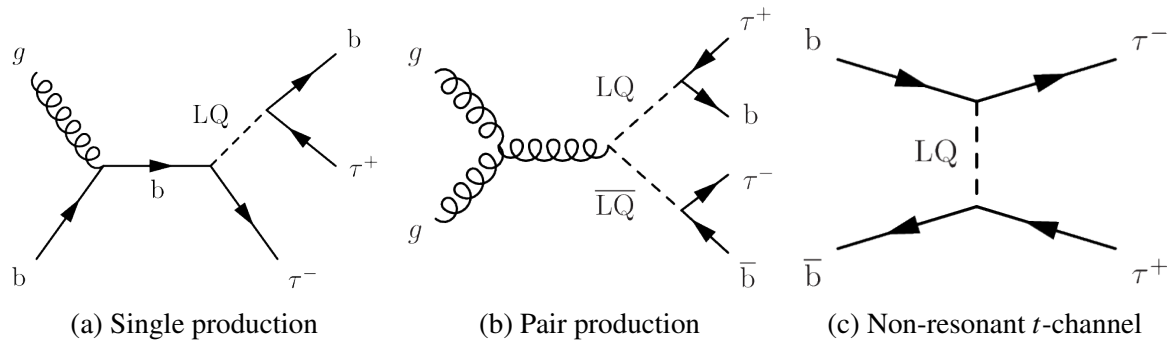


Fig. 3.2 Example Feynman diagrams of LQ production and exchange process for $\tau\tau(+b)$ searches.

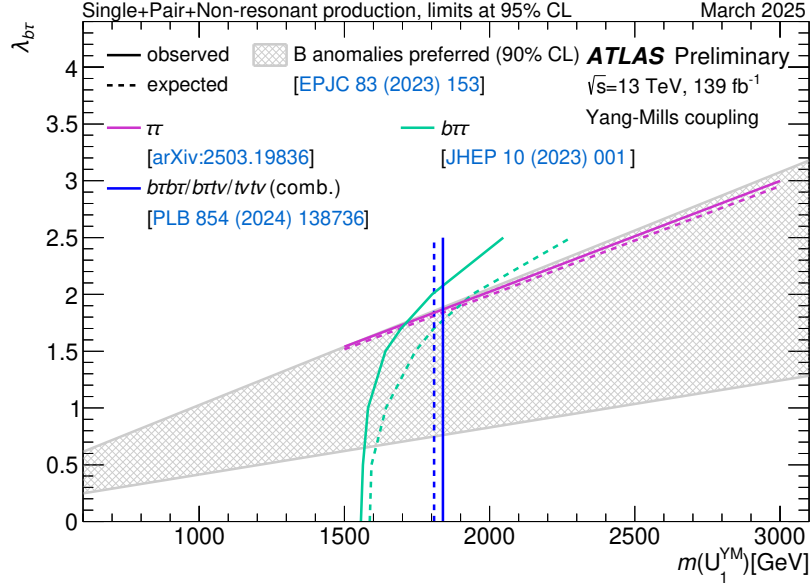


Fig. 3.3 Summary of the observed and expected 95% CL constraints on the U_1 vector-leptoquark coupling $\lambda_{b\tau}$ as a function of the leptoquark mass in the Yang–Mills coupling scenario, with $\beta_R^{33} = 0$. Pair-production searches dominate the direct mass constraint at lower masses, while single-production and $\tau\tau$ searches provide coupling-dependent sensitivity at larger masses. Here $\lambda_{b\tau} = g_U/\sqrt{2}$ for $\beta_L^{33} = 1$ [62].

On the other hand, the existing $\tau\tau(+b)$ searches generally assume β_L^{23} to be small or neglect its contribution, motivated by the flavour assumptions discussed in Sec. 3.1.3. Consequently, scenarios with a sizeable $c\nu_\tau$ coupling, and in particular the large- β_L^{23} region, have not been systematically explored at the LHC.

The $\tau\nu$ final state provides a complementary probe of the $b\tau$ and $c\nu_\tau$ coupling. CMS has performed an inclusive search for events with a hadronically decaying τ -lepton and large missing transverse momentum, including an interpretation in terms of leptoquark exchange [63], but without an explicit b -jet requirement. This thesis presents a dedicated search in the $\tau\nu(+b)$ final state. Prior to the analysis, no dedicated ATLAS or CMS leptoquark search had targeted this topology. The $\tau\nu(+b)$ search presented here is therefore complementary to the existing $\tau\tau(+b)$ searches: it probes both the $b\tau$ and $c\nu_\tau$ couplings and extends the collider interpretation to large values of β_L^{23} . The corresponding benchmark and target signatures are introduced in the following subsection.

3.1.5 Signal model and target signatures

In this thesis, we focus on the simplified U_1 model introduced in Sec. 3.1.2 in the Yang–Mills scenario, corresponding to $\kappa_c = \kappa_Y = 0$ in Eq. 3.1. The third-generation coupling is normalised to $\beta_L^{33} = 1$, while the primary interpretation assumes a pure-left-handed scenario ($\beta_R^{33} = 0$). An additional interpretation including a right-handed coupling is presented with the final results. Therefore, the signal is characterised by (M_U, g_U, β_L^{23}) .

Figure 3.4 shows representative U_1 signal diagrams targeted by the analysis. Both resonant single production and non-resonant t -channel exchange are considered, and each is divided according to the presence or absence of a b -tagged jet. This results in four target signatures: resonant $\tau\nu + b$, non-resonant $\tau\nu + b$, resonant $\tau\nu + 0b$, and non-resonant $\tau\nu + 0b$.

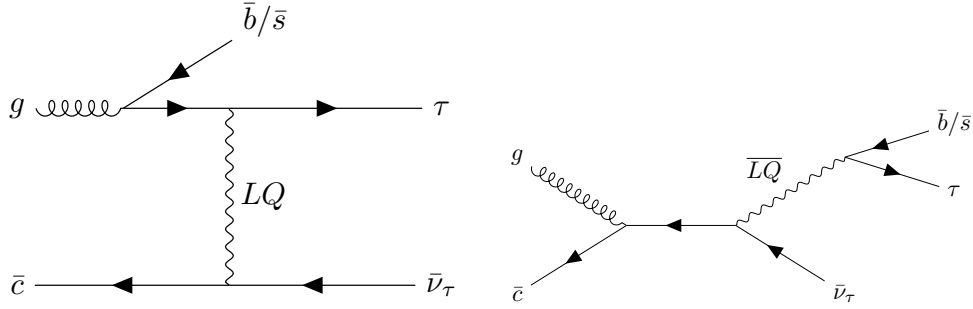


Fig. 3.4 Signal diagrams targeted in this analysis: non-resonant t -channel exchange (left) and resonant single production (right). The final-state jet can originate from a b - or s -quark.

The b -jet requirement provides strong discrimination against the dominant SM $\tau\nu + \text{jets}$ background, since genuine SM production of a $\tau\nu$ system in association with a b -quark is CKM suppressed [64]. The $\tau\nu + b$ categories are therefore particularly sensitive to the combination of the $b\tau$ and $c\nu_\tau$ interactions.

The zero- b categories extend the sensitivity to the large β_L^{23} region explored in this thesis. As β_L^{23} increases, the $s\tau$ interaction also becomes sizeable, producing signal events without a final-state b -quark. This contribution is particularly relevant for resonant production, where an on-shell U_1 can decay into $s\tau$. The four categories therefore provide complementary sensitivity to the different flavour and production configurations of the U_1 signal.

Chapter 4

LHC-ATLAS Experiment

4.1 Large Hadron Collider

The European Organisation for Nuclear Research (CERN) is an international particle-physics laboratory located near Geneva on the border between France and Switzerland. Since its foundation in 1954, CERN has developed and operated a chain of accelerators and experimental facilities to study the fundamental constituents of matter and their interactions [65]. The Large Hadron Collider (LHC) is the largest and highest-energy particle collider in the CERN accelerator complex. It is installed in the former Large Electron–Positron Collider (LEP) tunnel, which has a circumference of about 27 km and lies between approximately 45 m and 170 m below the surface [66, 67]. A schematic overview of the LHC ring and its interaction points is shown in Fig. 4.1.

The LHC was proposed as the next major collider after LEP. The first circulating beams were achieved in 2008, and the first physics run began in 2010. The discovery of a new boson compatible with the Standard Model (SM) Higgs boson by the ATLAS and CMS experiments in 2012 marked one of the central achievements of the LHC physics programme [11, 12]. Beyond Higgs-boson measurements and precision studies of SM processes, a major purpose of the LHC is to search for new particles and interactions beyond the SM.

The LHC collides protons at high centre-of-mass energies in order to extend the kinematic reach for the production of heavy particles. For two proton beams with equal beam energy E_{beam} , the proton–proton centre-of-mass energy is approximately $\sqrt{s} = 2E_{\text{beam}}$. However, the hard-scattering process in a proton–proton (pp) collision occurs between partons, namely quarks and gluons, which carry only fractions of the proton momenta. The partonic centre-of-mass energy therefore varies from event to event and is generally lower than the nominal pp collision energy. Increasing $\sqrt{s}$ consequently extends the accessible mass range for new heavy particles.

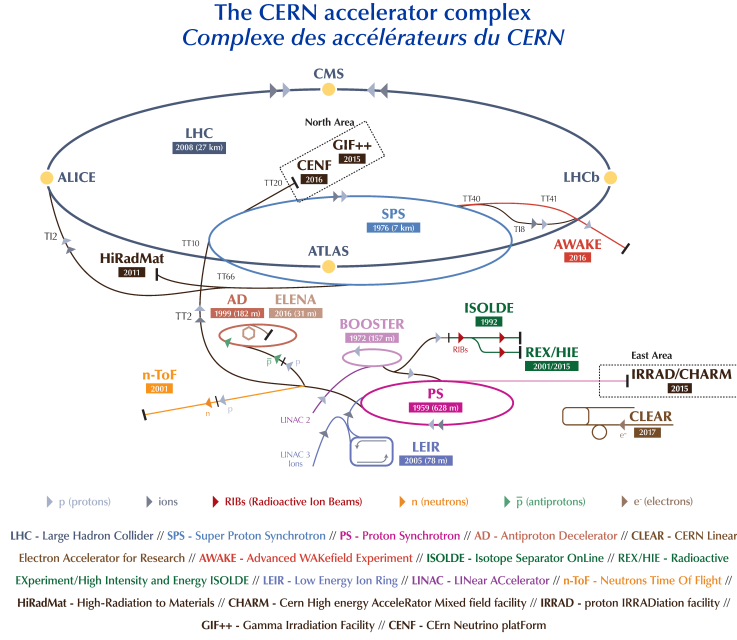


Fig. 4.1 Schematic layout of the CERN accelerator complex, including the LHC ring and its main interaction regions [68].

Another key machine parameter is the luminosity, which quantifies the collision rate per unit cross section. For a process with cross section σ , the production rate is

$$R = \mathcal{L}\sigma, \quad (4.1)$$

where $\mathcal{L}$ is the instantaneous luminosity. For two Gaussian bunches colliding with frequency f_{coll} , the luminosity can be expressed as [69]

$$\mathcal{L} = f_{\text{coll}} \frac{n_1 n_2}{4\pi\sigma_x^* \sigma_y^*} \mathcal{F}, \quad (4.2)$$

where n_1 and n_2 are the numbers of particles in the two colliding bunches, σ_x^* and σ_y^* are the RMS transverse beam sizes in the horizontal and vertical directions at the interaction point, and $\mathcal{F}$ is a geometrical reduction factor accounting for effects such as the crossing angle. The integrated luminosity,

$$\mathcal{L}_{\text{int}} = \int \mathcal{L} dt, \quad (4.3)$$

determines the total expected number of produced events by $N = \sigma \mathcal{L}_{\text{int}}$. High instantaneous luminosity increases the event rate, while high integrated luminosity improves the sensitivity to rare processes with small production cross sections.

Four major experiments are installed around the LHC ring. ATLAS, located at Point 1, and CMS, located at Point 5, are general-purpose detectors designed for a broad physics programme including Higgs-boson measurements, precision SM studies, and searches for new particles. ALICE, located at Point 2, is optimised for heavy-ion physics and the study of strongly interacting matter at high energy density. LHCb, located at Point 8, is designed for precision studies of heavy-flavour hadrons and flavour-changing processes [66]. The analysis presented in this thesis uses data recorded by the ATLAS detector.

The LHC physics programme is organised into run periods. Run 1 took place from 2010 to 2012, with pp collision data recorded at $\sqrt{s} = 7$ TeV and $\sqrt{s} = 8$ TeV. Run 2 took place from 2015 to 2018 at $\sqrt{s} = 13$ TeV. After standard data-quality selections, the full ATLAS Run 2 pp dataset corresponds to an integrated luminosity of $140.1 \pm 1.2 \text{ fb}^{-1}$, with a relative uncertainty of 0.83% [70]. Run 3 began in 2022 after the second long shutdown and operated at the higher centre-of-mass energy of $\sqrt{s} = 13.6$ TeV [71]. The Run 2 dataset is used for the leptoquark search presented in this thesis.

An important consequence of high-luminosity operation at the LHC is pile-up. The proton beams are divided into bunches, and a bunch crossing occurs when bunches from the two beams pass through the interaction point. Several pp interactions can occur in a single bunch crossing. The interaction corresponding to the hard-scattering process of interest is referred to as the hard-scatter interaction, while additional inelastic pp interactions in the same bunch crossing constitute in-time pile-up. Detector signals may also originate from interactions in neighbouring bunch crossings, depending on the detector response time. This contribution is referred to as out-of-time pile-up.

The mean number of inelastic pp collisions per bunch crossing is commonly denoted by μ . During Run 2, the nominal bunch spacing was 25 ns, corresponding to a maximum bunch-crossing frequency of 40 MHz. The luminosity-weighted mean value over the full Run 2 dataset was $\langle \mu \rangle = 33.7$, corresponding to approximately 34 interactions per bunch crossing [72].

Pile-up has an important impact on event reconstruction. It produces extra charged-particle tracks, calorimeter energy deposits, and reconstructed jets. These contributions can affect the reconstruction of the primary vertex, jet energies, hadronically decaying τ -leptons, flavour-tagged jets, and missing transverse momentum. Pile-up effects must therefore be modelled in simulated samples and mitigated through dedicated reconstruction and calibration procedures.

4.2 ATLAS experiment

4.2.1 Overview

ATLAS, which stands for A Toroidal LHC ApparatuS, is a detector located at Point 1 of the Large Hadron Collider. It was designed to exploit the full physics potential of the LHC, including precision measurements of SM processes, studies of the Higgs boson, and searches for new physics beyond the SM. The detector surrounds the nominal interaction point and provides nearly full solid-angle coverage, allowing the reconstruction of a wide range of final-state particles produced in proton–proton collisions.

The ATLAS detector is approximately 44 m long, 25 m in diameter, and has a total weight of about 7000 tonnes [73]. It has a cylindrical, forward–backward symmetric layout centred on the interaction point. A schematic view of the detector is shown in Figure 4.2.

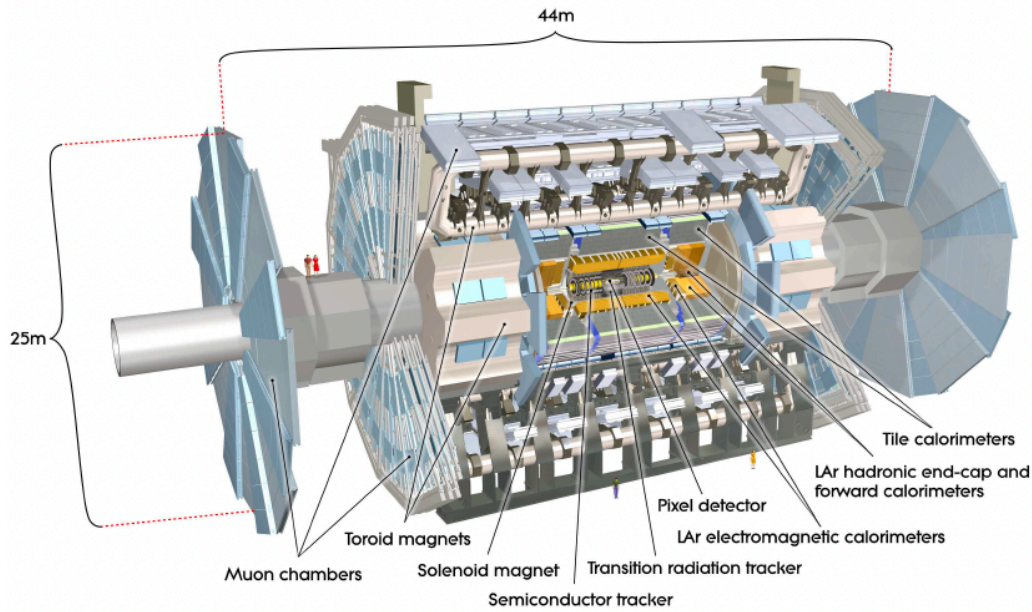


Fig. 4.2 Schematic view of the ATLAS detector and its subdetector systems [73].

Starting from the beam line and moving outwards, the main detector systems are the Inner Detector (ID), the calorimeters, and the Muon Spectrometer (MS):

- **Inner Detector (ID)** is immersed in a 2 T axial magnetic field provided by a thin superconducting solenoid and is used to reconstruct charged-particle trajectories close to the interaction point.
- **Calorimeters** surround the ID and measure the energies of electrons, photons, hadrons, hadronically decaying τ -leptons, and jets.

- ***Muon Spectrometer (MS)*** forms the outermost major detector system and measures muons in the magnetic field generated by large superconducting toroids.

The magnetic fields provided by the solenoid and toroids allow the momentum and charge of charged particles to be determined from the curvature of their trajectories. The trigger and data acquisition system selects potentially interesting events from the high LHC bunch-crossing rate for permanent storage and offline reconstruction.

The detector configuration described in this chapter corresponds primarily to Run 2. Several detector and trigger upgrades were installed for Run 3 to cope with higher luminosity conditions, but these upgrades are not used in the analysis presented in this thesis [74].

4.2.2 Coordinate system

ATLAS uses a right-handed coordinate system centred at the nominal interaction point. The z -axis is defined along the beam direction, the x -axis points from the interaction point towards the centre of the LHC ring, and the y -axis points upwards [73]. Cylindrical coordinates (r, ϕ) are used in the plane transverse to the beam, where r is the distance from the beam axis and ϕ is the azimuthal angle around the z -axis.

The polar angle θ is measured with respect to the positive z -axis. The pseudorapidity η is defined as

$$\eta = -\ln \tan \left(\frac{\theta}{2} \right). \quad (4.4)$$

The transverse momentum is defined as $p_T = p \sin \theta$, and the transverse energy is defined similarly. The angular separation between two reconstructed objects is commonly expressed as

$$\Delta R = \sqrt{(\Delta \eta)^2 + (\Delta \phi)^2}. \quad (4.5)$$

These quantities are used throughout this thesis to describe object kinematics, angular separations, overlap removal, and event selections. Figure 4.3 illustrates the coordinate system used in ATLAS.

4.2.3 Magnet system

The ATLAS magnet system consists of four large superconducting magnets: a central solenoid, one barrel toroid, and two end-cap toroids, as shown in Figure 4.4. The central solenoid surrounds the Inner Detector with a length of approximately 5.3 m and provides an

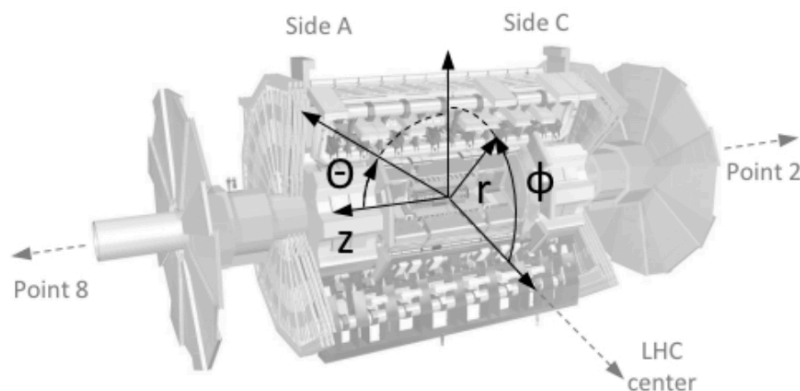


Fig. 4.3 The coordinate system used in ATLAS [75].

axial magnetic field of 2 T for charged-particle momentum measurements in the tracking system. The solenoid is designed to minimise the amount of inactive material in front of the electromagnetic calorimeter.

The toroid system provides magnetic fields of approximately 0.5 T and 1 T in the barrel and end-cap regions of the Muon Spectrometer, respectively. The barrel toroid surrounds the calorimeter system in the central region, while the two end-cap toroids provide bending power in the forward regions. The toroidal field bends muon trajectories, allowing their momenta to be measured using the precision tracking chambers of the MS. The air-core design of the toroids reduces multiple scattering and helps preserve the momentum resolution for high- p_T muons [73]. The combination of the solenoid and toroid systems allows charged-particle momenta to be measured independently in the ID and, for muons, in the MS.

4.2.4 Inner Detector

The Inner Detector (ID) is the innermost tracking system of ATLAS and provides tracking for charged particles within $|\eta| < 2.5$ [73]. It is immersed in the 2 T magnetic field of the central solenoid and records local position measurements, called hits, along their trajectories. During Run 2, it consisted of the pixel detector, the Semiconductor Tracker (SCT), and the Transition Radiation Tracker (TRT). The Insertable B-Layer (IBL) is installed before Run 2 as the innermost pixel layer [76]. Figure 4.5 shows a transverse view of the Inner Detector.

- *Insertable B-Layer (IBL) and Pixel detector*

The pixel detector provides high-granularity measurements close to the beam line. Its precise spatial resolution is essential for reconstructing tracks near the interaction point and for identifying displaced vertices from the decays of long-lived particles such

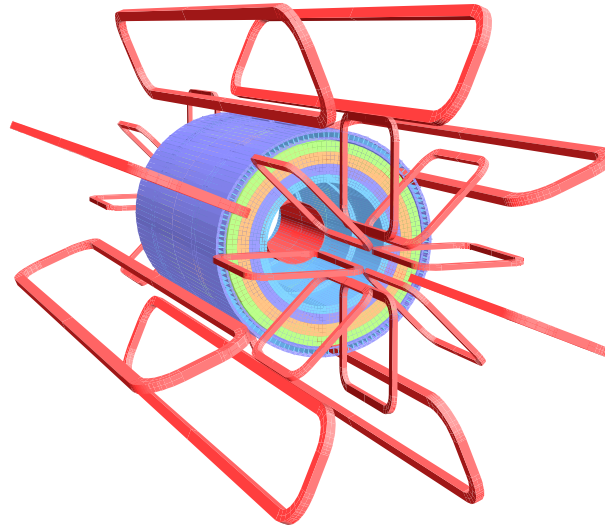


Fig. 4.4 Layout of the ATLAS superconducting magnet system, including the central solenoid, barrel toroid, and end-cap toroids [73].

as b - and c -hadrons. The IBL further improves the impact-parameter resolution and enhances the robustness of tracking and flavour tagging under high pile-up conditions.

- ***Semiconductor Tracker (SCT)***

The SCT surrounds the pixel detector and forms the intermediate precision-tracking system of the ID. It is a silicon microstrip detector covering $|\eta| < 2.5$, consisting of four cylindrical barrel layers and two end-caps with nine disks on each side. Each module contains silicon strip sensors arranged back-to-back with a small stereo angle, allowing two-dimensional position information to be reconstructed. The SCT typically provides four space-point measurements per track, improving pattern recognition, momentum reconstruction, and vertex reconstruction when combined with the high-granularity pixel measurements.

- ***Transition Radiation Tracker (TRT)***

The TRT is the outermost tracking subsystem of the ID and provides tracking information at larger radii, covering approximately $|\eta| < 2.0$. It is based on thin straw-tube drift chambers, which provide a large number of measurements along the trajectory of each charged particle. Although the spatial precision of each TRT hit is lower than that of the silicon detectors, the large number of straw hits and the longer lever arm improve the momentum measurement and help stabilise track reconstruction.

Together, these subsystems provide the tracking information needed in ATLAS.

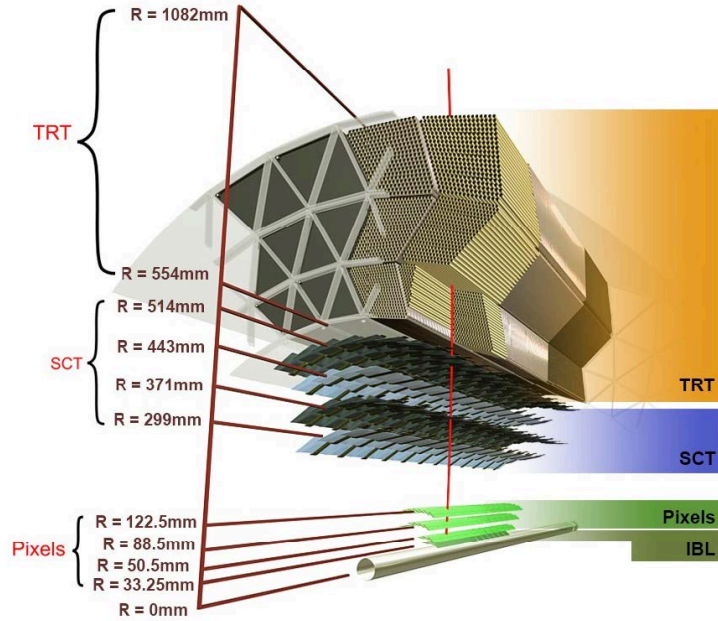


Fig. 4.5 Transverse view of the ATLAS Inner Detector, showing the IBL, pixel detector, SCT, and TRT subsystems [76].

4.2.5 Calorimeter

The ATLAS calorimeter system surrounds the Inner Detector and measures the energies and directions of particles [73]. It consists of electromagnetic (EM) and hadronic calorimeters, as shown in Figure 4.6. Specifically, it provides measurements of charged particles, photons and jets, and covers the region up to $|\eta| < 4.9$. It also provides the shower information to the Level-1 trigger system.

- **Electromagnetic (EM) calorimeter**

The EM calorimeter uses liquid argon (LAr) as the active medium and lead absorbers arranged in an accordion geometry, as shown in Figure 4.7. The accordion geometry provides continuous azimuthal coverage and fine segmentation of electromagnetic showers. The EM calorimeter is divided into a barrel section covering $|\eta| < 1.475$ and two end-cap sections covering $1.375 < |\eta| < 3.2$. It is used to reconstruct and identify electrons and photons and also contributes to the reconstruction of hadronically decaying τ -leptons, particularly through the measurement of electromagnetic energy deposits from neutral-pion decays, which don't leave the charged-particle tracks in the ID.

The EM calorimeter consists of a presampler followed by three longitudinal sampling layers, each segmented into readout cells, as illustrated in Figure 4.8. The presampler

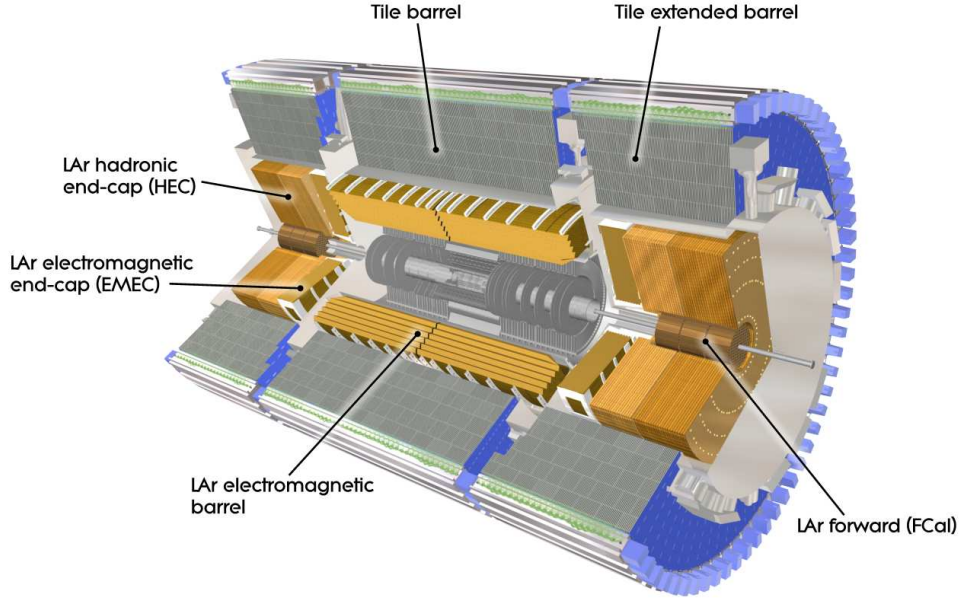


Fig. 4.6 Schematic view of the ATLAS calorimeter system [73].

is installed for $|\eta| < 1.8$ and is used to correct for energy lost in material upstream of the calorimeter. The first sampling layer has fine segmentation in η , which is useful for resolving the transverse shape of electromagnetic showers, while the middle layer collects the largest fraction of the shower energy. The back layer provides an additional measurement of the longitudinal shower development.

The energy resolution of the electromagnetic calorimeter can be parameterised as

$$\frac{\sigma_E}{E} = \frac{a}{\sqrt{E}} \oplus b \oplus \frac{c}{E}, \quad (4.6)$$

where a is the intrinsic sampling term, b is the constant term, and c represents the contribution from electronic and pile-up noise [77].

- **Hadronic calorimeter**

The hadronic calorimeter uses different structure for different η regions. In the region of $|\eta| < 1.7$, the Tile Calorimeter uses steel absorbers and scintillating tiles as the active medium. It measures the energies of hadrons and jets. In the end-cap and forward regions, liquid-argon calorimeters are used. The hadronic end-cap calorimeter consists of two wheels per end-cap, covers the range $1.5 < |\eta| < 3.2$, and uses copper absorbers. The forward calorimeter extends the coverage to $3.1 < |\eta| < 4.9$ and uses

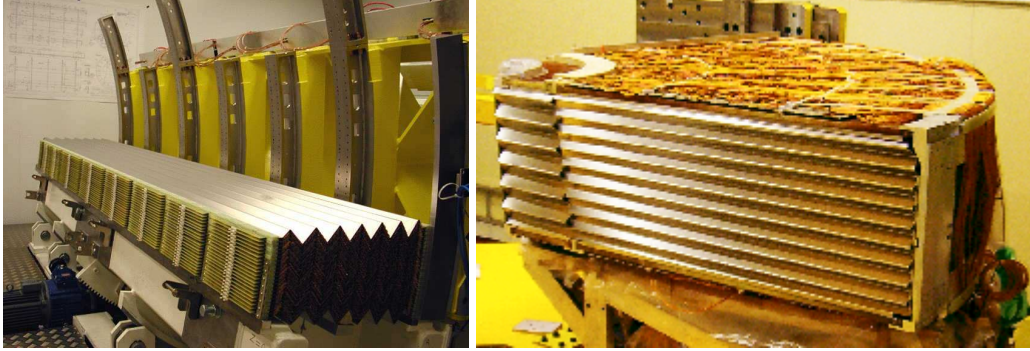


Fig. 4.7 Photograph of the EM barrel (left) and end-cap (right) LAr modules. The accordion absorber structure is visible [73].

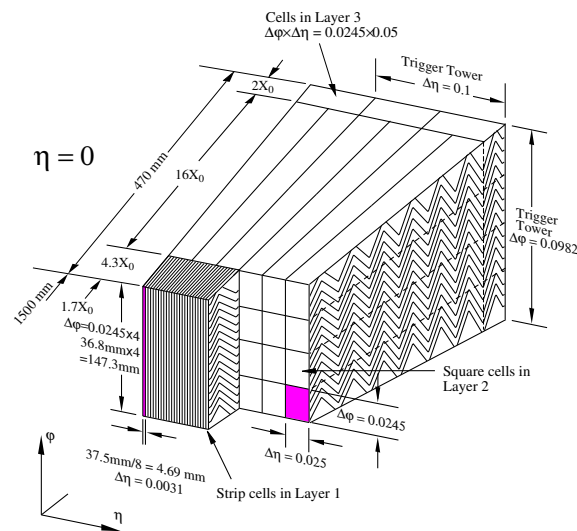


Fig. 4.8 Cut-away view of an ATLAS LAr electromagnetic calorimeter module, showing the longitudinal sampling layers and the segmentation of the readout cells [73].

1153 dense absorber materials suitable for the high-radiation environment close to the beam
1154 line.

1155 4.2.6 Muon Spectrometer

1156 The Muon Spectrometer (MS) forms the outermost major detector system of ATLAS. It is
1157 embedded in the magnetic field produced by the barrel and end-cap air-core toroid magnets.
1158 The bending of muon trajectories in this field is measured by precision chambers, and the
1159 track sagitta is used to determine the muon momentum. The MS provides precision muon
1160 measurements in the range $|\eta| < 2.7$, while the dedicated muon trigger chambers cover the
1161 region up to approximately $|\eta| < 2.4$ [73].

1162 During Run 2, the MS consisted of Monitored Drift Tubes (MDTs), Cathode Strip
1163 Chambers (CSCs), Resistive Plate Chambers (RPCs) and Thin Gap Chambers (TGCs). A
1164 view of these chambers is shown in Figure 4.9.

1165 • *Monitored Drift Tubes (MDTs)*

1166 The MDT chambers provide precision tracking over most of the MS acceptance. They
1167 consist of aluminium drift tubes with a diameter of about 30 mm, each containing a
1168 central sense wire. A muon crossing a tube ionises the gas, and the drift time of the
1169 electrons to the wire is measured to determine the track position. Several tube layers
1170 are grouped into chamber stations, providing precise measurements in the bending
1171 plane of the toroidal magnetic field. MDT chambers are used over most of the region
1172 $|\eta| < 2.7$, except for the innermost forward stations at $2.0 < |\eta| < 2.7$, where CSCs
1173 are used.

1174 • *Cathode Strip Chambers (CSCs)*

1175 The CSCs are multi-wire proportional chambers used in the innermost forward muon
1176 stations, where the particle rates are highest. They cover approximately $2.0 < |\eta| < 2.7$
1177 and provide precision measurements in the forward region. Their small drift distance
1178 and segmented cathode-strip readout allow them to operate efficiently in the high-rate
1179 environment close to the beam line.

1180 • *Resistive Plate Chambers (RPCs)*

1181 The RPCs provide fast muon trigger information in the barrel region, covering approx-
1182 imately $|\eta| < 1.05$. They are gaseous parallel-plate detectors operated in avalanche
1183 mode. Their fast time response allows the correct bunch crossing to be identified
1184 for each muon candidate and provides the Level-1 muon trigger in the barrel. The

RPCs also provide coordinate measurements complementary to the precision MDT chambers.

- *Thin Gap Chambers (TGCs)*

The TGCs are multi-wire proportional chambers designed to provide fast signals for the Level-1 trigger. They extend to approximately $|\eta| < 2.7$ and provide trigger coverage in the end-cap region up to $|\eta| < 2.4$. The TGC system identifies high- p_T muon candidates and assigns them to the correct bunch crossing. It also provides a second coordinate measurement, complementing the precision tracking information from MDTs and CSCs.

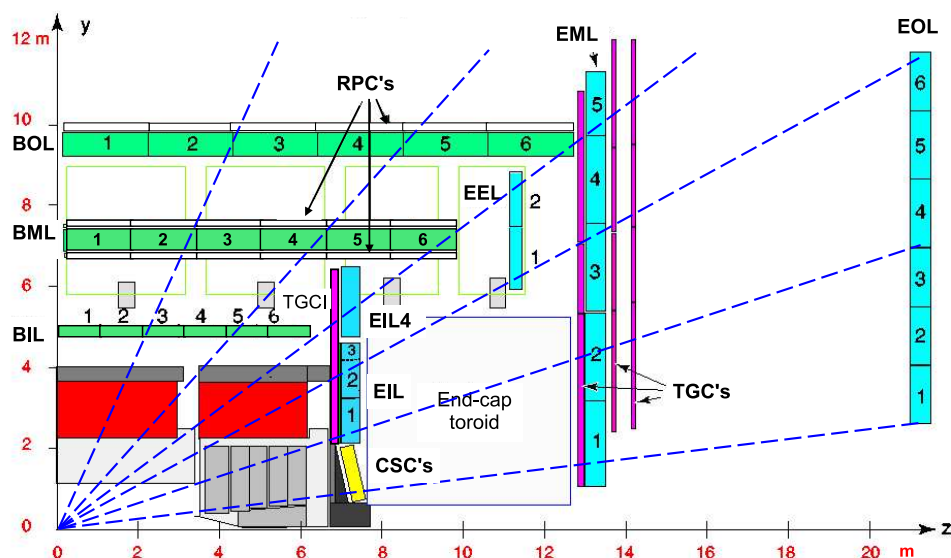


Fig. 4.9 Cross-section of the ATLAS Muon Spectrometer, showing the barrel and end-cap muon chambers. The green and cyan chambers represent the MDT system, while the dashed lines indicate infinite-momentum muon trajectories [73].

During the second long shutdown, the innermost end-cap muon stations were replaced by the New Small Wheels (NSWs) for Run 3 operation. This upgrade was introduced to maintain good muon trigger and reconstruction performance under higher luminosity and pile-up conditions. The NSW detectors are therefore not relevant to the Run 2 dataset analysed in this thesis.

4.2.7 Data Acquisition and Trigger System

At the LHC, proton bunches can cross at a rate of up to 40 MHz for a bunch spacing of 25 ns. This rate is far above the available detector readout and permanent-storage bandwidth. The ATLAS trigger and data acquisition (TDAQ) system is therefore designed to select potentially interesting events online and reduce the event rate to a manageable level. ATLAS uses a two-level trigger system consisting of a hardware-based Level-1 (L1) trigger and a software-based High-Level Trigger (HLT) [78].

Level-1 (L1) trigger

The L1 trigger is the first stage of the ATLAS online event selection. It is implemented in custom hardware and uses reduced-granularity detector information to make a fast decision with a latency of approximately $2.5 \mu\text{s}$. It reduces the input rate from up to 40 MHz to at most 100 kHz. The L1 trigger was based on information from the calorimeter and muon trigger systems, together with topological selections applied to L1 trigger objects. A schematic overview of the Run 2 TDAQ system is shown in Figure 4.10.

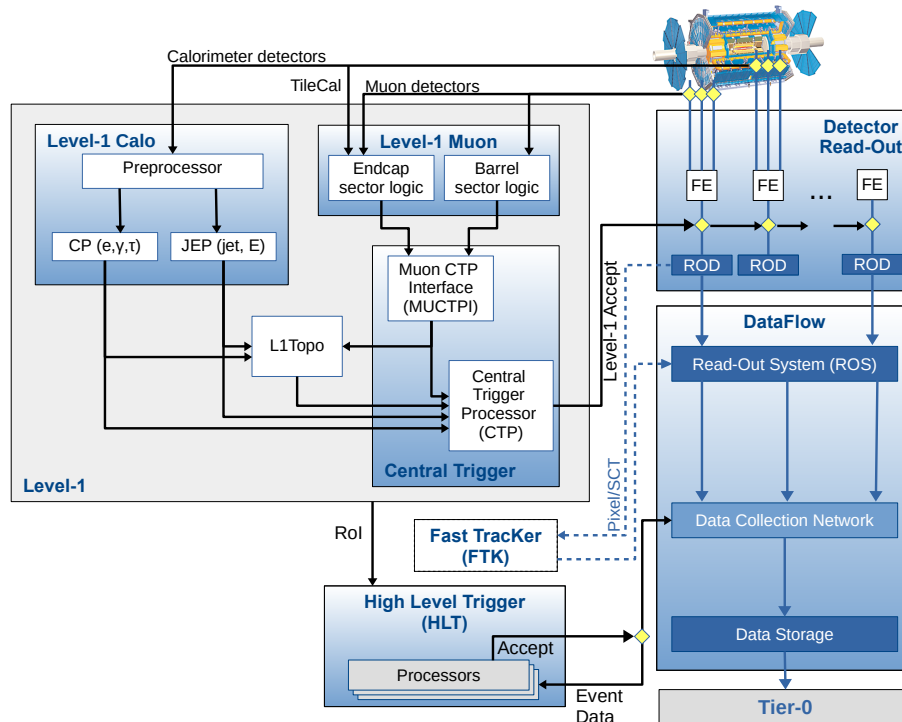


Fig. 4.10 Schematic overview of the ATLAS Run-2 trigger and data acquisition system [78].

• *L1 calorimeter trigger (L1Calo)*

L1Calo uses reduced-granularity information from the electromagnetic and hadronic calorimeters to reconstruct energy deposits within the Level-1 latency. Signals from neighbouring calorimeter cells are combined into trigger towers in the (η, ϕ) plane, and dedicated hardware algorithms search for local maxima in the transverse-energy distribution by comparing the energies of neighbouring towers. The same calorimeter information is also summed over larger regions to reconstruct jet-like energy deposits and global quantities such as the total and missing transverse energy.

• *L1 muon trigger (L1Muon)*

L1Muon identifies muon candidates using fast signals from the RPCs and the TGCs. Hits in different trigger-chamber layers are required to form spatial coincidence patterns consistent with a muon trajectory, while their timing information is used to assign the candidate to the correct bunch crossing. The amount of bending in the magnetic field is used to determine whether the candidate satisfies predefined transverse-momentum thresholds.

• *L1 topological trigger (L1Topo)*

L1Topo receives compact trigger-object information reconstructed by L1Calo and L1Muon, including their transverse momentum or energy and (η, ϕ) positions. Dedicated FPGA hardware combines these objects and calculates event-level topological quantities. And these quantities are compared with predefined selection requirements.

The trigger decisions from L1Calo, L1Muon, and L1Topo are then sent to the Central Trigger Processor (CTP), which combines them according to the predefined L1 trigger menu.

If at least one enabled trigger item is satisfied, the CTP issues a Level-1 Accept (L1A) signal for the corresponding bunch crossing. The L1A signal is distributed to the detector front-end electronics and initiates the transfer of the detector data for that bunch crossing from the front-end buffers to the readout system.

For accepted events, the L1 trigger also provides Regions of Interest (RoIs), which identify detector regions containing the trigger objects responsible for the L1 decision. These RoIs provide seed information for the HLT and allow detailed reconstruction to be performed initially in limited detector regions, reducing the required processing time.

High-Level Trigger

The HLT is the second stage of the ATLAS trigger system. It is implemented in a large computing farm and runs software algorithms using detector information with substantially

1246 finer granularity than is available at L1. The HLT reconstruction is guided by RoIs identified
1247 by the L1 trigger, while full-event reconstruction can also be used for signatures such
1248 as jets and missing transverse momentum. The HLT algorithms are designed to provide
1249 reconstruction performance close to that of the offline algorithms while remaining sufficiently
1250 fast for online operation. As a result, the HLT reduced the L1 output rate to an average
1251 recording rate of approximately 1 kHz for permanent storage.

1252 The trigger menu comprises the active L1 and HLT selections used during data taking with
1253 prescale factors, which are applied to selected trigger items to control their rates according to
1254 the instantaneous luminosity and available bandwidth.

Chapter 5

Data and Monte Carlo Simulations

Monte Carlo (MC) simulation is used to model the signal and Standard Model background processes and to predict their event yields and kinematic distributions in this search. Event generation begins by sampling the incoming partons and their momentum fractions from proton PDFs and evaluating the matrix element for the hard-scattering process. The outgoing partons are then evolved through the parton shower and hadronisation, followed by the decays of unstable particles. Finally, pile-up interactions and the ATLAS detector response are simulated, after which the events are reconstructed with the same algorithms used for collision data.

This chapter describes the collision data and simulated event samples used in the search. The Run 2 dataset is presented first, followed by the simulated signal and Standard Model background samples. The trigger selections are summarised at the end of the chapter.

5.1 Dataset selection

The search uses the full LHC Run 2 proton–proton collision dataset recorded by the ATLAS detector between 2015 and 2018 at a centre-of-mass energy of $\sqrt{s} = 13$ TeV. Only data recorded during stable-beam periods and passing the data-quality requirements are retained [72]. These periods are defined by the ATLAS good-runs lists (GRLs).

The selected dataset corresponds to an integrated luminosity of $140.1 \pm 1.2 \text{ fb}^{-1}$, with a relative uncertainty of 0.83% [70]. The integrated luminosity recorded in each year is summarised in Table 5.1.

The mean pile-up during Run 2 was $\langle\mu\rangle \simeq 34$, as described in Section 4.1.

Table 5.1 Integrated luminosity of the Run 2 proton–proton collision data used in this analysis.

Year	Integrated luminosity [fb^{-1}]
2015	3.24 ± 0.04
2016	33.40 ± 0.30
2017	44.63 ± 0.50
2018	58.79 ± 0.64
Total	140.07 ± 1.17

5.2 Signal samples

The signal samples implement the U_1 vector-leptoquark benchmark defined in Section 3.1.5. The Yang–Mills scenario with $\kappa_c = \kappa_Y = 0$ and $\beta_L^{33} = 1$ is used. The simulated signal covers both the pure-left-handed scenario with $\beta_R^{33} = 0$ and the right-handed scenario with $|\beta_R^{33}| = 1$. Apart from their coupling parameters, the same event-generation, parton-shower and detector-simulation settings are used for both scenarios.

Signal events are generated at leading order (LO) in QCD using MADGRAPH5_AMC@NLO 3.3.1 [79], interfaced to PYTHIA 8.308 [80]. The hard-scattering process produces a τ -lepton, a ν_τ and up to two additional partons, which can be gluons or u -, d -, s -, c - or b -quarks. The NNPDF3.0nlo PDF set [81] is used for the hard-scattering calculation.

The generation is performed in the five-flavour scheme, in which the b -quark is treated as an active parton in the proton. The proton PDFs therefore allow processes with an initial-state b -quark. All allowed U_1 diagrams contributing to this final state are retained in the matrix-element calculation. The resonant single-production and non-resonant t -channel processes shown in Figure 3.4 are therefore included coherently and cannot be separated at the generation stage. Other allowed contributions, including leptoquark pair production where kinematically accessible, are also retained.

PYTHIA subsequently simulates the parton shower and hadronisation. It also models the underlying event using the A14 tune [82] and the NNPDF2.3lo PDF set [83]. The CKKW–L merging procedure [84] prevents the same radiation from being included in both the matrix-element and parton-shower calculations. The Durham $k_\perp$ merging scale is set to 15 GeV and the D parameter to 0.4. Together, they determine whether an emission is treated as an additional matrix-element parton or as part of the shower.

For the pure-left-handed interpretation, dedicated signal samples are generated for 152 combinations of (M_U, g_U, β_L^{23}) satisfying

$$1.5 \text{ TeV} \leq M_U \leq 3.0 \text{ TeV}, \quad (5.1)$$

$$0.5 \leq g_U \leq 3.0, \quad (5.2)$$

$$0 \leq \beta_L^{23} \leq 2.2, \quad (5.3)$$

with $\beta_L^{33} = 1$ and $\beta_R^{33} = 0$. The lower boundary of the mass scan covers the region around the exclusion limits obtained from previous leptoquark pair-production searches [57, 58]. For $M_U > 3 \text{ TeV}$, the selected signal is dominated by non-resonant exchange, where the sensitivity is governed mainly by the ratio g_U/M_U . The primary mass scan is therefore restricted to $M_U \leq 3 \text{ TeV}$.

For the right-handed interpretation with $|\beta_R^{33}| = 1$, one baseline sample is generated at each of the six mass points $M_U \in \{1.5, 2.0, 2.5, 3.0, 4.0, 5.0\} \text{ TeV}$ with $g_U = 1$ and $\beta_L^{23} = 0.2$. Predictions at other values of g_U and β_L^{23} are obtained from these baseline samples using corrected MADGRAPH5_AMC@NLO internal event weights, rather than from separately generated samples at every parameter point. The correction includes an adjustment of the normalisation and a two-dimensional truth-level shape correction based on the leading-jet p_T and $m_{b\tau}$. Details of the correction and its validation are given in Appendix A. This strategy avoids repeating event generation and detector simulation for every coupling configuration, thereby reducing the required computing resources.

The U_1 amplitude can interfere with the SM amplitude for production of the same $\tau\nu_\tau + \text{jets}$ final state. The total matrix element can be expressed as

$$|\mathcal{M}_{\text{SM}} + \mathcal{M}_{U_1}|^2 = |\mathcal{M}_{\text{SM}}|^2 + |\mathcal{M}_{U_1}|^2 + 2 \text{Re}(\mathcal{M}_{\text{SM}}\mathcal{M}_{U_1}^*). \quad (5.4)$$

The first term represents the SM contribution, the second represents the pure U_1 contribution, corresponding to the BSM-only component, and the third represents their interference. Unlike the BSM-only component, the interference term can be negative and can reduce the predicted yield relative to the SM-only expectation in some regions of phase space.

To separate the BSM-only and interference terms in Eq. (5.4), the two contributions are generated independently with MADGRAPH5_AMC@NLO using the configuration described above. The BSM-only component is selected by requiring exactly two new-physics couplings at the amplitude level, denoted by NP==2, whereas the interference component is isolated at the squared-matrix-element level by requiring $\text{NP}^2 = 2$. The interference sample therefore contains only the third term in Eq. (5.4).

All τ -lepton decay modes are retained in the signal simulation. Although the signal regions target hadronically decaying τ -leptons, leptonic decays can contribute to regions containing electrons or muons. The inclusive decay configuration allows this signal contamination to be evaluated.

The decays of bottom- and charm-hadrons are simulated with EVTGEN [85]. All signal samples used in this thesis are processed with the ATLAS fast detector simulation ATLFAT3 [86], which provides a computationally efficient simulation of the calorimeter response.

5.3 Background samples

The simulated SM backgrounds considered in this analysis are $W + \text{jets}$, $Z + \text{jets}$, $t\bar{t}$, single-top and diboson production. The dominant contribution is $W \rightarrow \tau\nu + \text{jets}$ production. The QCD multijet background is negligible after event cleaning and is therefore neglected, as discussed in Appendix B.

The $W/Z + \text{jets}$ samples are generated with SHERPA 2.2.11 [87], except for $Z \rightarrow \tau\tau$, which uses version 2.2.14. Matrix elements include up to two additional partons at next-to-leading order (NLO) in QCD and up to five at LO. They are merged with the SHERPA parton shower [88] using the MEPS@NLO prescription [89–92].

Only a small fraction of inclusive $W \rightarrow \tau\nu$ events passes the high- E_T^{miss} requirement, leaving limited MC statistics. Two dedicated filtered samples are therefore used to reduce the MC statistical uncertainty in the high- E_T^{miss} tail. Their respective generator-level requirements are

$$\begin{aligned} 70 \text{ GeV} < E_T^{\text{miss}} < 150 \text{ GeV} \quad \text{and} \quad p_T^\tau < 150 \text{ GeV}, \\ E_T^{\text{miss}} > 150 \text{ GeV} \quad \text{or} \quad p_T^\tau > 150 \text{ GeV}. \end{aligned} \tag{5.5}$$

Top-quark pair and single-top production are generated at NLO accuracy in QCD using POWHEG BOX v2 [93–96], interfaced to PYTHIA 8.230 [97] for parton showering and hadronisation. The single-top samples include s -channel, t -channel and associated Wt production. The predicted $t\bar{t}$ yield is normalised to the inclusive production cross-section calculated at next-to-next-to-leading order (NNLO) in QCD with next-to-next-to-leading-logarithmic (NNLL) soft-gluon resummation [98]. A dilepton filter is applied to the $t\bar{t}$ sample, requiring both W bosons to decay leptonically.

Diboson production is also modelled with SHERPA.

The hard-scattering calculations use the NNPDF3.0nnlo PDF set for $W/Z + \text{jets}$ and NNPDF3.0nlo for $t\bar{t}$ and single-top production [81]. The underlying event is modelled using the A14 tune for PYTHIA and the default tune for SHERPA. For samples not generated with SHERPA, the decays of bottom- and charm-hadrons are modelled with EVTGEN [85]. The full ATLAS detector simulation based on GEANT4 [99, 100] is used for all background samples.

The principal MC generator configurations are summarised in Table 5.2.

Table 5.2 Summary of the principal MC generator configurations used for the signal and SM background processes. The quoted perturbative order refers to the hard-scattering matrix-element calculation. For the $W/Z + \text{jets}$ samples, matrix elements with up to two additional partons are calculated at NLO and those with up to five additional partons at LO. The symbol ℓ denotes an electron or muon.

Process	Generator	ME order	Parton shower	ME PDF set	Tune
U_1 signal	MADGRAPH5_AMC@NLO 3.3.1	LO	PYTHIA 8	NNPDF3.0nlo	A14
$W \rightarrow \ell\nu + \text{jets}$	SHERPA 2.2.11	NLO	SHERPA	NNPDF3.0nnlo	SHERPA default
$W \rightarrow \tau\nu + \text{jets}$	SHERPA 2.2.11	NLO	SHERPA	NNPDF3.0nnlo	SHERPA default
$Z \rightarrow ee/\mu\mu/\nu\nu + \text{jets}$	SHERPA 2.2.11	NLO	SHERPA	NNPDF3.0nnlo	SHERPA default
$Z \rightarrow \tau\tau + \text{jets}$	SHERPA 2.2.14	NLO	SHERPA	NNPDF3.0nnlo	SHERPA default
$t\bar{t}$	POWHEG BOX v2	NLO	PYTHIA 8	NNPDF3.0nlo	A14
Single- t	POWHEG BOX v2	NLO	PYTHIA 8	NNPDF3.0nlo	A14
Diboson	SHERPA	NLO	SHERPA	NNPDF3	SHERPA default

5.4 Triggers

The ATLAS Run 2 trigger architecture was described in Chapter 4. This analysis uses only single-electron, single-muon and E_T^{miss} triggers.

The signal-region event selection uses the lowest-threshold unprescaled E_T^{miss} trigger, which is fully efficient for at least offline $E_T^{\text{miss}} > 200$ GeV applied in signal regions and the corresponding validation regions.

Single-electron and single-muon triggers are used in regions containing an electron or muon. The lowest online thresholds range from 20 to 26 GeV [101, 102].

Table 5.3 lists the single-electron, single-muon and E_T^{miss} trigger chains used in this analysis.

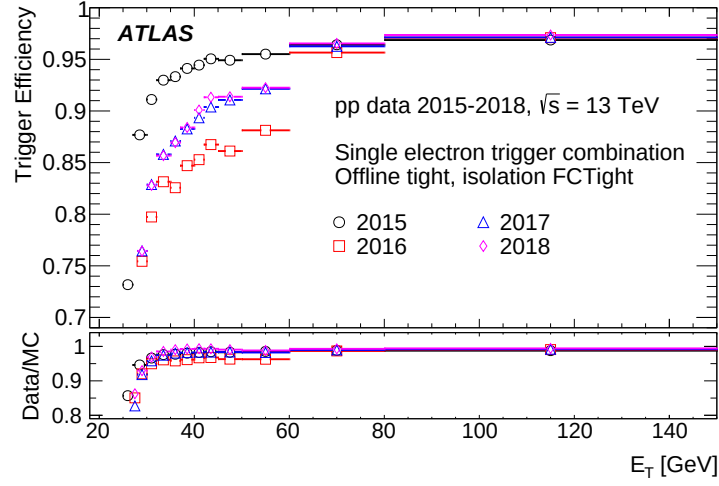
In an HLT chain name, e, mu and xe represent an electron, a muon and E_T^{miss} , respectively. The following number gives the online threshold in GeV. An explicit L1 suffix identifies the first-level trigger that seeds the chain. The electron labels lhloose, lhmedium and lhtight specify likelihood-based identification requirements, while nod0 indicates that transverse-impact-parameter information is omitted. The labels iloose, ivarloose and ivarmedium

specify isolation requirements. For E_T^{miss} , mht denotes a calculation based on the negative vector sum of jet transverse momenta, whereas pufit denotes a calorimeter-based calculation with a fit-based pile-up correction [103].

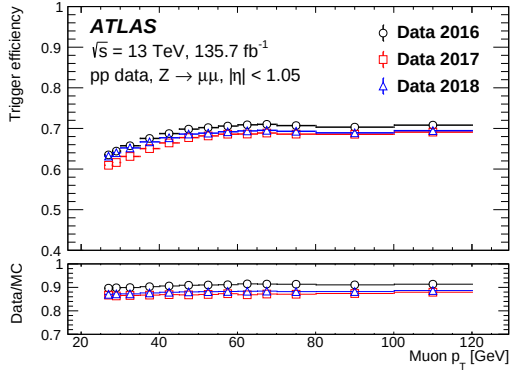
Figures 5.1 and 5.2 show representative Run 2 trigger-efficiency measurements. Differences in trigger efficiency between collision data and simulation are corrected using data-to-simulation efficiency scale factors.

Table 5.3 Single-electron, single-muon and E_T^{miss} trigger chains used in the Run 2 analysis. Single-lepton chains listed for the same flavour and year are combined with a logical OR, while the E_T^{miss} chain is selected according to the data-taking period. The extra xe70 or xe65 in the 2018 names denotes an additional calorimeter-cell E_T^{miss} requirement of 70 or 65 GeV.

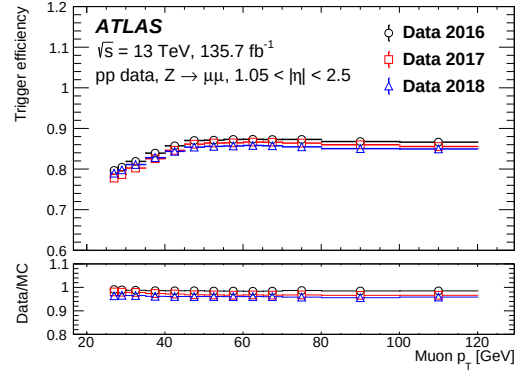
Trigger	Year	HLT chain
Single-electron	2015	HLT_e24_lhmedium_L1EM20VH
		HLT_e60_lhmedium
		HLT_e120_lhloose
	2016–2018	HLT_e26_lhtight_nod0_ivarloose HLT_e60_lhmedium_nod0 HLT_e140_lhloose_nod0
Single-muon	2015	HLT_mu20_iloose_L1MU15
		HLT_mu40
	2016–2018	HLT_mu26_ivarmedium HLT_mu50
E_T^{miss}	2015	HLT_xe70_mht
	2016	HLT_xe90_mht_L1XE50
		HLT_xe100_mht_L1XE50
		HLT_xe110_mht_L1XE50
	2017	HLT_xe110_pufit_L1XE55
		HLT_xe110_pufit_L1XE50
	2018	HLT_xe110_pufit_xe70_L1XE50
		HLT_xe110_pufit_xe65_L1XE50



(a) Single-electron trigger combination.



(b) Single-muon triggers, barrel.



(c) Single-muon triggers, end-cap.

Fig. 5.1 Single-lepton trigger efficiencies measured in Run 2 $Z \rightarrow \ell\ell$ events. (a) Electron efficiency versus offline E_T for electrons with Tight identification and FCTight isolation [101]. (b,c) Muon efficiency versus offline p_T in the barrel and end-cap, respectively [102]. Lower panels give data-to-simulation efficiency ratios. Electron uncertainties include statistical and systematic contributions, while muon uncertainties are statistical only.

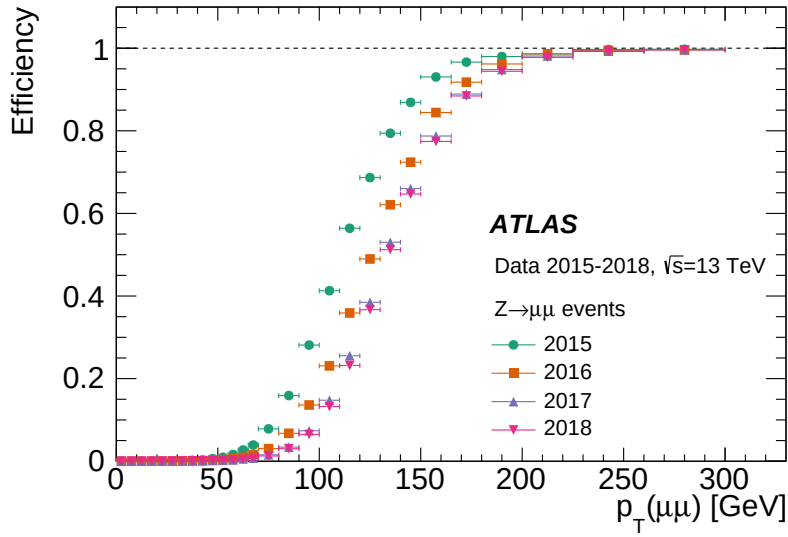


Fig. 5.2 Combined L1 and HLT E_T^{miss} trigger efficiencies in $Z \rightarrow \mu\mu$ events for each year of Run 2 [103]. Since muons are treated as invisible by these calorimeter-based triggers, the dimuon transverse momentum $p_T(\mu\mu)$ provides a reference for the missing transverse momentum measured by the trigger. The curves use the lowest unprescaled trigger as it changes within each year. Error bars show statistical uncertainties.

Chapter 6

Object Reconstruction and Identification

Hadronically decaying τ -leptons, electrons, muons and jets are used in this analysis. This chapter describes their reconstruction, identification and calibration, followed by their definitions in the analysis, overlap removal and the reconstruction of E_T^{miss} .

These objects are reconstructed and identified using detector measurements. Tracks are used to determine charged-particle momenta and production vertices, while energy deposits are measured in the calorimeters. The reconstructed energies and momenta are calibrated to correct biases in the detector response. Simulated event weights are further corrected using efficiency scale factors.

6.1 Tracks

6.1.1 Reconstruction

A track is reconstructed by fitting the trajectory of a charged particle to hits in the ID. Its momentum and charge are determined from the curvature in the magnetic field, while its position near the beam line is used to assign it to a production vertex. Track reconstruction is performed in the following stages [104]:

1. **Space-point formation.** Signals in neighbouring channels of a silicon sensor are grouped into clusters. The coordinates provided by Pixel clusters and SCT clusters are converted into three-dimensional space points.
2. **Seeding and track finding.** Combinations of space points are used to form track seeds. These seeds are extended through the silicon layers with a combinatorial Kalman filter, which is used to refine the track parameters.

3. **Ambiguity resolution.** Track candidates are ranked using their fit quality, momentum and hit variables. Meanwhile, candidates that share the same clusters are marked as "shared hits". Candidates fail the track-quality requirement are then rejected to suppress duplicate and fake tracks.
4. **TRT extension.** Tracks are extended with additional TRT hits. The extended track is retained if its quality score is higher than that of the original track reconstructed using only pixel and SCT hits.

6.1.2 Track parameters and quality

The fitted trajectory is described by five parameters, as illustrated in Figure 6.1:

- d_0 : distance in the $x - y$ plane between the beam line and the perigee point, defined as the point on the track closest to the beam line.
- z_0 : the z coordinate of the perigee point relative to the reference plane at $z = 0$.
- ϕ and θ : the azimuthal and polar angles of the momentum, respectively.
- q/p : the electric charge divided by the momentum magnitude.

Two quantities are used to test whether a track is consistent with production at the primary vertex [105]:

- **Transverse impact-parameter significance**, d_0/σ_{d_0} : the transverse distance from the beam line in units of its uncertainty σ_{d_0} .
- **Projected longitudinal distance**, $\Delta z_0 \sin \theta = (z_0 - z_{PV}) \sin \theta$: the longitudinal distance between the track perigee and the primary vertex. Here, z_{PV} is the z coordinate of the primary vertex.

Small absolute values of both quantities suggest that the track is from the primary vertex. A large longitudinal distance can also indicate a track from another pp collision. Prompt electron and muon candidates are therefore selected by requiring small values of both quantities, and displaced candidates with large values can originate from a separate decay vertex, for example from a b - or c -hadron decay.

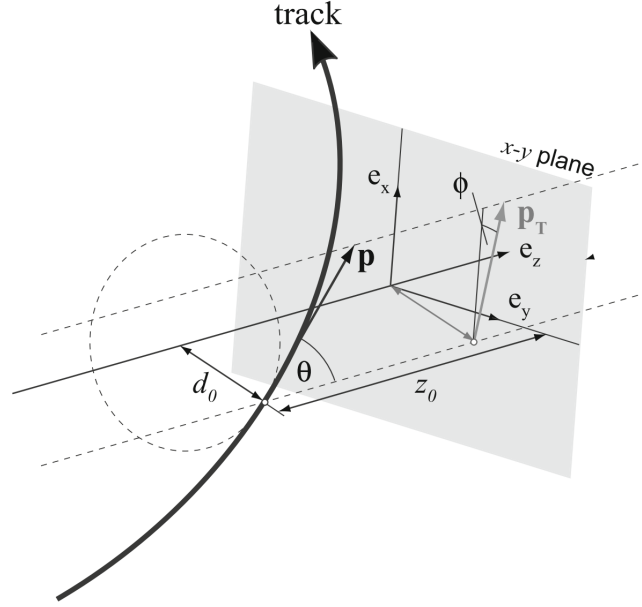


Fig. 6.1 Geometry of the Track parameters. The grey plane represents the reference $z = 0$ plane [106].

6.2 Primary vertex

A primary vertex (PV) is the reconstructed position of a pp collision. It is used as the reference for prompt-lepton selection, isolation, pile-up jet rejection and the track contribution to E_T^{miss} . Its position is determined from tracks originating from a common point. Several PVs can be reconstructed in an event because of pile-up. The vertex with the largest sum of squared transverse momenta of its tracks is selected as the PV of the hard scattering [107].

6.3 Electrons

6.3.1 Reconstruction

The electron object is reconstructed from an ID track matched to an electromagnetic shower in the calorimeter. Bremsstrahlung photons emitted in the upstream material can spread the shower energy over several clusters and alter the curvature of the electron track. Both effects are treated in the reconstruction [108]:

1. **Topo-cluster reconstruction.** Neighbouring cells are grouped into topological clusters according to their signal significance above the expected noise. Clusters with a substantial electromagnetic energy fraction are used as electron seeds.

2. **Track refitting and matching.** Electron-track candidates are refitted with a Gaussian-sum filter, in which a mixture of Gaussian components is used to describe radiative energy losses. The refitted tracks are extrapolated to the second layer of the EM calorimeter and matched to clusters using their differences in η and ϕ . If several tracks match the same cluster, the ΔR between each extrapolated track position and the cluster centre is used together with the pixel and SCT hits to rank the tracks.

3. **Supercluster formation.** Nearby clusters are added to the seed to recover energy from bremsstrahlung. Additional clusters sharing the matched electron track can be collected over a wider region in ϕ .

The cluster association is illustrated in Figure 6.2. The wider collection in ϕ helps recover energy separated from the electron by its bending in the magnetic field.

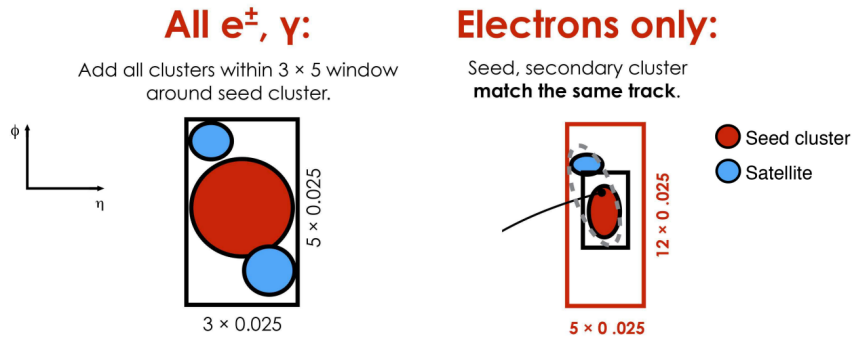


Fig. 6.2 Cluster association for electron superclusters. Seed and additional clusters are shown in red and blue, respectively. Geometrically nearby clusters are collected [108].

6.3.2 Identification

Electron candidates are distinguished from hadronic showers and converted photons using a likelihood discriminant [109]. Its inputs include the shower shapes in the EM calorimeter, hit variables, and the agreement between the track and calorimeter measurements. The Loose, Medium and Tight working points are defined and provide increasing background rejection with the different likelihood discriminant threshold. The Tight likelihood working point, denoted TightLH, is used for all electron candidates in this analysis. Its typical prompt-electron identification efficiency is about 80% [110].

6.3.3 Isolation

Electrons from hadron decays can pass the electron identification requirements. These electrons are usually accompanied by other particles from the same jet, while prompt electrons typically have smaller nearby track p_T and calorimeter E_T sums. Isolation requirements are therefore used to suppress this background.

Two isolation variables are calculated using tracks and calorimeter clusters within cones around the electron direction [108]:

- **Track isolation** is performed using the p_T sum of tracks in the cone. At high electron p_T , other decay products can lie close to the electron due to the boost effect, so the cone radius is reduced.
- **Calorimeter isolation** follows the same principle as track isolation, but uses cluster E_T within a fixed cone of $\Delta R = 0.2$. The corrected sum estimates the E_T from other particles near the electron.

In the analysis, baseline electrons are required to pass Loose_VarRad isolation while signal electrons must additionally satisfy HighPtCaloOnly. The Loose_VarRad working point requires the track and calorimeter isolation sums to be below 15% and 20% of the electron p_T , respectively. The HighPtCaloOnly working point requires the calorimeter isolation sum to be below $\max(0.015 p_T, 3.5 \text{ GeV})$. At high E_T , the prompt-electron efficiencies are about 99% for Loose_VarRad and 90–95% for HighPtCaloOnly [110].

6.3.4 Calibration

The electron energy is calibrated for losses in upstream material, energy outside the cluster and variations in detector response [108]. $Z \rightarrow ee$ events are used for the calibration. Reconstruction, identification and isolation efficiencies are measured with the tag-and-probe method: one electron (the tag) is required to pass strict identification cuts, while the other is used as the probe. After background subtraction, the efficiency is the fraction of probes that pass the selection. For a selection efficiency ε , the scale factor is defined as

$$\text{SF} = \frac{\varepsilon_{\text{data}}}{\varepsilon_{\text{MC}}}. \quad (6.1)$$

The scale factors are then applied to simulated event weights for the electron working points used in this analysis.

6.4 Muons

6.4.1 Reconstruction

Muons are reconstructed using tracks in the ID and MS, together with their energy deposits in the calorimeters [105]. The main reconstruction categories are:

- **Combined muons** are reconstructed by matching and fitting ID and MS tracks, including the energy loss between the two tracking systems.
- **Segment-tagged muons** are reconstructed by matching an ID track to one or more local track segments in the MS when a complete MS track is unavailable.
- **Calorimeter-tagged muons** are reconstructed by matching an ID track to calorimeter deposits compatible with a minimum-ionising particle. Acceptance is recovered in regions with limited MS coverage.
- **MS-extrapolated muons** are reconstructed by extrapolating an MS track towards the interaction region without a matched ID track.

Combined muons are used in this analysis.

6.4.2 Identification

Muon identification tests whether the detector hits and the ID and MS momentum measurements are consistent with a single muon trajectory [105]. Decays of charged mesons in flight can change the trajectory and momentum, leading to a disagreement between the momentum measurements. For muon candidates, this agreement is quantified by the q/p significance,

$$S_{q/p} = \frac{|(q/p)_{\text{ID}} - (q/p)_{\text{MS}}|}{\sqrt{\sigma_{\text{ID}}^2 + \sigma_{\text{MS}}^2}}, \quad (6.2)$$

where $(q/p)_{\text{ID}}$ and $(q/p)_{\text{MS}}$ are the charge-to-momentum ratios determined from the ID and MS tracks, respectively. σ_{ID} and σ_{MS} denote the uncertainties. Large values indicate poor agreement and help reject incorrectly matched candidates.

In the analysis, baseline muons are required to pass Loose identification while signal muons must additionally satisfy the Medium working point. The Medium working point requires combined candidates within the ID acceptance to satisfy $S_{q/p} < 7$ and, outside $|\eta| < 0.1$, have at least two MS precision stations with at least three MDT or CSC hits each.

The Loose working point applies the same requirements and additionally accepts segment-tagged and calorimeter-tagged candidates in $|\eta| < 0.1$, where MS coverage is limited. The typical prompt-muon efficiencies are about 99% for Loose and 97% for Medium.

6.4.3 Isolation

Muons from heavy-flavour decays are often accompanied by other particles inside a jet. They are suppressed using track and calorimeter isolation, defined by the surrounding activity after excluding the muon's own contribution [105]. Baseline muons are required to pass Loose_VarRad isolation, and signal muons must additionally satisfy Tight_VarRad. Both working points use a track-cone radius that decreases as the muon p_T increases and a fixed calorimeter cone of $\Delta R = 0.2$. The track and calorimeter isolation sums must be below 15% and 30% of the muon p_T for Loose_VarRad, respectively. These limits are reduced to 4% and 15% for Tight_VarRad, allowing less activity around the muon and rejecting more muons produced inside jets. The typical prompt-muon isolation efficiencies are about 95% for Loose_VarRad and 86% for Tight_VarRad [111].

6.4.4 Calibration

The muon momentum scale and resolution are calibrated using resonance decays, e.g., $Z \rightarrow \mu\mu$ and $J/\psi \rightarrow \mu\mu$ [112]. The reconstructed mass distributions constrain biases in the muon momentum measurement. The efficiency scale factors are measured with tag-and-probe samples as for electrons and applied to simulated event weights for the selected working points [105].

6.5 Jets

6.5.1 Reconstruction

Jets are reconstructed from the collimated hadrons produced through parton showering and hadronisation. In this analysis, the constituents used for jet clustering are first reconstructed with the particle-flow (PFlow) algorithm, which combines ID track momenta with calorimeter energy measurements [113]. The main steps are:

1. Calorimeter topological clusters are initially calibrated for electromagnetic showers and matched to ID tracks.

2. The expected calorimeter energy deposited by the matched charged particles is subtracted from the clusters, avoiding double counting when the track momenta are used.
3. Vertex association is used to suppress charged-particle contributions from pile-up. The selected tracks and remaining calorimeter contributions form the PFlow constituents.

These PFlow constituents are then grouped into jets using the anti- k_t algorithm with $R = 0.4$ [114]. Pairs are selected for combination according to their angular separation and p_T . Their four-momenta are added, and the merged object is used in the next clustering step. Low- p_T constituents are preferentially clustered around nearby high- p_T constituents, so isolated jets have approximately circular boundaries with radius R .

6.5.2 Identification and pile-up rejection

Jets from the hard scattering are selected after jet cleaning and pile-up rejection.

- **Jet cleaning** is performed with the requirements on calorimeter pulse shapes and the distribution of energy among calorimeter layers to reject noise and non-collision backgrounds. For example, a jet with $|\eta| < 2$ is rejected if more than 99% of its energy is deposited in a single calorimeter layer. The efficiency for collision jets exceeds 99.5% [115].
- **Pile-up rejection** uses the vertex origins and p_T of the jet tracks. The jet-vertex tagger (JVT) is constructed from two variables [116]. The first is based on the ratio of the p_T sum of tracks from the PV to that of tracks from all vertices. The second is the same PV-track p_T sum divided by the reconstructed jet p_T . Large values of both variables favour jets from the hard scattering. The neural-network JVT (NNJVT) is used in this analysis with the FixedEffPt working point [117], whose target efficiency for hard-scatter jets is 88–98% [118].

6.5.3 Calibration

Jet energies are calibrated to correct the jet energy scale (JES) and improve the jet energy resolution (JER). The JES characterises the average energy response, while the JER quantifies the spread of reconstructed energies at a fixed particle-level energy. The following corrections are applied [119]:

1. **Pile-up subtraction** removes the estimated energy deposited by pile-up particles. The corresponding correction to the jet p_T is estimated as ρA , where ρ is the event p_T density and A is the jet area.

- 1588 2. **Energy and direction calibration** corrects the differences in energy and η between
1589 reconstructed and matched particle-level jets. Particle-level jets are built from stable
1590 simulated particles, excluding muons and neutrinos, and therefore provide a reference
1591 without detector effects.
- 1592 3. **Global sequential calibration** corrects the remaining response differences between
1593 jets with different particle compositions or shower shapes. Track multiplicity and
1594 energy fractions in calorimeter layers are used to reduce these differences, improving
1595 the resolution while preserving the average energy scale.
- 1596 4. **Residual in situ calibration** uses well-measured Z bosons and photons as references to
1597 check the jet energy response. For higher- p_T jets, the vector sum of several calibrated
1598 lower- p_T jets are used as the reference. The average ratio of jet p_T to reference p_T
1599 is compared between data and simulation to determine the final correction, and then
1600 applied only to data.

1601 6.6 Jet flavour tagging

1602 Jet flavour tagging is used to distinguish b -quark originating jets (b -jets) from other jet
1603 flavours.

1604 b -jets are identified through displaced b -hadron decays, which have a lifetime of about
1605 1.5 ps. Therefore a boosted b -hadron can travel several millimetres before decaying. The
1606 decay tracks therefore typically have large impact parameters and originate from a secondary
1607 vertex displaced from the PV. Charm hadrons produced in the b -hadron decay can further
1608 decay at an additional displaced vertex. These features are used for jet flavour tagging.

1609 The tagger uses the GN2v01 algorithm, which is based on a transformer neural network
1610 trained on simulated events [120]. Its inputs include jet and track kinematics, track impact
1611 parameters, and hit variables. These variables are first encoded as a numerical vector for
1612 each track. Within the transformer, each vector is updated through self-attention, using
1613 information from the other tracks. Correlations between tracks can thereby be used to
1614 recognise a common displaced origin without requiring explicitly reconstructed secondary
1615 vertices as separate inputs. The network then calculates a weight for each track using its
1616 updated vector and adds the weighted vectors together, which is called attention pooling.

1617 The resulting vector is used for the main task of jet-flavour classification. It is also
1618 combined with the individual track vectors for two auxiliary tasks: identifying track origins
1619 and predicting whether pairs of tracks originate from the same vertex. The three tasks share

the transformer and are trained jointly, helping it learn the decay structure. The network and its three tasks are illustrated in Figure 6.3.

The jet-flavour classifier produces four outputs for the b -jet, c -jet, hadronic- τ and light-jet hypotheses, where light jets originate from u , d or s quarks or gluons. These outputs are combined into a single discriminant, D_b , which compares the b -jet output with a weighted combination of the other three outputs. Larger values favour the b -jet hypothesis. The GN2v01 85% working point is used in this analysis, corresponding to a nominal b -jet tagging efficiency of 85%.

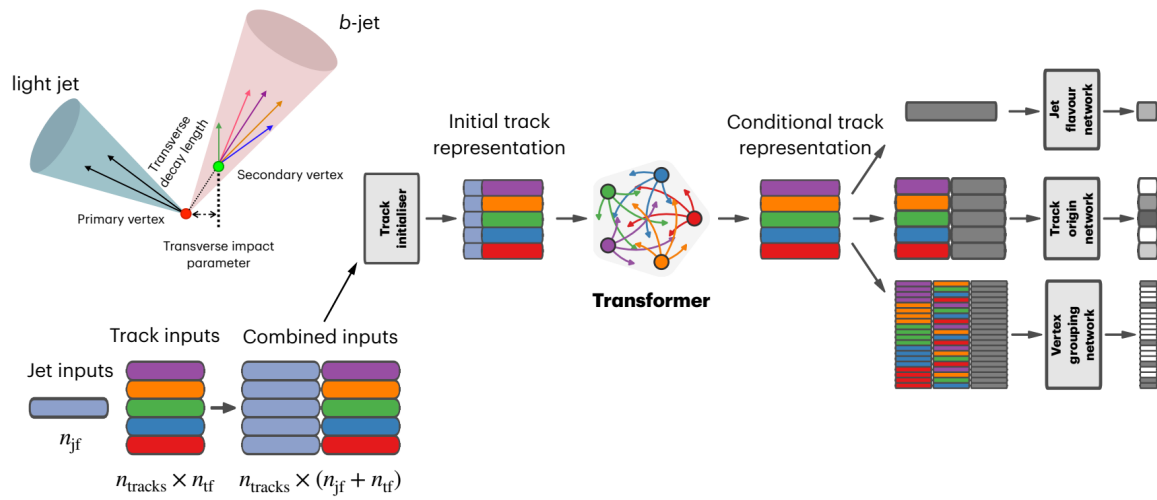


Fig. 6.3 GN2 flavour-tagging architecture. The jet variables are appended to the input vector of each track before processing by the track initialiser and transformer. The resulting track vectors and their weighted combination are passed to the jet-flavour, track-origin and vertex-grouping networks. The symbols n_{jf} , n_{tf} and n_{tracks} denote the numbers of jet variables, track variables and tracks, respectively [120].

6.6.1 Calibration

The b -jet tagging efficiency is measured in dileptonic $t\bar{t}$ events with an oppositely charged electron–muon pair and two jets [120]. The c -jet mistagging efficiency, the probability for a c -jet to pass the b -tagging requirement, is measured in semileptonic $t\bar{t}$ events using charm jets from hadronic W decays. The corresponding light-jet mistagging efficiency is measured in Z +jets events with $Z \rightarrow ee$ or $Z \rightarrow \mu\mu$ decays.

These efficiencies are measured in jet p_T bins and compared with simulation to obtain flavour-dependent scale factors, as defined in Eq. 6.1. Corrections are then applied to the simulated event weights for both tagged and untagged jets.

6.7 Hadronic tau leptons

6.7.1 Reconstruction

τ leptons can decay both leptonically and hadronically. Approximately 65% of τ decays are hadronic, with the major modes containing charged and possible neutral pions [121]. In this analysis we only consider the hadronic decay tau. It is denoted by τ_{had} . Here the τ -neutrino is excluded in the reconstruction.

Hadronic tau candidates are first seeded by anti- k_t jets with $R = 0.4$. The production vertex is then chosen as the vertex with the largest fraction of the p_T sum of tracks within $\Delta R < 0.2$ of the seed axis. This vertex provides the reference for track association and impact-parameter calculations. The tracks are then classified by a recurrent neural network (RNN) to select tau-decay tracks. The number of selected tau-decay tracks defines the prong multiplicity: candidates with one or three such tracks are called one-prong or three-prong candidates, respectively.

6.7.2 Identification

Hadronic tau decays are identified using an RNN discriminant to distinguish them from quark- and gluon-initiated jets [122, 121]. Track momenta and calorimeter shower properties are used as inputs, since tau decays typically have narrower showers and lower charged multiplicity than these jets. Track displacements and, for three-prong candidates, the significance of the reconstructed decay-vertex displacement are also included as inputs sensitive to the finite tau lifetime. Nearby activity is also included in the RNN inputs, so isolation information is used as part of the identification.

Baseline tau candidates are required to pass Loose RNN identification, and signal candidates must additionally satisfy Tight. The higher RNN-score threshold for Tight rejects more jets at the cost of tau efficiency. For reconstructed one-prong (three-prong) candidates, the target identification efficiencies are 85% (75%) for Loose and 60% (45%) for Tight.

Electrons can be misidentified as hadronic tau candidates because their narrow showers and tracks are similar to the signature of a one-prong tau decay [121]. To suppress this background, both baseline and signal tau candidates are required to pass Loose electron rejection. This working point requires a minimum score from a separate RNN using shower and track information, including the relation between calorimeter energy and track momentum. The tau efficiencies are approximately 95% and 98% for one-prong and three-prong candidates, respectively. The corresponding electron rejection efficiencies are approximately 99.7% and 98.8%.

6.7.3 Calibration

The tau energy scale (TES) calibration is derived from MC samples [121]. In these samples, the generated visible tau p_T , excluding the decay neutrino, is used as the reference for correcting the reconstructed p_T . The visible p_T is reconstructed in two ways: from calorimeter energy deposits and shower directions, and from a combination of tracking and calorimeter measurements. In the latter, charged-hadron momenta are measured from track curvature in the magnetic field, while neutral particles, which leave no tracks, are measured through their calorimeter deposits. The calorimeter deposits associated with charged hadrons are subtracted to avoid double counting. These two measurements are combined in a weighted average. The averaged p_T is then corrected using a boosted regression tree (BRT) trained on these MC samples.

In addition to the momentum calibration, differences in tau reconstruction, identification and electron-rejection efficiencies between data and simulation must be correcte. For this purpose, data-to-simulation scale factors are applied to simulated event weights. In this analysis, for signal candidates passing Tight tau identification, the electron-veto scale factors for Medium tau identification are used because no corresponding factors were available for Tight.

6.8 Overlap removal

Different reconstructed objects can share the same detector measurements. ΔR is calculated using rapidity for overlap removal. An ordered overlap-removal procedure is therefore applied to resolve these ambiguities before event selection. Overlap removal is applied to the baseline objects in the order listed below.

1. **Electron–electron:** the lower- p_T electron is removed if two electrons share an ID track, so that a single track is not counted as two electrons.
2. **Tau–electron:** a tau candidate within $\Delta R < 0.2$ of an electron is removed, since an electron shower and its track can also be reconstructed as a tau candidate.
3. **Tau–muon:** a tau candidate within $\Delta R < 0.2$ of a muon is removed to reject muons misidentified as hadronic tau decays.
4. **Calorimeter-tagged muon–electron:** a calorimeter-tagged muon sharing the electron ID track is removed. The electron interpretation is preferred because the calorimeter tag has no independent MS confirmation.

- 1701 5. **Electron–muon**: an electron sharing an ID track with a remaining muon is removed.
- 1702 6. **Jet–electron**: a jet within $\Delta R < 0.2$ of an electron is removed because the electron’s
- 1703 EM shower can also be reconstructed as a jet.
- 1704 7. **Electron–jet**: an electron within $\Delta R < 0.4$ of a remaining jet is removed to suppress
- 1705 electrons from hadron decays inside jets.
- 1706 8. **Jet–muon**: a jet with fewer than three tracks is removed if the muon is ghost-associated
- 1707 with it.¹
- 1708 9. **Muon–jet**: a muon within $\Delta R < 0.4$ of a remaining jet is removed to suppress muons
- 1709 from hadron decays inside jets.
- 1710 10. **Jet–tau**: a jet within $\Delta R < 0.2$ of a remaining tau candidate is removed, so that the
- 1711 visible tau-decay products are not also counted as a separate jet.

1712 6.9 Missing transverse momentum

1713 The incoming partons at the LHC carry little transverse momentum, so the vector sum of the
 1714 transverse momenta of all the particles is expected to be approximately zero. An imbalance
 1715 in the visible transverse momentum can therefore indicate undetected particles, such as
 1716 neutrinos or weakly interacting BSM particles. In this analysis, the neutrino produced in the
 1717 LQ process and the tau-decay neutrino can together give rise to sizeable missing transverse
 1718 momentum ($\vec{p}_T^{\text{miss}}$). $\vec{p}_T^{\text{miss}}$ is reconstructed as the negative vector sum of the calibrated visible
 1719 contributions, with its magnitude denoted by E_T^{miss} [123]:

$$1720 \quad \vec{p}_T^{\text{miss}} = - \left(\sum_e \vec{p}_T^e + \sum_\mu \vec{p}_T^\mu + \sum_{\tau_{\text{had}}} \vec{p}_T^\tau + \sum_j \vec{p}_T^j + \vec{p}_T^{\text{soft}} \right), \quad (6.3)$$

$$1721 \quad E_T^{\text{miss}} = \left| \vec{p}_T^{\text{miss}} \right|. \quad (6.4)$$

1722 Two types of contribution are included in the reconstruction:

- 1723 • **hard term**: The electron, muon, hadronic-tau and jet contributions selected for $\vec{p}_T^{\text{miss}}$
- 1724 reconstruction. The jet contribution is selected using the Tight $\vec{p}_T^{\text{miss}}$ working point to
- 1725 suppress pile-up jets. This requires jet $p_T > 20$ GeV for $|\eta| < 2.5$ and $p_T > 30$ GeV
- 1726 for $2.5 \leq |\eta| < 4.5$.

¹Ghost association means that a copy of the muon’s ID track is made, keeping its direction but reducing its momentum to a negligible value. This “ghost” is clustered together with the normal jet inputs. If the clustering algorithm places the ghost in a particular jet, the muon is associated with that jet.

- **track soft term:** The vector sum of transverse momenta of selected PV tracks not used in the hard term.

6.10 Object definitions in the analysis

Baseline and signal objects are defined to provide two levels of selection in the analysis. After overlap removal as described in Section 6.8, baseline objects are used for object counting and lepton vetoes. Signal objects are selected from these baseline collections with tighter identification or isolation requirements, as summarised in Table 6.1.

Table 6.1 Baseline object definitions and additional signal requirements used in the analysis. A dash in the signal column indicates that the requirement depends on the analysis region.

Object	Requirement	Baseline	Additional signal requirement
Electron	p_T	$> 10 \text{ GeV}$	—
	$ \eta $	$(\eta < 1.37) \text{ } (1.52 < \eta < 2.47)$	
	$ d_0 /\sigma_{d_0}$		< 5
	$ \Delta z_0 \sin \theta $		$< 0.5 \text{ mm}$
	Identification		TightLH
	Isolation	Loose_VarRad	HighPtCaloOnly
Muon	p_T	$> 10 \text{ GeV}$	—
	$ \eta $		< 2.5
	$ d_0 /\sigma_{d_0}$		< 3
	$ \Delta z_0 \sin \theta $		$< 0.5 \text{ mm}$
	Identification	Loose	Medium
	Isolation	Loose_VarRad	Tight_VarRad
Hadronic tau	$ \eta $	$(\eta < 1.37) \text{ } (1.52 < \eta < 2.5)$	
	Tracks		1, 3
	Charge		$ Q = 1$
	Energy calibration		MVA
	p_T	$> 20 \text{ GeV}$	—
	Jet rejection	RNN, Loose	RNN, Tight
	Electron rejection		RNN, Loose
Jet	Algorithm		Anti- k_t
	Radius		$R = 0.4$
	Inputs		PFlow
	Cleaning		LooseBad
	p_T	$> 20 \text{ GeV}$	—
	$ \eta $	< 2.5	—
	NNJVT		FixedEffPt
	b -tagging	GN2v01, 85% b -jet efficiency	

Chapter 7

Event Selection

This chapter defines the signal regions (SRs) targeting the processes shown in Figure 3.4. The analysis variables and preselection are introduced first, followed by the optimisation of the one- b -jet regions, the definition of the zero- b -jet regions, and a study of $\tau\tau$ signal contamination in the SRs.

7.1 Preselection and variables definition

7.1.1 Analysis variables

The transverse mass m_T of the τ and $\vec{p}_T^{\text{miss}}$ system is defined as

$$m_T = \sqrt{2p_T^\tau E_T^{\text{miss}} [1 - \cos \Delta\phi(\tau, E_T^{\text{miss}})]}. \quad (7.1)$$

Here $\Delta\phi(\tau, E_T^{\text{miss}})$ is the azimuthal separation between the τ momentum and $\vec{p}_T^{\text{miss}}$. The variable m_T is used to probe the high-energy tail of the $\tau\nu$ final state, which is particularly sensitive to non-resonant leptoquark contributions [64].

The invariant masses $m_{b\tau}$ and $m_{j\tau}$ are defined as:

$$m_{b\tau}^2 = (p_b + p_\tau)^2, \quad m_{j\tau}^2 = (p_{j_1} + p_\tau)^2, \quad (7.2)$$

where p_b , p_{j_1} and p_τ are four-momenta, and j_1 is the jet with the largest p_T . The zero- b -jet regions contain no b -tagged jet, so $m_{j\tau}$ is used in place of $m_{b\tau}$.

7.1.2 Multijet suppression

Multijet events can enter the SR and CR selections when a jet is misidentified as a τ and a jet-energy mismeasurement produces large E_T^{miss} . A high E_T^{miss} threshold rejects most of these events, but their large production cross section makes additional suppression necessary. In the simulated dijet events that survive the kinematic requirements, the apparent missing momentum arises from the mismeasurement of an energetic jet. If its momentum is underestimated, the reconstructed momentum imbalance points approximately towards that jet. This instrumental contribution is referred to as fake E_T^{miss} . In signal events, the neutrinos produce genuine missing momentum with no corresponding tendency to align with a jet. Figure 7.1 illustrates this difference.

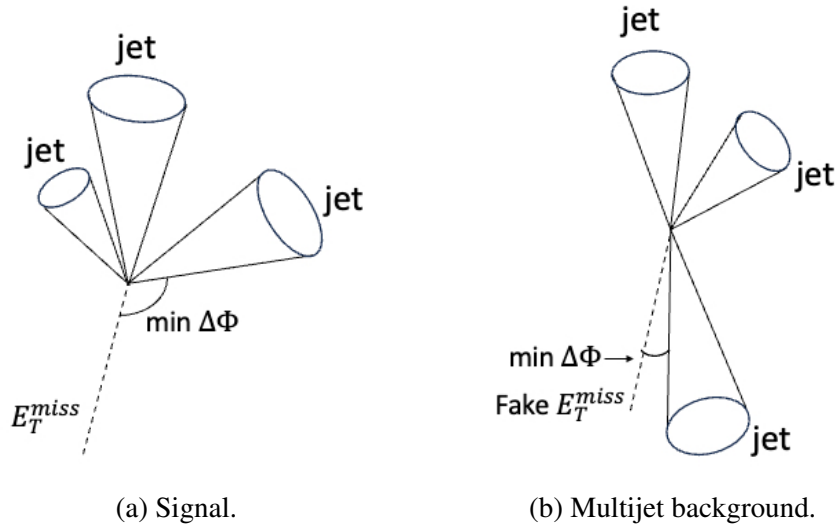


Fig. 7.1 Schematic transverse-plane configurations illustrating (a) genuine missing momentum from neutrinos in signal events and (b) fake missing momentum from a jet-energy mismeasurement in multijet events. Underestimating a jet momentum produces an imbalance approximately aligned with that jet, giving a small minimum jet- $\vec{p}_T^{\text{miss}}$ azimuthal separation.

The minimum jet- $\vec{p}_T^{\text{miss}}$ azimuthal separation tests this alignment. The ratio of E_T^{miss} to the p_T of the closest jet in azimuth provides a second discriminating variable:

$$\Delta\phi_{\min} = \min_j \Delta\phi(\vec{p}_T^{\text{miss}}, \vec{p}_T^j), \quad R_{\text{miss}/j} = \frac{E_T^{\text{miss}}}{p_T^{j_{\min}}}, \quad (7.3)$$

where $j_{\min}$ is the jet with the smallest azimuthal separation from $\vec{p}_T^{\text{miss}}$. It need not be the leading jet. Mismeasured multijet events tend to have small $\Delta\phi_{\min}$ and a modest $R_{\text{miss}/j}$. By contrast, the neutrinos in signal events can produce large E_T^{miss} even when a much softer jet happens to lie nearby in azimuth.

Events are retained if

$$\Delta\phi_{\min} > 0.4 \quad \text{or} \quad R_{\text{miss}/j} > 6. \quad (7.4)$$

The second condition retains signal events that would be lost with the angular requirement alone. The distributions, threshold scan, signal retention and residual multijet contribution are studied in Appendix B.

7.1.3 Preselection

The preselection establishes the $\tau\nu$ + jets final state before the kinematic optimisation. Events must pass the lowest-threshold unprescaled $E_{\text{T}}^{\text{miss}}$ trigger for the relevant data-taking period, contain exactly one selected τ , and contain no selected electron or muon. The multijet suppression in Eq. 7.4 is applied at this stage. The data-quality requirements of Chapter 5 are applied. The baseline for the one- b -jet optimisation additionally requires $N_b = 1$ with $p_{\text{T}}^b > 50$ GeV and $E_{\text{T}}^{\text{miss}} > 200$ GeV. The latter requirement selects the trigger-efficiency plateau discussed in Section 5.4. The zero- b -jet category replaces the tagged jet with at least one untagged jet, as described in Section 7.3. The baseline requirements and final kinematic selections are collected together in Table 7.3.

7.2 One b -jet signal region (SR1b) optimisation and definition

The SR1b regions are defined by requiring exactly one b -tagged jet to target the $b\tau$ interaction in the U_1 model and suppress inclusive W +jets production. The SM subprocesses $gu \rightarrow b\tau\nu$ and $gc \rightarrow b\tau\nu$ are suppressed by the small CKM elements V_{ub} and V_{cb} , respectively [64]. Competing SM $\tau\nu$ + jets events without a genuine b -jet enter this category through mistagging.

The b -jet p_{T} is used to distinguish resonant $U_1 \rightarrow b\tau$ decays, which produce an energetic decay jet, from non-resonant production, where the radiated jet is typically softer. The sample is divided at $p_{\text{T}}^b = 250$ GeV, with $p_{\text{T}}^b > 250$ GeV defining SR1b-Res and $50 < p_{\text{T}}^b < 250$ GeV defining SR1b-NonRes.

7.2.1 Scan variables and optimisation criterion

The τ p_{T} spectrum from heavy-leptoquark production extends to higher values than the bulk of the SM background. A lower threshold on p_{T}^{τ} is imposed to select energetic τ leptons and improve the signal-to-background ratio.

A high E_T^{miss} threshold is imposed to suppress the large contribution from resonant SM W production. The leptoquark signal extends to larger E_T^{miss} , motivating a lower threshold in both categories.

High values of m_T are selected to distinguish non-resonant leptoquark exchange from the falling SM background. A tighter m_T threshold is therefore imposed in SR1b-NonRes than in SR1b-Res, where the $b\tau$ mass is also used.

The invariant mass $m_{b\tau}$ is used to probe resonant $U_1 \rightarrow b\tau$ decays, while the SM backgrounds have no corresponding heavy resonance. The signal distribution is shifted below M_U and broadened by the τ -decay neutrino, with additional smearing from the jet and τ energy resolutions. A lower mass threshold is imposed to suppress the SM backgrounds while retaining this broad signal.

The separation $\Delta\phi(\tau, E_T^{\text{miss}})$ is used to select the approximately back-to-back τ and $\vec{p}_T^{\text{miss}}$ configuration of the non-resonant signal, which helps suppress top-quark backgrounds. Resonant leptoquark production is also suppressed by a strong back-to-back requirement, because the transverse momentum balance is modified by the decay jet.

The jet multiplicity $N_j - N_b$ tends to be larger in $t\bar{t}$ events because the two top-quark decays produce additional jets. An upper limit is imposed to suppress this background while retaining signal events with fewer jets.

The kinematic thresholds are scanned together because the discriminating variables are correlated. For example, increasing E_T^{miss} also raises m_T at fixed p_T^τ and $\Delta\phi(\tau, E_T^{\text{miss}})$, so the effect of an additional m_T requirement depends on the other cuts.

Lower thresholds on p_T^τ , E_T^{miss} and m_T are scanned in both categories, together with an upper limit on $N_j - N_b$. A lower threshold on $\Delta\phi(\tau, E_T^{\text{miss}})$ is tested for the non-resonant topology. For the resonant topology, upper thresholds on this angle and a lower threshold on $m_{b\tau}$ are tested, as summarised in Table 7.1.

Table 7.1 Thresholds tested in the SR1b optimisation. Each set lists the trial values for the indicated inequality. A zero mass threshold or an angular upper threshold of π represents no effective requirement on that variable.

Variable	SR1b-Res	SR1b-NonRes
$\Delta\phi(\tau, E_T^{\text{miss}})$	$< \{0.3, 1.2, 2.4, 2.6, \pi\}$	$> \{0.3, 1.2, 2.4, 2.6\}$
$m_{b\tau}$ [GeV]	$> \{0, 800\}$	Not scanned
E_T^{miss} [GeV]	$> \{200, 250, 300, 350, 400, 500, 600\}$	
p_T^τ [GeV]	$> \{200, 250, 300, 350, 400, 500, 600\}$	
m_T [GeV]	$> \{100, 200, 300, 400, 500, 600, 700, 800, 900, 1000\}$	
$N_j - N_b$		$\leq \{1, 2, 3\}$

The mass scan uses only $m_{b\tau} > 0$ and 800 GeV. The broad $m_{b\tau}$ distribution and limited simulated background statistics in its high-mass tail favour an inclusive mass requirement over a finely scanned mass window.

The resonant optimisation targets $M_U = 1.5$ TeV, where on-shell production is important. The non-resonant optimisation uses $(M_U, g_U, \beta_L^{23}) = (2.5 \text{ TeV}, 2.5, 1.0)$, where virtual exchange provides the dominant signal contribution.

The expected sensitivity of each cut combination is ranked using an approximate counting-experiment significance [124]:

$$Z = \sqrt{2 \left[n \ln \left(\frac{n(b + \sigma_b^2)}{b^2 + n\sigma_b^2} \right) - \frac{b^2}{\sigma_b^2} \ln \left(\frac{b^2 + n\sigma_b^2}{b(b + \sigma_b^2)} \right) \right]}, \quad (7.5)$$

where s and b are the expected signal and background event yields, $n = s + b$, and σ_b is the absolute uncertainty in the background expectation. A fractional background uncertainty of 30%, corresponding to $\sigma_b = 0.30b$, is assumed for this scan. The scan uses the pure BSM signal yield. The effect of interference on the final signal prediction is discussed below. The quantity Z is used to compare selections, while the final confidence limits are obtained with the likelihood analysis in Chapter 10.

Cut combinations with $b \leq 2$ are discarded to avoid selecting apparently background-free regions caused by limited MC statistics. If a particular simulated background has a negative weighted yield after a trial selection, its contribution is set to zero when evaluating the scan criterion. This is a prescription for comparing trial cuts in sparse simulated samples. It does not imply a negative physical event rate.

7.2.2 Interference and signal efficiency

The interference term in Eq. 5.4 changes the signal contribution remaining after the selection. If S denotes the pure BSM yield and I the interference yield, the excess above the SM prediction is $S + I$ at the nominal signal normalisation. Destructive interference gives $I < 0$, so a selection ranked using S alone can overestimate the final sensitivity. Its effect is therefore checked after the optimisation and included in the final statistical model.

The requirements on p_T^τ , E_T^{miss} and m_T suppress the low-energy phase space in which interference with SM W production is most important. In particular, the non-resonant requirements $p_T^\tau > 200$ GeV, $E_T^{\text{miss}} > 400$ GeV and $m_T > 600$ GeV retain the hard tail while reducing the destructive contribution. The residual interference has a negligible effect on the sensitivity of the one- b -jet regions in the parameter range probed by the analysis. This

conclusion does not apply to the zero- b -jet regions, where the interference contribution must be retained, as discussed in Section 7.3.

The signal efficiency is monitored through the cumulative cutflow in Table 7.2. For $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$, the resonant selection retains 4.2 of the 35.2 expected signal events passing the one- b -jet baseline, or approximately 12%. For the non-resonant optimisation benchmark, it retains 6.6 of 111.0 events, or approximately 6%. The background decreases from approximately 4027 baseline events to 2.4 and 2.1 events, respectively. These fractions describe signal retention during the cut optimisation and use the baseline sample as their denominator. The acceptance times selection efficiency, $A\epsilon$, instead uses the total generated signal as its reference and includes the kinematic acceptance and reconstruction and selection losses. The cutflow uses the pure BSM contribution because the signed interference yield is not an event-selection efficiency. The tabulated numbers describe the optimisation study before the background fit, while the final selections are chosen with the statistical uncertainty in the background prediction kept below 30%.

Table 7.2 Cumulative signal (S) and background (B) yields in the one- b -jet optimisation. The resonant and non-resonant signal benchmarks are $(M_U [\text{TeV}], g_U, \beta_L^{23}) = (1.5, 1.5, 0.6)$ and $(2.5, 2.5, 1.0)$, respectively. These are simulated expectations before the background fit.

Resonant			Non-resonant		
Cumulative requirement	S	B	Cumulative requirement	S	B
Baseline	35.2	4027.3	Baseline	111.0	4027.3
$p_T^\tau > 200 \text{ GeV}$	19.9	193.1	$p_T^\tau > 200 \text{ GeV}$	70.5	193.1
$E_T^{\text{miss}} > 200 \text{ GeV}$	19.9	193.1	$E_T^{\text{miss}} > 400 \text{ GeV}$	26.4	15.1
$m_T > 200 \text{ GeV}$	19.4	143.6	$m_T > 600 \text{ GeV}$	24.0	6.4
$m_{b\tau} > 800 \text{ GeV}$	4.9	5.2	$\Delta\phi(\tau, E_T^{\text{miss}}) > 1.2$	23.8	6.4
$N_j - N_b \leq 3$	4.7	4.3	$N_j - N_b \leq 1$	17.2	2.8
$p_T^b > 250 \text{ GeV}$	4.2	2.4	$50 < p_T^b < 250 \text{ GeV}$	6.6	2.1

7.2.3 Selected regions and background composition

The selected SR1b-Res requirements are $p_T^\tau > 200 \text{ GeV}$, $E_T^{\text{miss}} > 200 \text{ GeV}$, $m_T > 200 \text{ GeV}$ and $m_{b\tau} > 800 \text{ GeV}$, with at most three jets that are not b -tagged. No additional $\Delta\phi(\tau, E_T^{\text{miss}})$ requirement is retained. The region is treated as a single event-counting bin because the available background simulation does not support a finely binned fit to the broad $m_{b\tau}$ distribution.

The selected SR1b-NonRes requirements are $p_T^\tau > 200 \text{ GeV}$, $E_T^{\text{miss}} > 400 \text{ GeV}$, $m_T > 600 \text{ GeV}$ and $\Delta\phi(\tau, E_T^{\text{miss}}) > 1.2$, with at most one jet that is not b -tagged. No $m_{b\tau}$ require-

ment is imposed because the b -jet and τ do not reconstruct an on-shell leptoquark in the target topology. This region also contributes a single event count to the fit.

The loose $N - 1$ distributions in Figures 7.2 and 7.3 illustrate the variables entering the optimisation. An $N - 1$ distribution omits the requirement on the plotted variable while retaining the other requirements. To make the shapes visible with the available simulation, the resonant m_T and p_T^τ thresholds are lowered by 100 GeV. For the non-resonant region, the m_T , E_T^{miss} and p_T^τ thresholds are each lowered by 100 GeV before the plotted requirement is removed. The trigger-plateau requirement $E_T^{\text{miss}} > 200$ GeV is retained. These distributions therefore illustrate a larger phase space than the final SRs. The resonant signal has a broad $m_{b\tau}$ distribution, while the non-resonant signal extends into the high- m_T and high- E_T^{miss} tails.

The remaining background in both one- b -jet regions is dominated by W +jets, principally $W \rightarrow \tau\nu$. The tagged jet can originate from heavy-flavour production or a misidentified light-flavour jet. In the final background prediction, after the control-region constraints described in Chapter 8, W +jets contributes approximately 70% in each region. In SR1b-Res, diboson production contributes about 15%, followed by $t\bar{t}$ and single-top production, including Wt . In SR1b-NonRes, $t\bar{t}$ contributes about 14%, $Z \rightarrow \tau\tau$ +jets about 9%, and diboson production about 5%. Top-quark events can pass the selection through $t\bar{t} \rightarrow b\tau\nu\bar{b}\tau\nu$ or $Wt \rightarrow b\tau\nu\tau\nu$ when only one τ is selected and the remaining decay products satisfy the jet and lepton requirements. The $Z \rightarrow \tau\tau$ contribution similarly requires only one selected τ , with neutrinos from the decays producing E_T^{miss} . Among events with a jet misidentified as a τ , $Z \rightarrow \nu\nu$ +jets is the main source because the invisible Z decay supplies genuine missing momentum. Multijet production is negligible after the cleaning and full SR requirements.

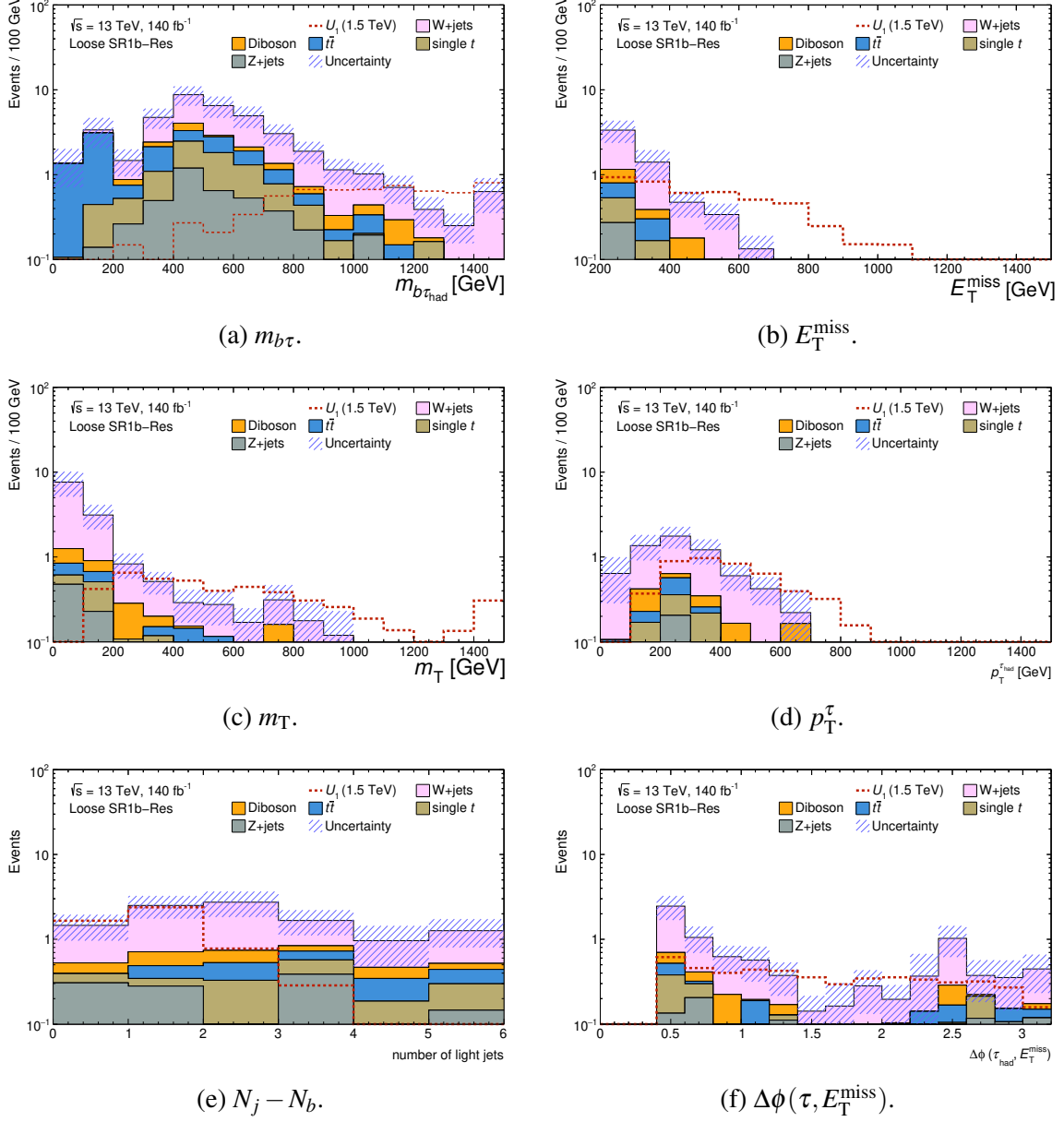


Fig. 7.2 Distributions of the six scan variables in the loose resonant one- b -jet region. The m_T and p_T^τ thresholds are reduced to 100 GeV, and the requirement on each plotted variable is removed, retaining the baseline $E_T^{\text{miss}} > 200$ GeV. The stacked backgrounds are shown before the fit. The dotted signal curve is the pure BSM prediction at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$. The hatched band represents the background uncertainty.

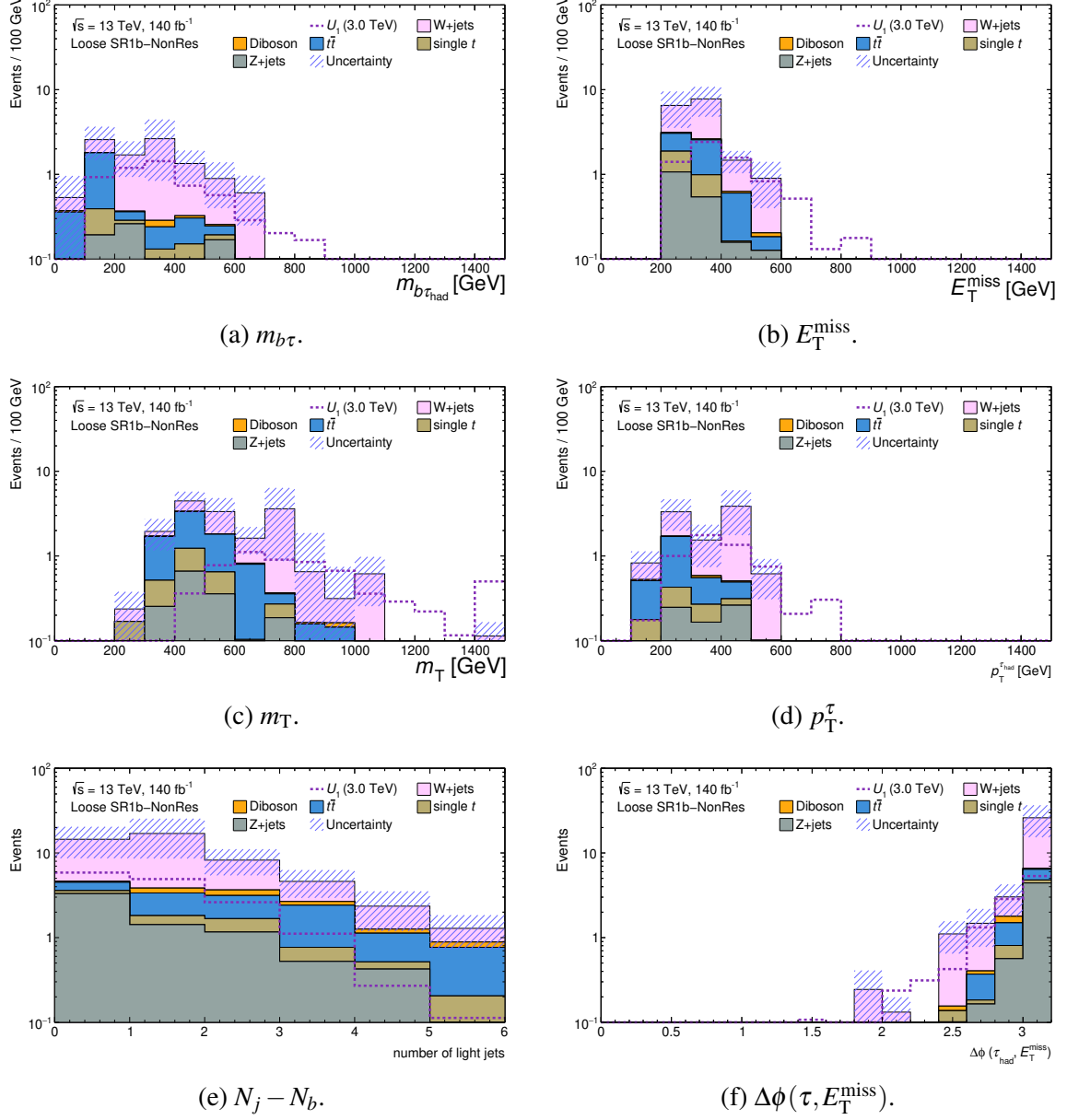


Fig. 7.3 Distributions of the six scan variables in the loose non-resonant one- b -jet region. Before removing the plotted requirement, the thresholds are lowered to $m_T > 500$ GeV, $E_T^{\text{miss}} > 300$ GeV and $p_T^\tau > 100$ GeV. The baseline $E_T^{\text{miss}} > 200$ GeV is retained. The stacked backgrounds are shown before the fit, and the hatched band represents their uncertainty. The dotted signal curve is the pure BSM prediction at $(M_U, g_U, \beta_L^{23}) = (3.0 \text{ TeV}, 2.5, 1.0)$.

7.3 b-veto signal region definition (SR0b)

The SR0b regions select signal events with no b -tagged jet and extend the sensitivity to large β_L^{23} . As shown by the flavour current and coupling texture in Eqs. 3.6 and 3.7, β_L^{23} controls the $s\tau$ interaction and contributes $V_{cs}\beta_L^{23}$ to the $c\nu_\tau$ interaction. The resulting zero- b -jet topologies were introduced in Section 3.1.5. In particular, $U_1 \rightarrow s\tau$ produces a resonant signal with a genuine non- b jet, so SR0b retains a physically distinct signal contribution in addition to events in which a b -jet is not tagged or reconstructed. A leading jet in this category is not explicitly identified as a strange-quark jet. The truth-level study also identifies a substantial contribution with a leading c jet. At $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$, approximately 30% and 60% of the selected truth-level resonant-category events have a leading s or c jet, respectively. The s -jet contribution contains the $s\tau$ resonance, whereas the c -jet contribution can contain an on-shell $U_1 \rightarrow c\nu_\tau$ decay. For the latter, the resonance is in the $c\nu_\tau$ system rather than the jet- τ system. In the non-resonant truth-level category, approximately 33% and 11% of the events have a radiated leading s or c jet, with most of the remainder containing a gluon jet in $sc \rightarrow \tau\nu_\tau$ production. These fractions describe the chosen benchmark and truth selection, rather than a flavour measurement of the reconstructed jets.

The two SR0b regions are defined by replacing the one- b -jet requirement with $N_b = 0$ and using the leading jet in the topology-dependent selections. The resonant region requires $p_T^{j_1} > 250 \text{ GeV}$ and $m_{j\tau} > 800 \text{ GeV}$, while the non-resonant region requires $50 < p_T^{j_1} < 250 \text{ GeV}$. The p_T^τ , E_T^{miss} , m_T and $\Delta\phi(\tau, E_T^{\text{miss}})$ thresholds follow the corresponding one- b -jet regions. The upper limits are imposed on the total jet multiplicity: $N_j \leq 4$ for SR0b-Res and $N_j \leq 2$ for SR0b-NonRes. These preserve the total jet limits used in SR1b after replacing the tagged jet with an untagged leading jet.

The $E_T^{\text{miss}} > 200 \text{ GeV}$ requirement in SR0b-Res is retained to keep its kinematic selection aligned with SR1b-Res. A supplementary study of tighter E_T^{miss} thresholds is presented in Appendix C. The SR0b-NonRes threshold remains $E_T^{\text{miss}} > 400 \text{ GeV}$, which also maximises the significance in that study.

Signal efficiency in SR0b is evaluated for the pure BSM contribution, with $A\epsilon$ defined as in Section 7.2.2. For illustration, at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.0, 0.2)$ with $\beta_R^{33} = 0$, $A\epsilon$ is 0.30% and 0.13% in SR0b-Res and SR0b-NonRes, compared with 0.64% and 0.069% in the corresponding SR1b regions. These efficiencies depend on the mass and flavour couplings because the relative contributions of the production and decay channels change. The destructive interference in the final zero- b -jet selections reduces the nominal pure BSM yield by approximately 20% at $M_U = 1.5 \text{ TeV}$ and by about 30–40% at higher masses. This reduction is appreciable even after the high- m_T and E_T^{miss} requirements, so the final prediction

includes the signed interference contribution explicitly. The supplementary E_T^{miss} scan in Appendix C also shows that a higher E_T^{miss} threshold can suppress this contribution.

Figure 7.4 shows the $m_{j\tau}$ distribution before the mass requirement in SR0b-Res. It also shows the transverse-mass distribution above 600 GeV in SR0b-NonRes. The high- $m_{j\tau}$ requirement enhances the resonant signal contribution, while the high- m_T requirement selects non-resonant production. Together with the one- b -jet regions, these selections retain sensitivity to both $b\tau$ and $s\tau$ final states over the coupling range studied in this thesis.

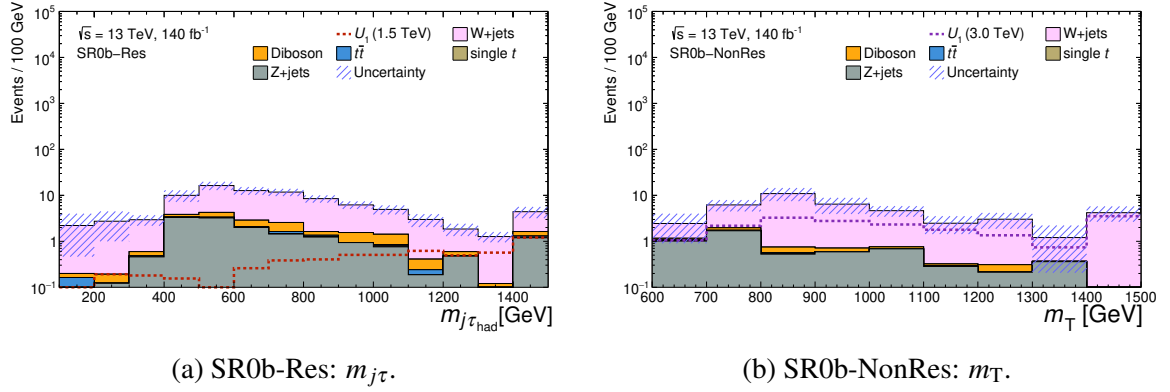


Fig. 7.4 Distributions of (a) $m_{j\tau}$ in SR0b-Res with the mass requirement removed and (b) m_T in SR0b-NonRes above 600 GeV. The other SR requirements are retained. The backgrounds are shown before the fit, and the hatched band represents their uncertainty. The dotted curves show pure BSM signals at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$ in (a) and $(3.0 \text{ TeV}, 2.5, 1.0)$ in (b).

The zero- b -jet background is dominated by $W \rightarrow \tau\nu$ +jets, since vetoing tagged b jets removes much of the top-quark background while retaining W production with light-flavour jets. In the final prediction after the control-region constraints, W +jets contributes approximately 76% in SR0b-Res and 82% in SR0b-NonRes. In the resonant region, diboson production contributes about 10%, $Z \rightarrow \nu\nu$ +jets about 8%, and $Z \rightarrow \tau\tau$ +jets about 5%. In the non-resonant region, the corresponding contributions are approximately 3%, 6% and 8%. The $Z \rightarrow \nu\nu$ events contain a jet misidentified as a τ , while $Z \rightarrow \tau\tau$ and diboson events can contain genuine τ leptons and neutrinos. The combined $t\bar{t}$ and single-top contribution is below 1% in each zero- b -jet region.

The final requirements for all four mutually exclusive SRs are summarised in Table 7.3. Each region contributes one event count to the simultaneous statistical analysis.

Table 7.3 Common baseline requirements and final signal-region selections. A dash denotes no additional requirement. The jet multiplicity includes b -tagged jets.

Baseline requirements				
Trigger	Lowest-threshold unprescaled E_T^{miss} trigger			
N_τ	1			
$N_\ell = N_e + N_\mu$	0			
Multijet suppression	$\Delta\phi_{\min} > 0.4$ or $R_{\text{miss}/j} > 6$			
Requirement	Resonant		Non-resonant	
	SR1b-Res	SR0b-Res	SR1b-NonRes	SR0b-NonRes
N_b	1	0	1	0
p_T^b [GeV]	> 250	–	$50 < p_T^b < 250$	–
p_T^{j1} [GeV]	–	> 250	–	$50 < p_T^{j1} < 250$
p_T^τ [GeV]			> 200	
E_T^{miss} [GeV]	> 200			> 400
m_T [GeV]	> 200			> 600
$m_{b\tau}$ [GeV]	> 800	–	–	–
$m_{j\tau}$ [GeV]	–	> 800	–	–
$\Delta\phi(\tau, E_T^{\text{miss}})$	–			> 1.2
N_j	≤ 4			≤ 2

7.4 Tau-pair signal contamination

Leptoquark production with a $\tau\tau$ final state can also populate the $\tau\nu$ + jets signal regions. Although two τ -leptons are produced, an event can satisfy the single- τ requirement when only one is reconstructed and selected. Neutrinos from the τ decays produce missing momentum. This is an additional BSM contribution from the same U_1 model, rather than an SM background.

The study uses simulated $\tau\tau$ samples generated with $\beta_L^{33} = 1$, $\beta_R^{33} = 0$ and $\beta_L^{23} = 0$. Their selected yields are compared with available $\tau\nu$ samples near the exclusion sensitivity, at $(M_U, g_U, \beta_L^{23}) = (1.8 \text{ TeV}, 3.0, 0.6)$ and $(3.0 \text{ TeV}, 2.0, 1.4)$. The corresponding $\tau\tau$ samples use $g_U = 2.8$ and 2.1 . The $\tau\tau$ production cross section is recalculated with MADGRAPH5_AMC@NLO at larger β_L^{23} , and the selected yields are scaled by the cross-section ratio to their generation point. This estimates the additional rate while retaining the acceptance and kinematic shapes of the available samples. It therefore provides a sensitivity study rather than a simulation of the full coupling dependence of the $\tau\tau$ acceptance.

The estimated additional yield is of order 35% of the nominal $\tau\nu$ yield in the one- b -jet regions and several percent in the zero- b -jet regions. To assess the effect on the search, the nominal signal predictions are increased by the factors shown in Table 7.4, and the exclusion fit is repeated.

Table 7.4 Additional signal normalisation used to estimate the sensitivity to $\tau\tau$ production. The factors multiply the nominal $\tau\nu$ signal prediction in the supplementary fit.

Region	Additional contribution	Applied factor
SR1b-Res	37%	1.37
SR1b-NonRes	33%	1.33
SR0b-Res	5%	1.05
SR0b-NonRes	7%	1.07

The expected upper limit on the signal strength improves by approximately 8–14% in this test. The corresponding change in the coupling limit is smaller because the signal rate depends strongly on g_U . At $M_U = 1.8 \text{ TeV}$ and $\beta_L^{23} = 0.6$, the expected upper limit on g_U changes from 2.18 to 2.07, an improvement of approximately 5%. At $M_U = 3.0 \text{ TeV}$ and $\beta_L^{23} = 1.4$, it changes from 1.96 to 1.91, or approximately 2.6%. The nominal interpretation retains the $\tau\nu$ signal model. The supplementary study indicates that the omitted positive $\tau\tau$ contribution could strengthen the expected coupling constraints by a few percent under the assumptions used for the rate estimate.

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Chapter 8

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Background Estimation

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The background prediction is obtained from the simulated samples introduced in Section 5.3, with the dominant $W + \text{jets}$ and top-quark contributions normalised to data in control regions (CRs). This chapter describes these constraints, the treatment of the small fake- τ contribution, and the validation of the prediction in regions outside the SRs. The object definitions and SR selections follow Chapters 6 and 7.

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8.1 Strategy overview

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The CRs provide data samples enriched in the backgrounds to be normalised, with little signal contamination. Four CRs are used: CRW-Res and CRW-NonRes constrain $W + \text{jets}$, while CRTop-Res and CRTop-NonRes constrain top-quark production. The total event count in each CR is used in a simultaneous fit. The fitted normalisation factors, λ_W^{Res} , $\lambda_W^{\text{NonRes}}$, λ_t^{Res} and $\lambda_t^{\text{NonRes}}$, multiply the corresponding simulated yields in the CRs, SRs and validation regions (VRs). The same top-quark factor is applied to $t\bar{t}$ and single-top production within each topology. Separate factors for the resonant and non-resonant categories allow the normalisation to differ between the high- and low-jet- p_T selections.

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The extrapolation from a CR to an SR is supplied by simulation: the fit fixes the background rate in the CR, while the simulated ratio of SR to CR yields determines the rate transferred to the SR. The $W + \text{jets}$ extrapolation therefore depends on the description of the lepton flavour and jet flavour, and the top-quark extrapolation also depends on the acceptance of the additional lepton and the relaxed CR requirements. The relevant uncertainties are described in Chapter 9. Diboson and $Z + \text{jets}$ production, together with the other small contributions, retain their simulated nominal yields and are allowed to vary within their assigned uncertainties in the fit. The fitted uncertainty parameters are propagated to the

SRs and VRs together with the normalisation factors. The multijet suppression is already described in Section 7.1.2 and Appendix B.

The VRs test these extrapolations without contributing event counts to the background-only fit. Their object multiplicities are summarised in Table 8.1. In particular, VRW tests the change from a b -jet veto to one tagged jet, and VRTop tests the prediction for a hadronic τ with no selected light lepton. The zero- b -jet regions containing a τ are the SR0b regions defined in Section 7.3. The likelihood construction and the final fitted normalisation factors are reserved for Chapter 10.

Table 8.1 Roles and object multiplicities of the CRs and nominal VRs. Each row represents separate resonant and non-resonant regions. The supplementary $1\tau 0\ell 1b$ regions are defined in Section 8.5.4.

Region	N_τ	N_ℓ	N_b	Purpose
CRW	0	1	0	Constrain $W + \text{jets}$ normalisation
CRTop	1	1	≥ 1	Constrain top-quark normalisation
VRW	0	1	1	Test the $0b \rightarrow 1b$ extrapolation
VRTop	1	0	2	Test the top prediction without a light lepton

8.2 $W + \text{jets}$ estimation

8.2.1 Control-region definition

The CRW regions select $W \rightarrow \ell \nu$ events by replacing the τ in the corresponding SR0b selection with an electron or muon. The b -jet veto suppresses top-quark production and provides a larger $W + \text{jets}$ sample than a tagged selection. Exactly one signal electron or muon and no selected τ are required. The signal-lepton requirements are those in Chapter 6. Single-lepton triggers are used, as listed in Section 5.4. This choice avoids the different response of the calorimeter-based E_T^{miss} trigger to electrons and muons, since the trigger-level missing momentum does not include the muon momentum.

The lepton replaces the τ in the transverse mass and angular separation, giving m_T^ℓ and $\Delta\phi(\ell, E_T^{\text{miss}})$ through the definitions in Section 7.1.1. The leading jet is used for the resonant/non-resonant categorisation and the invariant mass $m_{j\ell}$. Table 8.2 gives the selections together with those of VRW. The kinematic correspondence to the SRs restricts the extrapolation to similar hard-scattering configurations. The neutrino from a hadronic τ decay changes both the visible τ momentum and $\vec{p}_T^{\text{miss}}$, so the lepton replacement does not imply identical reconstructed distributions. These differences are retained in the simulated

acceptance ratio. The extrapolation is checked in the lower-mass $W \rightarrow \tau\nu$ sample described in Section 8.5.4. Both CRs have a $W + \text{jets}$ purity of approximately 70%. Diboson production is the largest secondary contribution in CRW-Res.

Table 8.2 Selections for CRW and VRW. Both require $N_\tau = 0$, $N_\ell = 1$ and a single-lepton trigger. The common event-quality and multijet requirements are retained. The selected jet is the leading jet in CRW and the tagged jet in VRW. Its momentum is denoted by p_T^J .

Requirement	Resonant	Non-resonant
N_b	0 in CRW, 1 in VRW	
p_T^ℓ [GeV]	> 200	
p_T^J [GeV]	> 250	$50 < p_T^J < 250$
E_T^{miss} [GeV]	> 200	> 400
m_T^ℓ [GeV]	> 200	> 600
$\Delta\phi(\ell, E_T^{\text{miss}})$	No requirement	> 1.2
$m_{j\ell}$ or $m_{b\ell}$ [GeV]	> 800	No requirement
N_j	≤ 4	≤ 2

8.2.2 Pre-fit comparisons

The pre-fit comparisons in Figures 8.1 and 8.2 test the simulated distributions before the CR normalisation is fitted. The $N - 1$ convention is the one introduced in Section 7.2. The electron and muon channels are shown separately to test their consistency before combining them. The plotted bands and the uncertainties in Table 8.3 are statistical only. No clear trend in the data-to-prediction ratio is observed across the displayed kinematic distributions, although their overall normalisations differ. The raw data-to-total-MC ratios are diagnostic quantities: they are not the fitted $W + \text{jets}$ normalisation factors, since the CRs also contain other backgrounds.

Table 8.3 Observed counts and data/total-MC ratios in the CRs before fitting. The uncertainties are statistical.

Region	Electron channel		Muon channel	
	Data	Data/MC	Data	Data/MC
CRW-Res	243	1.25 ± 0.09	219	1.08 ± 0.08
CRW-NonRes	474	1.00 ± 0.09	398	0.80 ± 0.08
CRTop-Res	30	0.73 ± 0.13	33	0.86 ± 0.15
CRTop-NonRes	240	0.81 ± 0.05	228	0.83 ± 0.06

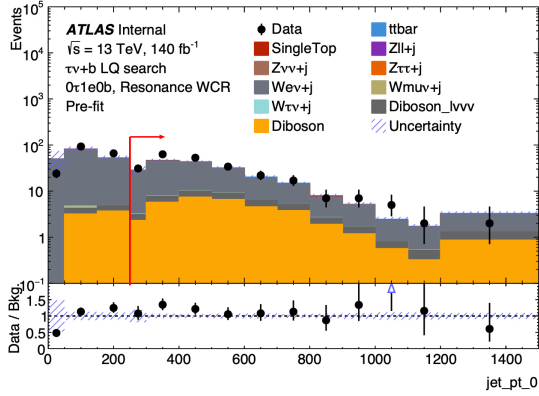
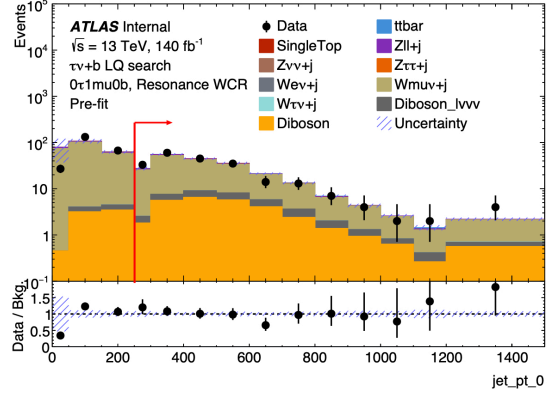
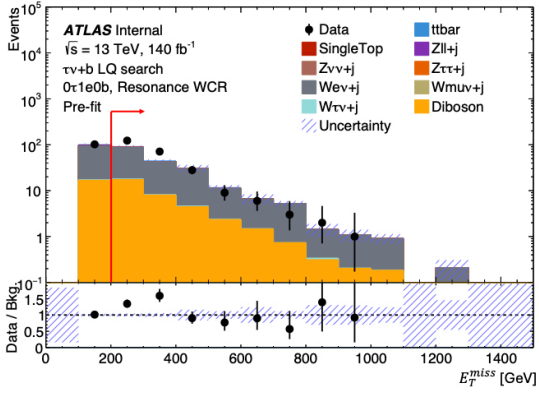
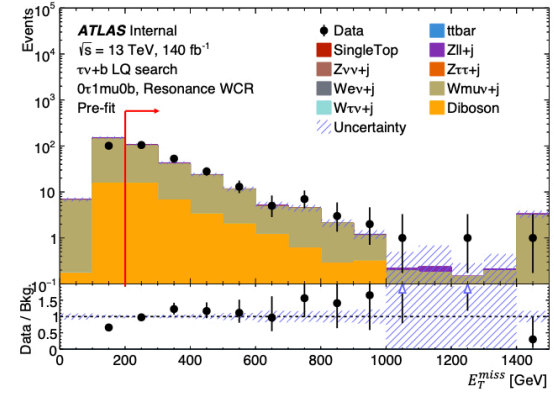
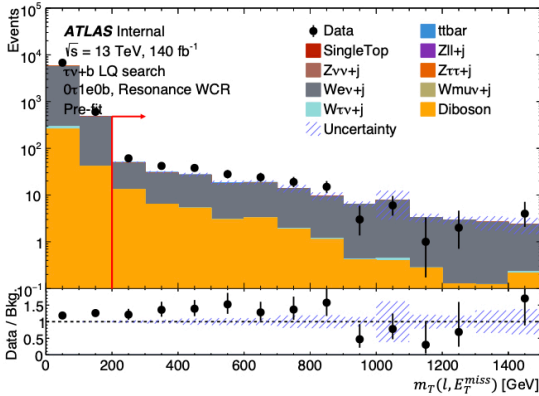
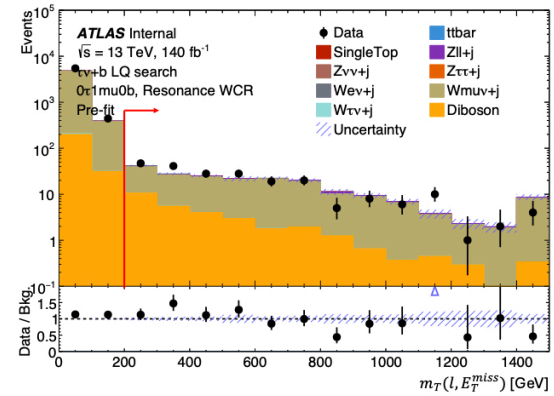
(a) p_T^j , e channel(b) p_T^j , μ channel(c) E_T^{miss} , e channel(d) E_T^{miss} , μ channel(e) m_T^ℓ , e channel(f) m_T^ℓ , μ channel

Fig. 8.1 Pre-fit $N - 1$ distributions in CRW-Res, with the electron channel in the left column and the muon channel in the right column. The stacked prediction contains all simulated backgrounds. The lower panels show data divided by the total prediction, and the hatched bands show the statistical uncertainty in that prediction. Arrows indicate the selection boundaries. The last bin includes the overflow.

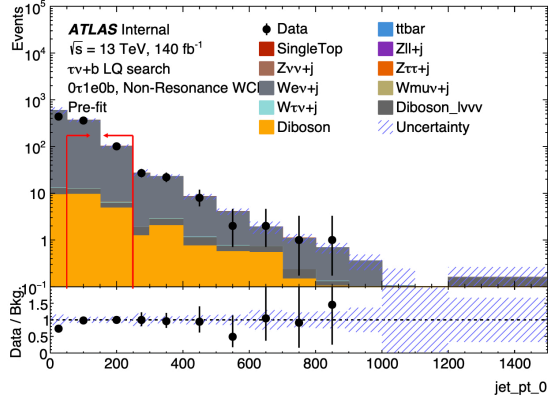
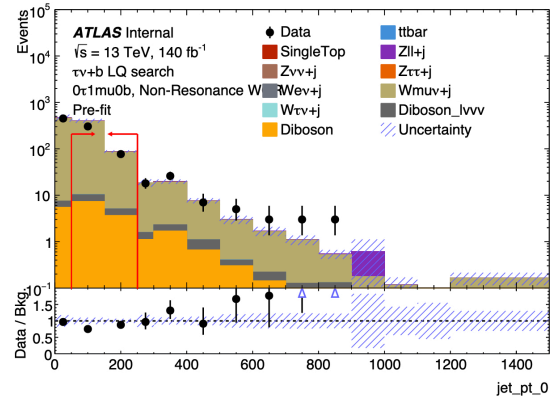
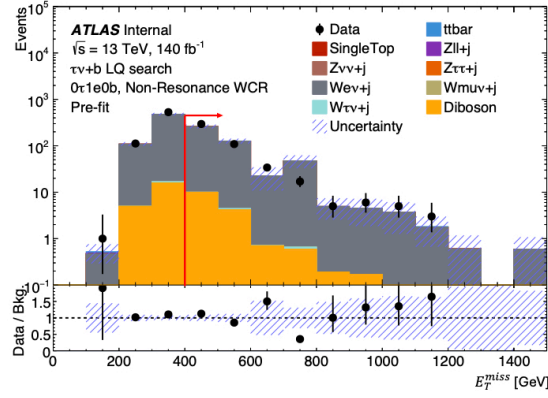
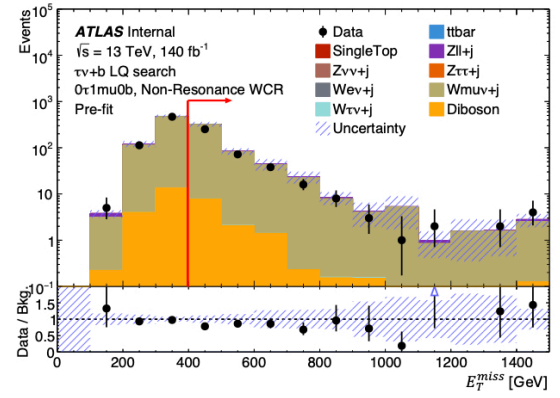
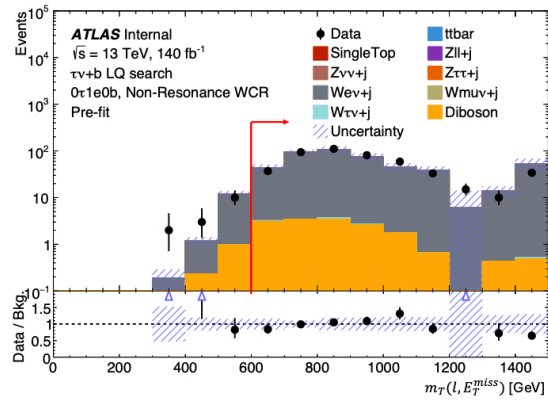
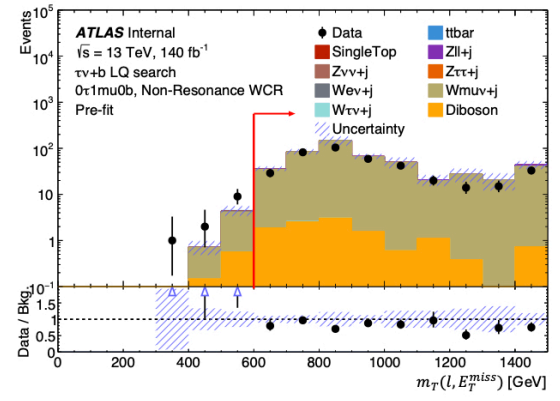
(a) p_T^{j1} , e channel(b) p_T^{j1} , μ channel(c) E_T^{miss} , e channel(d) E_T^{miss} , μ channel(e) m_T^l , e channel(f) m_T^l , μ channel

Fig. 8.2 Pre-fit $N - 1$ distributions in CRW-NonRes, with the electron channel in the left column and the muon channel in the right column. The stacked prediction contains all simulated backgrounds. The lower panels show data divided by the total prediction, and the hatched bands show the statistical uncertainty in that prediction. Arrows indicate the selection boundaries. The last bin includes the overflow.

8.2.3 Electron–muon consistency

The electron and muon channels give pre-fit data/MC ratios differing by 1.4σ in CRW-Res and 1.7σ in CRW-NonRes. The non-resonant difference is investigated through the corrections applied to simulation, the selection cutflow and the leading-jet p_T distribution. The lepton-dependent efficiency corrections, including the trigger scale factors, are checked separately in the two channels, with no incorrect application identified. The difference becomes visible after imposing $50 < p_T^{j_1} < 250$ GeV.

Figure 8.3 compares the electron and muon distributions separately in data and simulation. The electron data have an upward fluctuation around $p_T^{j_1} = 80\text{--}100$ GeV. The simulated event-weight distributions reveal a second effect: a small number of events carry weights of order 30. Most occur below 50 GeV, but one such event enters the selected muon sample above this threshold, with no corresponding event in the electron sample. A single simulated event can therefore increase the predicted muon yield by several tens of weighted events and lower its data/MC ratio.

These observations are consistent with statistical fluctuations in the electron data and the weighted muon simulation. They do not establish a lepton-dependent correction to the $W + \text{jets}$ rate. The two channels are therefore combined within each CR, and one $W + \text{jets}$ normalisation factor is used for each topology. The extrapolation of this combined constraint is checked in the VRs described in Section 8.5.

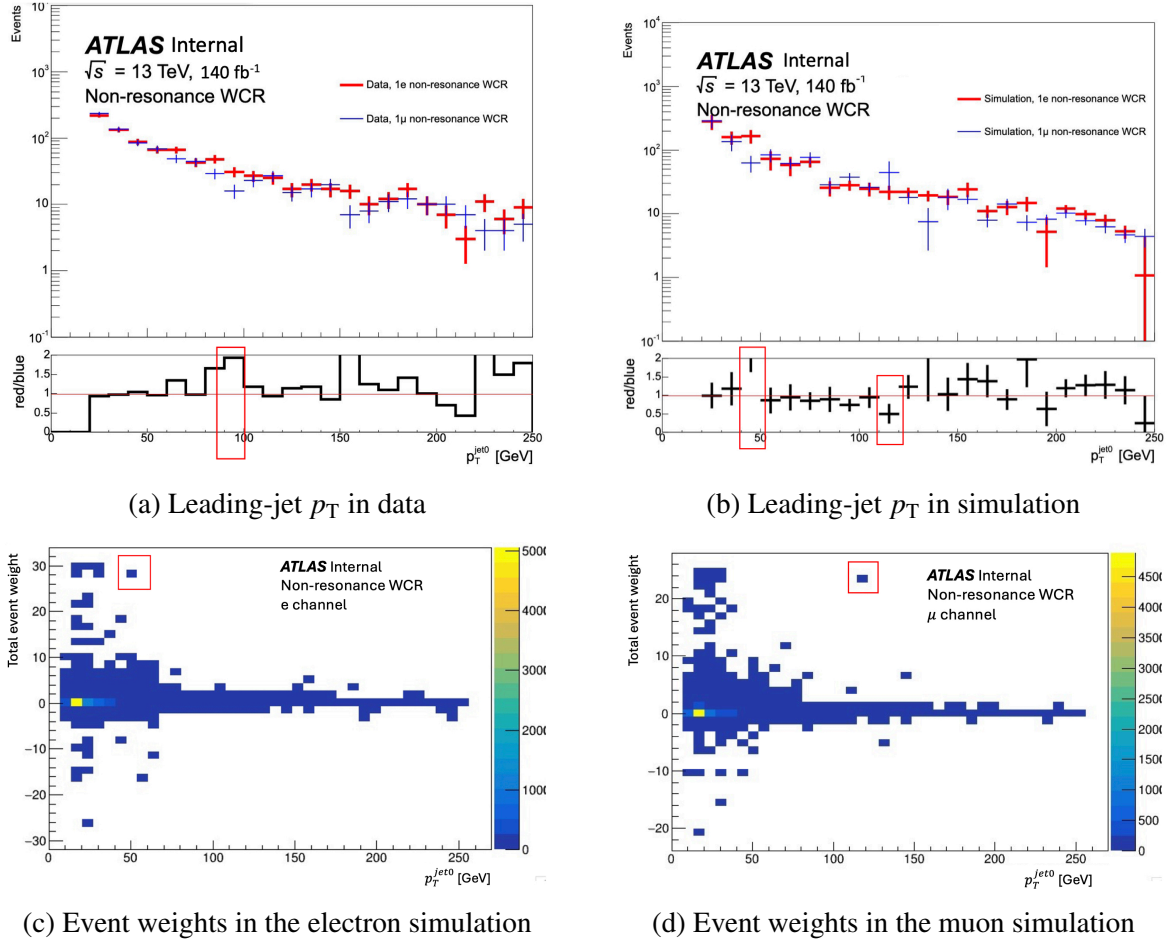


Fig. 8.3 Electron–muon consistency checks in CRW-NonRes. In (a) and (b), red and blue denote the electron and muon channels, and the lower panels show their ratio. Panels (c) and (d) show the simulated event weights against leading-jet p_T . The marked high-weight muon event illustrates the statistical effect discussed in the text.

8.3 Top-quark estimation

The $t\bar{t}$ and single-top backgrounds are constrained in the CRTop regions, which require one τ and an additional electron or muon. The relevant decays include $t\bar{t} \rightarrow b\tau\nu\bar{b}\ell\nu$ and $Wt \rightarrow b\tau\nu\ell\nu$. The additional lepton selects the second leptonic W decay and separates these regions from the lepton-vetoed SRs. Requiring at least one tagged b jet further suppresses $W + \text{jets}$ production. Inclusive $t\bar{t}$ and single-top samples are used. Within the $t\bar{t}$ and Wt samples, the large- E_T^{miss} selection favours decays in which both W bosons decay leptonically. The top-quark purity is approximately 90% after the full CR selections.

The relaxed thresholds in Table 8.4 increase the CR yields. Single-lepton triggers permit the lower $E_T^{\text{miss}} > 100$ GeV threshold. If more than one b jet is selected, the highest- p_T tagged jet defines the topology and the jet- τ observables. The resonant $m_{b\tau}$ requirement is removed.

Table 8.4 Selections for CRTop and VRTop. All regions require $N_\tau = 1$. Single-lepton triggers are used in CRTop and E_T^{miss} triggers in VRTop. The highest- p_T tagged jet is used when $N_b > 1$. Common event-quality and multijet requirements are retained.

Requirement	CRTop		VRTop	
	Res	NonRes	Res	NonRes
N_ℓ		1		0
N_b		≥ 1		2
p_T^ℓ [GeV]		> 28		—
p_T^τ [GeV]		> 100		
p_T^b [GeV]	> 250	(50, 250)	> 250	(50, 250)
E_T^{miss} [GeV]	> 100	> 100	> 200	> 200
m_T [GeV]	> 100	> 100	> 100	> 200
$m_{b\tau}$ [GeV]	—	—	> 400	—
$\Delta\phi(\tau, E_T^{\text{miss}})$	—	> 1.2	—	> 1.2
N_j	≤ 4	≤ 2	≤ 4	≤ 2

The pre-fit distributions in Figures 8.4 and 8.5 show comparable behaviour in the electron and muon channels. The data/MC ratios in Table 8.3 are near 0.8, with no clear dependence on m_T or E_T^{miss} resolved within the statistical precision. The combined top-quark yield is constrained by the fit, while the relative $t\bar{t}$ and single-top contributions are predicted separately in each region. Their ratio is consistent between the SRs and CRs within MC statistical uncertainties. The process-dependent modelling uncertainties are retained, as described in Chapter 9.

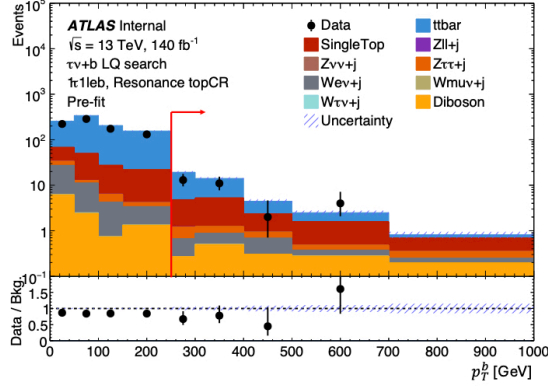
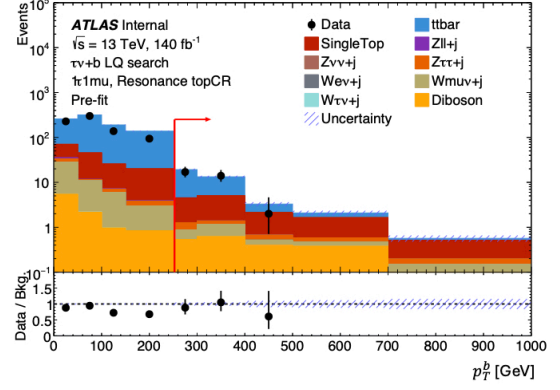
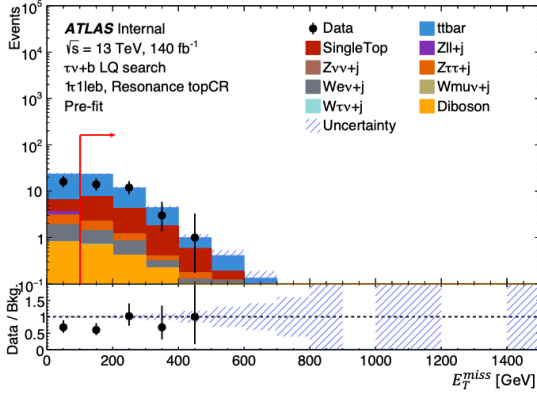
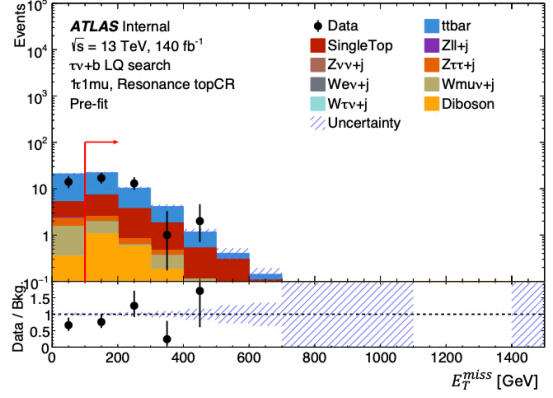
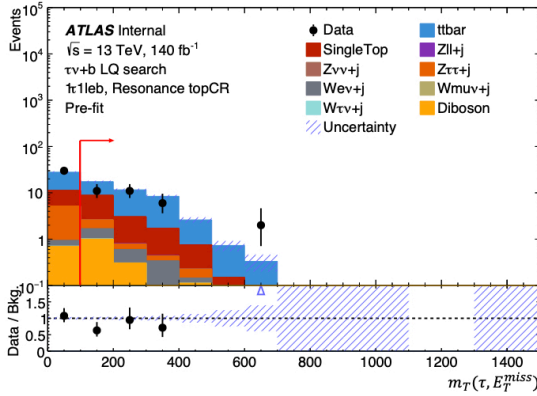
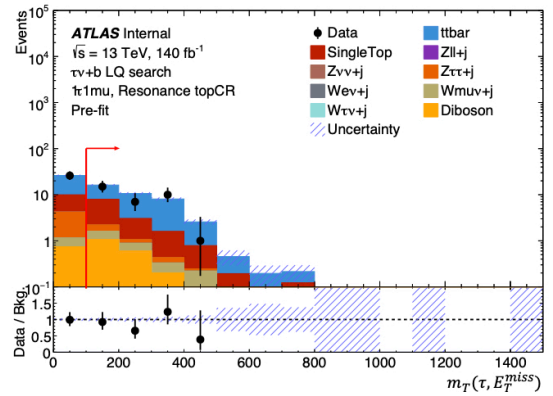
(a) p_T^b , e channel(b) p_T^b , μ channel(c) E_T^{miss} , e channel(d) E_T^{miss} , μ channel(e) m_T , e channel(f) m_T , μ channel

Fig. 8.4 Pre-fit $N - 1$ distributions in CRTop-Res, with the electron channel in the left column and the muon channel in the right column. The stacked prediction contains all simulated backgrounds. The lower panels show data divided by the total prediction, and the hatched bands show the statistical uncertainty in that prediction. Arrows indicate the selection boundaries. The last bin includes the overflow.

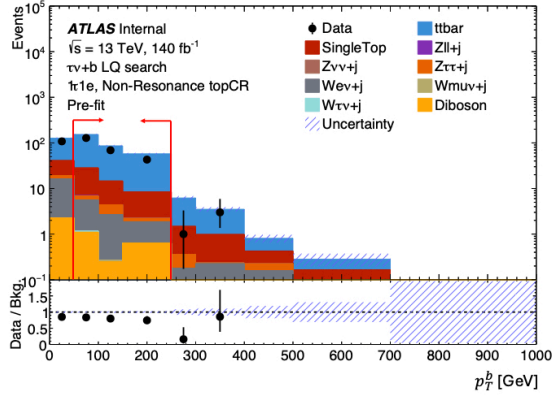
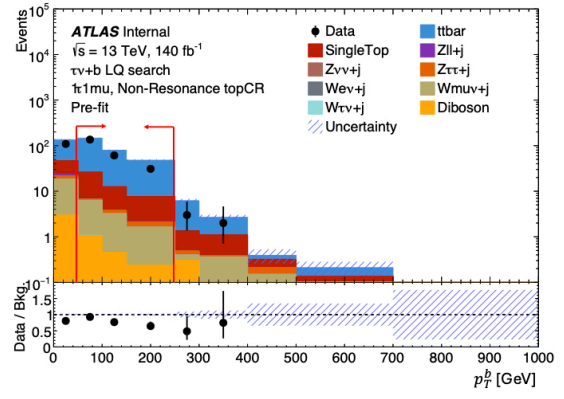
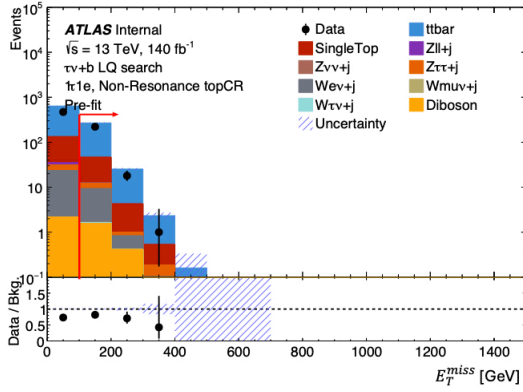
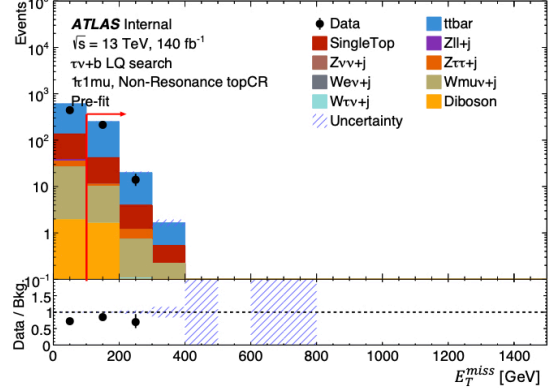
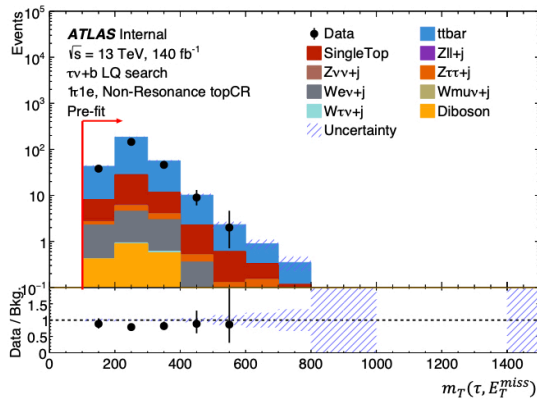
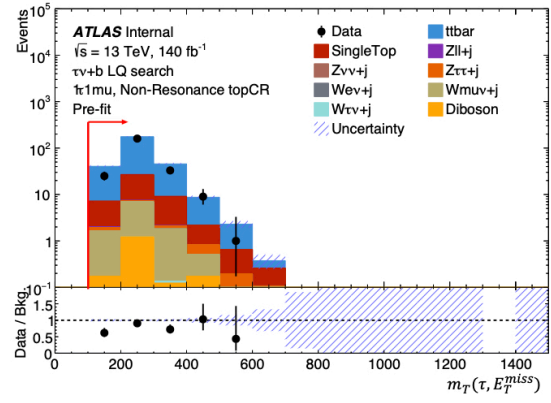
(a) p_T^b , e channel(b) p_T^b , μ channel(c) E_T^{miss} , e channel(d) E_T^{miss} , μ channel(e) m_T , e channel(f) m_T , μ channel

Fig. 8.5 Pre-fit $N - 1$ distributions in CRTop-NonRes, with the electron channel in the left column and the muon channel in the right column. The stacked prediction contains all simulated backgrounds. The lower panels show data divided by the total prediction, and the hatched bands show the statistical uncertainty in that prediction. Arrows indicate the selection boundaries. The last bin includes the overflow.

8.4 Fake- τ estimation

The fake- τ contribution is small because the selected SM background is dominated by events containing a genuine hadronically decaying τ . As discussed in Section 7.2, its principal source at large E_T^{miss} is $Z \rightarrow \nu\nu + \text{jets}$, where a jet passes the τ identification. This contribution is at most about 10% of the background in the SRs considered in the fake-rate study and is much smaller in the one- b -jet SRs. The nominal yield is taken from simulation, and a dedicated $Z \rightarrow \ell\ell$ study determines an uncertainty for the modelling of jets misidentified as τ candidates.

8.4.1 Validation with $Z \rightarrow \ell\ell$ events

The Z validation region (ZVR) selects a visible Z decay together with a τ candidate to test the simulated fake rate in a sample enriched in $Z + \text{jets}$. Two signal electrons or two signal muons are required, with $75 < m_{\ell\ell} < 105$ GeV, together with exactly one τ candidate and a b -jet veto. Events are required to satisfy $1 \leq N_j \leq 3$ and $p_T^\tau > 100$ GeV. Both lepton channels are used to increase the statistical precision. In the muon channel, events are selected using the E_T^{miss} trigger, whereas a single-electron trigger is used in the electron channel.

The quantity $|\vec{p}_T^{\ell\ell} + \vec{p}_T^{\text{miss}}|$ is required to exceed 200 GeV. Adding the reconstructed dilepton momentum to $\vec{p}_T^{\text{miss}}$ treats the visible leptons as unobserved and provides a recoil quantity analogous to the missing momentum in $Z \rightarrow \nu\nu$ events. An additional $p_T^{\ell\ell} > 130$ GeV requirement suppresses top-quark contamination. A transverse-mass threshold of 100 GeV is also applied in this study. The common multijet requirement is retained.

A total of 116 data events pass the ZVR selection, and the simulated $Z + \text{jets}$ purity is approximately 90%. Figure 8.6 shows the p_T^τ distribution with its threshold removed. The simulation overestimates the yield at high p_T^τ . Differences of up to 60% are observed in the overall yields in the validation comparisons. This comparison alone is sensitive to both the fake rate and the production kinematics of $Z + \text{jets}$. The RNN-score distributions in Figure 8.7 provide a further check using looser τ identification and a relaxed transverse-mass requirement. They compare samples with $p_T^\tau > 20$ GeV and $p_T^\tau > 100$ GeV. The better agreement at low RNN scores indicates that the discrepancy is mainly connected to the probability for a jet to pass the τ identification.

The small SR contribution and the precision of this check do not motivate a differential correction of the simulated fake rate. The nominal MC prediction for $Z \rightarrow \nu\nu + \text{jets}$ is therefore retained with an additional 100% normalisation uncertainty. This uncertainty covers the discrepancy observed in the ZVR. Changing the assigned uncertainty from 50% to 100% gives no appreciable change in the tested final results because this background is small. This

2109 additional uncertainty describes the fake- τ modelling in this component and is separate from
 2110 the uncertainties in genuine- τ reconstruction and identification. No independent fake-factor
 2111 prediction is used for the SR yield.

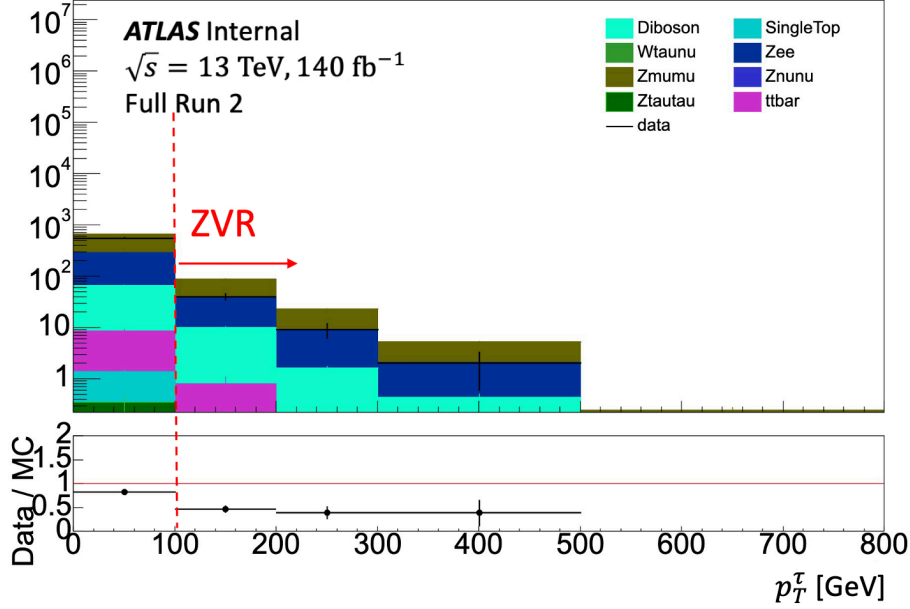


Fig. 8.6 Pre-fit $N - 1$ distribution of p_T^τ in the ZVR. The red arrow indicates the 100 GeV threshold. The lower panel compares data with the total simulation.

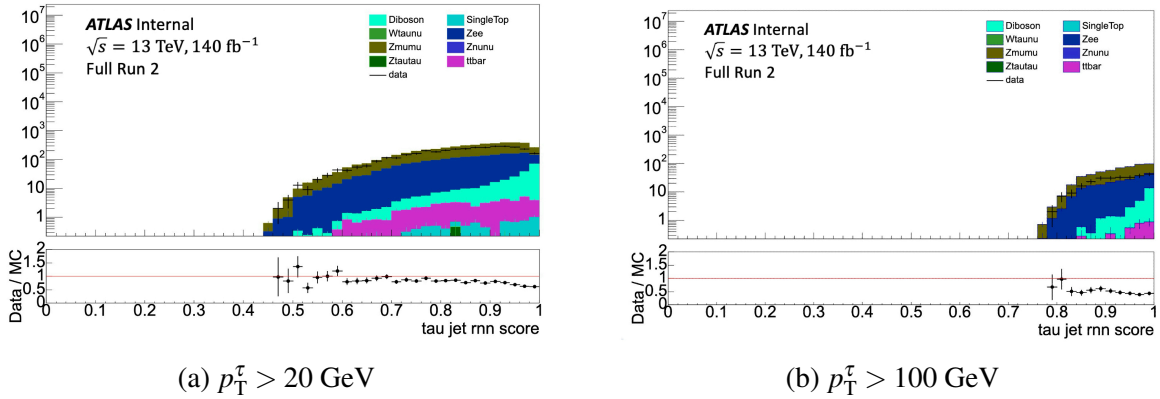


Fig. 8.7 The τ -identification RNN score in $Z + \text{jets}$ -enriched samples with (a) $p_T^\tau > 20$ GeV and (b) $p_T^\tau > 100$ GeV. Looser τ identification and a relaxed transverse-mass requirement are used to compare the low- and high-score regions. These selections differ from the nominal ZVR.

8.5 Validation regions

The VRs test the prediction in selections that isolate the main changes between the CRs and SRs. The nominal VRW and VRTop definitions are given in Tables 8.2 and 8.4. Their pre-fit distributions are used to inspect the simulation before fitting, while their post-fit yields test the extrapolation of the CR constraints.

8.5.1 One- b -jet validation with a light lepton

The VRW regions retain the single-electron or single-muon final state of CRW and require exactly one tagged b jet. The tagged jet replaces the leading untagged jet in the topology and invariant-mass selections. This change tests the prediction for the tagged sample, including the simulated heavy-flavour fraction of $W + \text{jets}$, before replacing the light lepton with a τ in the SR. The top-quark contribution increases under this selection and is predicted using the constraint from CRTop.

The flavour composition of vector-boson-plus-jets production depends on the kinematics and the simulation, as investigated in ATLAS modelling studies and measurements [125–127]. A 30% uncertainty is assigned to the ratio of W production with heavy-flavour hadrons, containing b or c quarks, to production with light-flavour hadrons in the tagged regions. This uncertainty covers possible differences in the flavour composition when transferring the b -vetoed CR constraint to a tagged sample. It is treated as uncorrelated between the resonant and non-resonant categories because they select different jet- p_T ranges. Its implementation and the remaining modelling uncertainties are described in Chapter 9.

The pre-fit $N - 1$ comparisons are shown in Figures 8.8 and 8.9. The electron and muon channels are combined in the plots, and their inclusive yields are listed separately in Table 8.5. The resonant region shows agreement within the statistical precision. The non-resonant region contains more data events than predicted in both lepton channels. The investigation of this excess and the additional check with a τ and one b jet are described below.

Table 8.5 Pre-fit background predictions and observed yields after the complete VRW selections. The prediction uncertainties are statistical only.

Region	Channel	Predicted background	Data
VRW-Res	e	17.0 ± 0.9	20
	μ	16.1 ± 1.0	17
VRW-NonRes	e	22.8 ± 4.4	33
	μ	23.8 ± 3.6	39

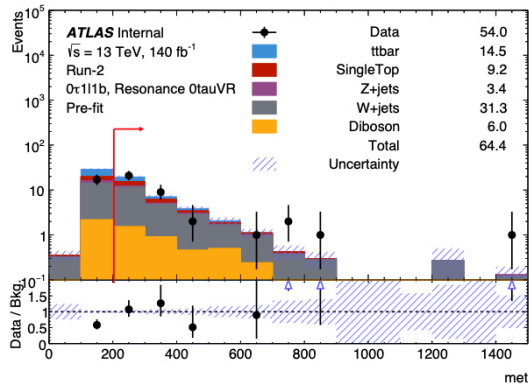
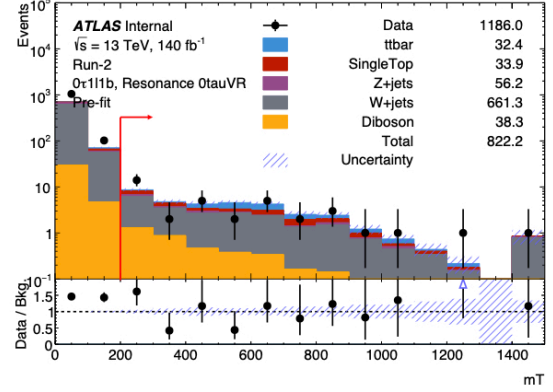
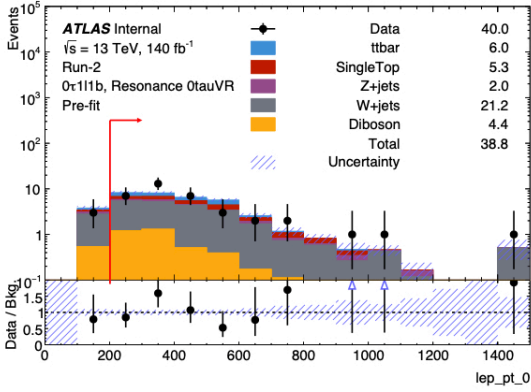
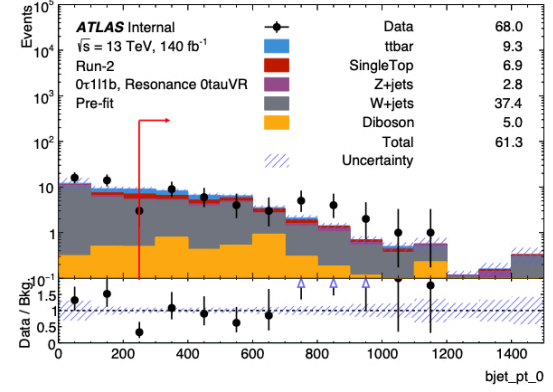
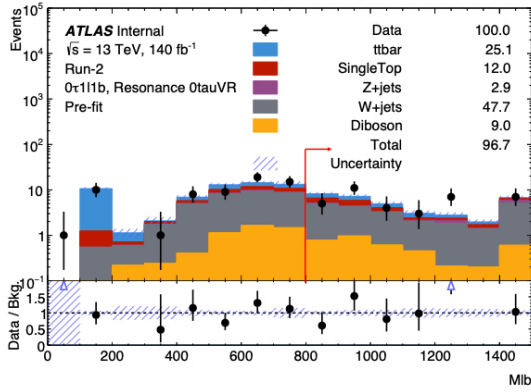
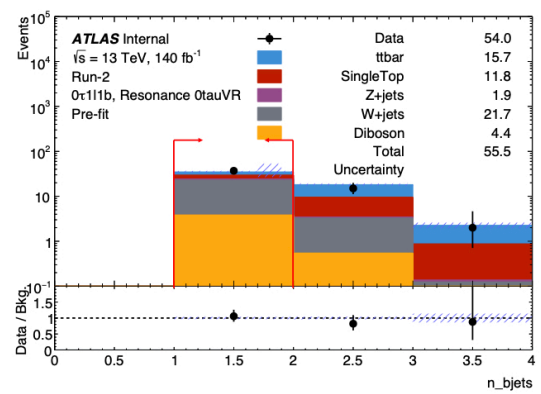
(a) E_T^{miss} (b) m_T^l (c) p_T^l (d) p_T^b (e) $m_{b\ell}$ (f) N_b

Fig. 8.8 Pre-fit $N - 1$ distributions in VRW-Res. The electron and muon channels are combined. The lower panels show data divided by the total simulated background. The hatched bands represent the statistical uncertainty in the prediction. The last bin includes the overflow. The original plot labels $0\tau\text{VR}$ denote VRW.

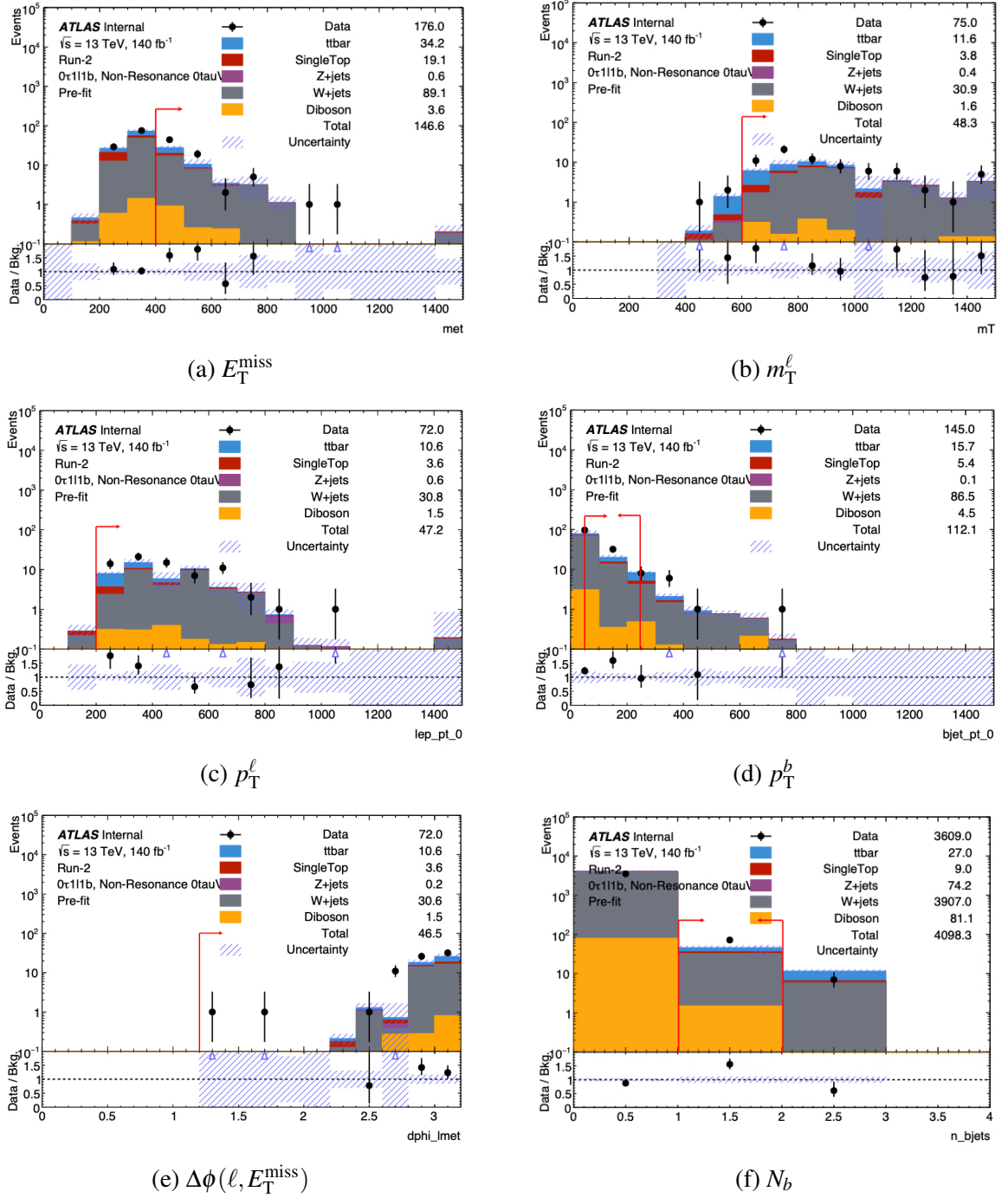


Fig. 8.9 Pre-fit $N - 1$ distributions in VRW-NonRes. The electron and muon channels are combined. The lower panels show data divided by the total simulated background. The hatched bands represent the statistical uncertainty in the prediction. The last bin includes the overflow. The original plot labels 0 τ VR denote VRW.

8.5.2 Top-quark validation without a light lepton

The VRTop regions require one τ , no electron or muon, and exactly two tagged b jets. The lepton veto reproduces the SR requirement, while the second tag increases the top-quark fraction. The lower p_T^τ and mass thresholds in Table 8.4 provide sufficient events for the check, and $E_T^{\text{miss}} > 200$ GeV keeps the selection on the E_T^{miss} -trigger plateau. The top-quark fractions are approximately 75% and 87% in the resonant and non-resonant regions, respectively.

Figures 8.10 and 8.11 show the pre-fit $N - 1$ distributions. The data are compatible with the simulated distributions within their statistical precision. Single-top production is more prominent in the resonant category, while $t\bar{t}$ dominates the non-resonant category. Together with VRW, these regions test the changes in tagged-jet multiplicity and selected-lepton multiplicity on which the SR prediction depends.

Signal contamination is checked using samples that include both hadronic and leptonic τ decays. The benchmarks are $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.2)$ for the resonant regions and $(2.5 \text{ TeV}, 1.5, 1.0)$ for the non-resonant regions. The contamination is about 1.3% and 0.7% in VRW, and 1.2% and 0.6% in VRTop, respectively.

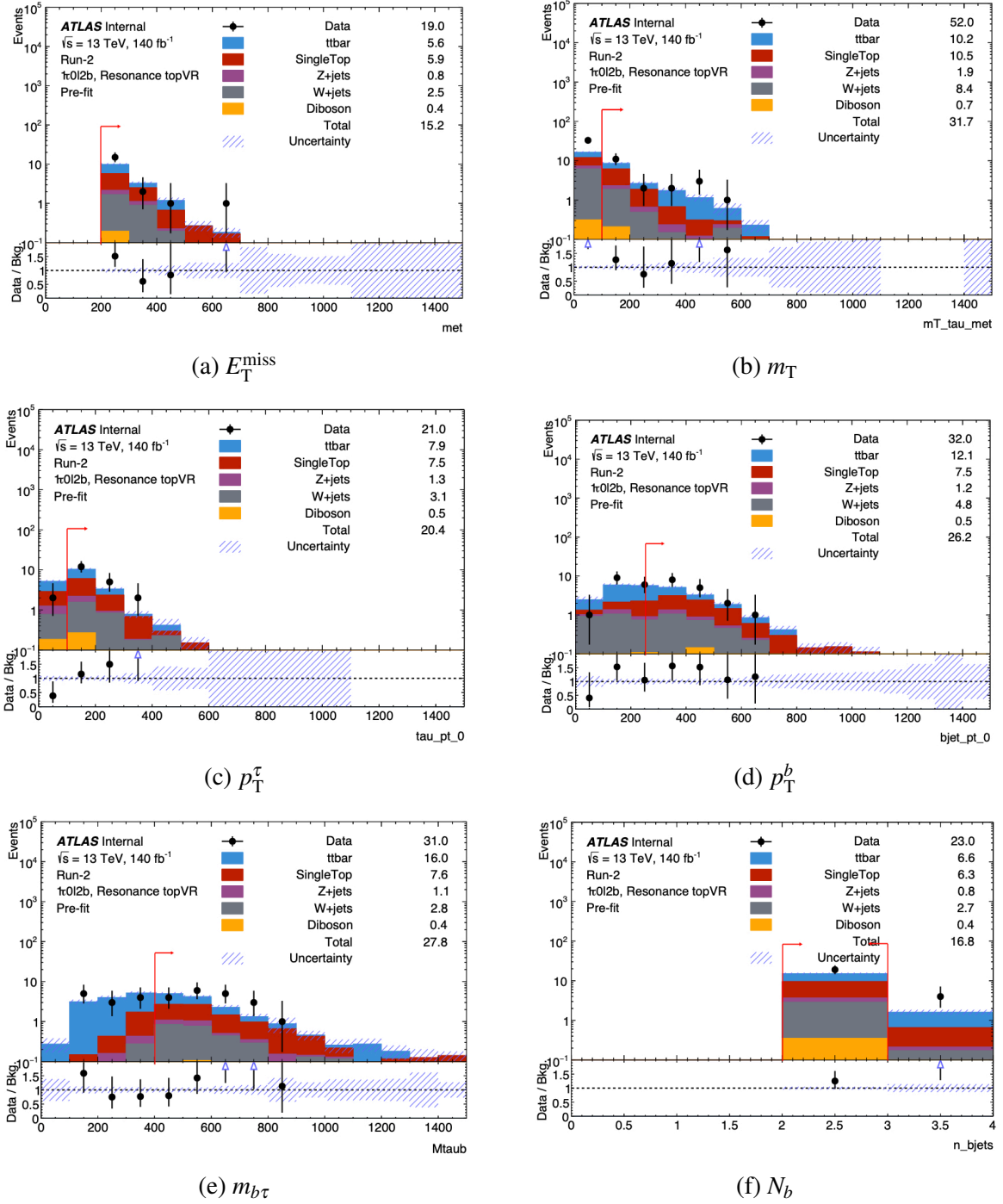


Fig. 8.10 Pre-fit $N - 1$ distributions in VRTop-Res. The lower panels show data divided by the total simulated background. The hatched bands represent the statistical uncertainty in the prediction. The last bin includes the overflow. The leading tagged jet is used for the jet-dependent variables.

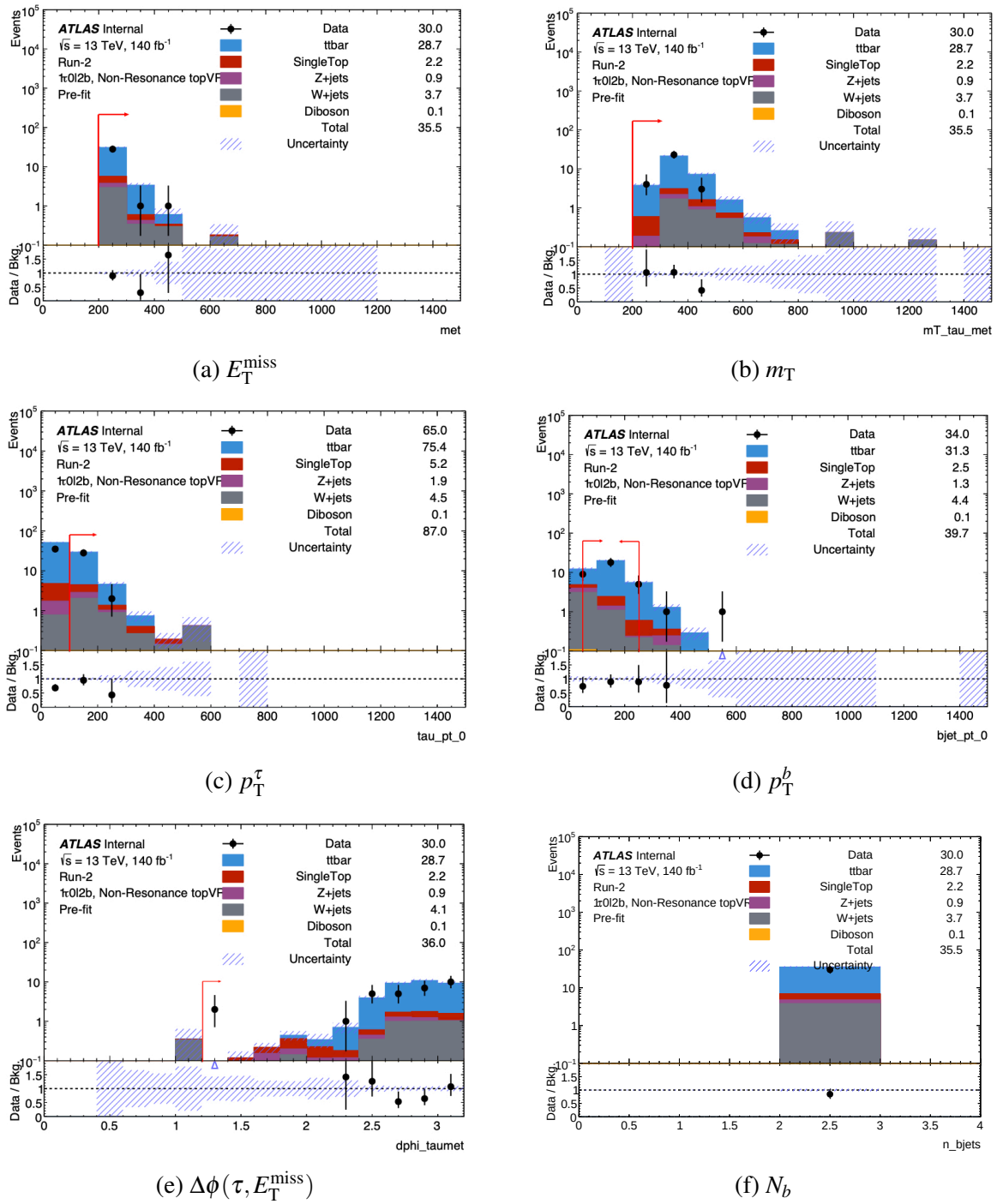


Fig. 8.11 Pre-fit $N - 1$ distributions in VRTop-NonRes. The lower panels show data divided by the total simulated background. The hatched bands represent the statistical uncertainty in the prediction. The last bin includes the overflow. The leading tagged jet is used for the jet-dependent variables.

8.5.3 Investigation of VRW-NonRes

The excess in VRW-NonRes remains a deviation of about 2σ after the background-only fit and the assigned uncertainties are included. Its origin is examined through the cutflow, nearby E_T^{miss} selections and the properties of the selected data events. The pre-fit ratio stays close to unity until the $E_T^{\text{miss}} > 400$ GeV requirement is imposed. A finer-binned comparison locates much of the excess in $425 < E_T^{\text{miss}} < 450$ GeV, as shown in Figure 8.12. Changing the E_T^{miss} threshold to 300 GeV or 500 GeV reduces the discrepancy in the statistical-only checks to 1.8σ and 1.5σ , respectively. This behaviour indicates a local excess rather than a persistent offset throughout the high- E_T^{miss} tail.

Within $425 < E_T^{\text{miss}} < 450$ GeV, the excess is concentrated in the $300 < p_T^\ell < 400$ GeV bin, above the 200 GeV lepton threshold. The 12 data events in this interval comprise seven electron and five muon events. Their lepton and b -jet positions show no concentration in a particular detector region. No specific detector problem is identified, and the neighbouring selections give no evidence for a broad failure of the background prediction. The observations are consistent with a statistical fluctuation, although these checks alone do not determine the origin of each excess event.

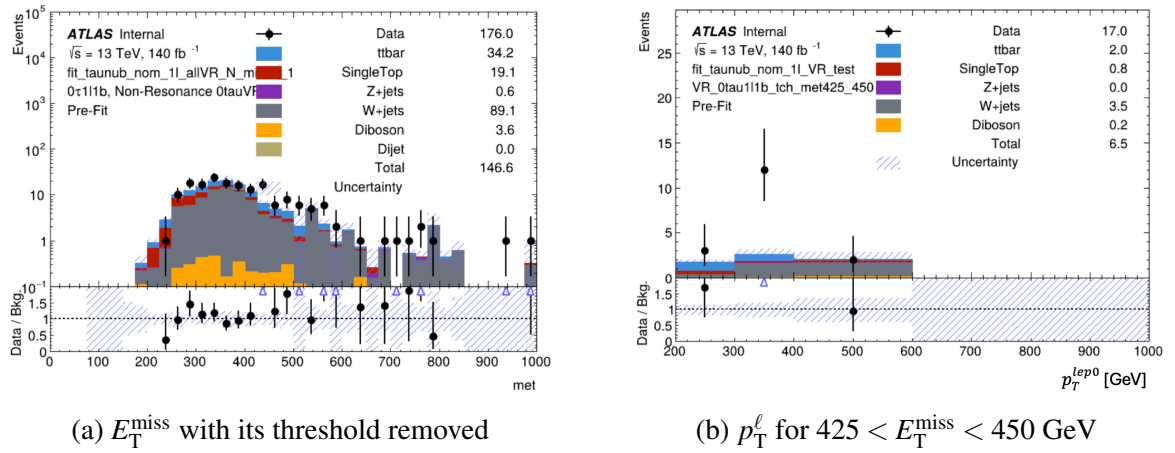


Fig. 8.12 Diagnostic pre-fit distributions for the excess in VRW-NonRes. The finer E_T^{miss} binning in (a) resolves the interval $425 < E_T^{\text{miss}} < 450$ GeV. Panel (b) shows the lepton p_T within that interval. The lower panels show data divided by the simulated prediction.

8.5.4 Additional $1\tau 0\ell 1b$ validation

The additional $1\tau 0\ell 1b$ regions test the background with the same object multiplicities as SR1b in adjacent mass intervals. They provide a direct check of the one- b -jet, hadronic- τ prediction following the excess in the light-lepton validation sample. The resonant region requires $400 < m_{b\tau} < 800$ GeV, and the non-resonant region requires $500 < m_T < 600$ GeV. These upper bounds keep the regions separate from the corresponding SRs. Further reductions in p_T^τ , E_T^{miss} and m_T increase the event yields, as summarised in Table 8.6. All other requirements follow the corresponding SR1b selections.

Table 8.6 Selections and yields in the supplementary $1\tau 0\ell 1b$ validation regions. Both use E_T^{miss} triggers and the remaining SR1b requirements. The background-only fit to the CRs is extrapolated to these regions. The quoted prediction uncertainties are those of this validation study, without an additional uncertainty for the changed mass intervals.

	Resonant	Non-resonant
p_T^τ [GeV]	> 100	> 100
E_T^{miss} [GeV]	> 200	> 200
m_T [GeV]	> 100	$500 < m_T < 600$
$m_{b\tau}$ [GeV]	$400 < m_{b\tau} < 800$	No requirement
Signal contamination	$\sim 2\%$	$\sim 7\%$
Post-fit background	25.4 ± 1.5	8.0 ± 1.7
Observed data	24	9

Signal contamination is evaluated before examining the data, using the same benchmarks as in Section 8.5.2. The estimated contributions are about 2% and 7% in the resonant and non-resonant regions, respectively. Background yields of 25.4 ± 1.5 and 8.0 ± 1.7 events are predicted after the CR fit, compared with 24 and 9 observed events. The agreement supports the transfer of the background normalisation to the $1\tau 0\ell 1b$ final state in these lower-mass intervals. No additional uncertainty for the changed $m_{b\tau}$ or m_T selections is included in this supplementary study, so its agreement is a cross-check rather than a separate determination of the uncertainty in the high-mass SR prediction.

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Chapter 9

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Systematics Uncertainties

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9.1 Experimental uncertainties

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9.2 Theoretical uncertainties

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Chapter 10

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Statistical Analysis and Results

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10.1 Statistical analysis

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10.1.1 Hypothesis test

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10.1.2 Likelihood function

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10.2 Results

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10.2.1 Background-only fit

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10.2.2 Unblinded S+B fit in CR+SRs

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10.2.3 Exclusion limits with toy models

2199 **Chapter 11**

2200 **Discussion and Conclusion**

References

- 2202 [1] Sheldon L. Glashow. Partial-symmetries of weak interactions. *Nucl. Phys.*, 22:
2203 579–588, 1961. doi: 10.1016/0029-5582(61)90469-2.
- 2204 [2] Steven Weinberg. A model of leptons. *Phys. Rev. Lett.*, 19:1264–1266, 1967. doi:
2205 10.1103/PhysRevLett.19.1264.
- 2206 [3] Abdus Salam. Weak and electromagnetic interactions. In N. Svartholm, editor,
2207 *Elementary Particle Theory: Relativistic Groups and Analyticity*, pages 367–377,
2208 Stockholm, 1968. Almqvist and Wiksell.
- 2209 [4] Wolfgang Pauli. The connection between spin and statistics. *Phys. Rev.*, 58:716–722,
2210 1940. doi: 10.1103/PhysRev.58.716.
- 2211 [5] Gerard 't Hooft. Renormalization of massless yang–mills fields. *Nucl. Phys. B*, 33:
2212 173–199, 1971. doi: 10.1016/0550-3213(71)90395-6.
- 2213 [6] Gerard 't Hooft and Martinus J. G. Veltman. Regularization and renormalization of
2214 gauge fields. *Nucl. Phys. B*, 44:189–213, 1972. doi: 10.1016/0550-3213(72)90279-9.
- 2215 [7] F. Englert and R. Brout. Broken symmetry and the mass of gauge vector mesons.
2216 *Phys. Rev. Lett.*, 13:321–323, 1964. doi: 10.1103/PhysRevLett.13.321.
- 2217 [8] Peter W. Higgs. Broken symmetries and the masses of gauge bosons. *Phys. Rev. Lett.*,
2218 13:508–509, 1964. doi: 10.1103/PhysRevLett.13.508.
- 2219 [9] G. S. Guralnik, C. R. Hagen, and T. W. B. Kibble. Global conservation laws and
2220 massless particles. *Phys. Rev. Lett.*, 13:585–587, 1964. doi: 10.1103/PhysRevLett.13.
2221 585.
- 2222 [10] Particle Data Group. Review of particle physics. *Prog. Theor. Exp. Phys.*, 2025.
- 2223 [11] ATLAS Collaboration. Observation of a new particle in the search for the standard
2224 model higgs boson with the atlas detector at the lh. *Phys. Lett. B*, 716:1–29, 2012.
2225 doi: 10.1016/j.physletb.2012.08.020.
- 2226 [12] CMS Collaboration. Observation of a new boson at a mass of 125 gev with the cms
2227 experiment at the lh. *Phys. Lett. B*, 716:30–61, 2012. doi: 10.1016/j.physletb.2012.
2228 08.021.
- 2229 [13] ATLAS and CMS Collaborations. Combined measurement of the higgs boson mass in
2230 pp collisions at $\sqrt{s} = 7$ and 8 tev with the atlas and cms experiments. *Phys. Rev. Lett.*,
2231 114:191803, 2015. doi: 10.1103/PhysRevLett.114.191803.

- [14] Nicola Cabibbo. Unitary symmetry and leptonic decays. *Phys. Rev. Lett.*, 10:531–533, 1963. doi: 10.1103/PhysRevLett.10.531.
- [15] Makoto Kobayashi and Toshihide Maskawa. Cp-violation in the renormalizable theory of weak interaction. *Prog. Theor. Phys.*, 49:652–657, 1973. doi: 10.1143/PTP.49.652.
- [16] Super-Kamiokande Collaboration. Evidence for oscillation of atmospheric neutrinos. *Phys. Rev. Lett.*, 81:1562–1567, 1998. doi: 10.1103/PhysRevLett.81.1562.
- [17] SNO Collaboration. Direct evidence for neutrino flavor transformation from neutral-current interactions in the sudbury neutrino observatory. *Phys. Rev. Lett.*, 89:011301, 2002. doi: 10.1103/PhysRevLett.89.011301.
- [18] Planck Collaboration. Planck 2018 results. vi. cosmological parameters. *Astron. Astrophys.*, 641:A6, 2020. doi: 10.1051/0004-6361/201833910.
- [19] A. D. Sakharov. Violation of cp invariance, c asymmetry, and baryon asymmetry of the universe. *Pisma Zh. Eksp. Teor. Fiz.*, 5:32–35, 1967.
- [20] M. B. Gavela, P. Hernandez, J. Orloff, and O. Pene. Standard model cp-violation and baryon asymmetry. *Mod. Phys. Lett. A*, 9:795–810, 1994. doi: 10.1142/S0217732394000629.
- [21] Andrew G. Cohen, D. B. Kaplan, and A. E. Nelson. Progress in electroweak baryogenesis. *Ann. Rev. Nucl. Part. Sci.*, 43:27–70, 1993. doi: 10.1146/annurev.ns.43.120193.000331.
- [22] C. A. Baker et al. An improved experimental limit on the electric dipole moment of the neutron. *Phys. Rev. Lett.*, 97:131801, 2006. doi: 10.1103/PhysRevLett.97.131801.
- [23] LHCb Collaboration. Test of lepton universality in $b \rightarrow s\ell^+\ell^-$ decays. *Phys. Rev. Lett.*, 131:051803, 2023. doi: 10.1103/PhysRevLett.131.051803.
- [24] LHCb Collaboration. Measurement of lepton universality parameters in $b^+ \rightarrow k^+\ell^+\ell^-$ and $b^0 \rightarrow k^{*0}\ell^+\ell^-$ decays. *Phys. Rev. D*, 108:032002, 2023. doi: 10.1103/PhysRevD.108.032002.
- [25] Heavy Flavor Averaging Group (HFLAV). Average of $R(D)$ and $R(D^*)$ for CKM 2025. <https://hflav-eos.web.cern.ch/hflav-eos/semi/ckm25/html/RDsDsstar/RDRDs.html>, 2025. Last update: 28 September 2025.
- [26] BaBar Collaboration. Evidence for an excess of $\bar{B} \rightarrow D^{(*)}\tau^-\bar{\nu}_\tau$ decays. *Phys. Rev. Lett.*, 109:101802, 2012. doi: 10.1103/PhysRevLett.109.101802.
- [27] BaBar Collaboration. Measurement of an Excess of $\bar{B} \rightarrow D^{(*)}\tau^-\bar{\nu}_\tau$ Decays and Implications for Charged Higgs Bosons. *Phys. Rev. D*, 88(7):072012, 2013. doi: 10.1103/PhysRevD.88.072012.
- [28] Belle Collaboration. Measurement of the branching ratio of $\bar{B} \rightarrow D^{(*)}\tau^-\bar{\nu}_\tau$ relative to $\bar{B} \rightarrow D^{(*)}\ell^-\bar{\nu}_\ell$ decays with hadronic tagging at Belle. *Phys. Rev. D*, 92(7):072014, 2015. doi: 10.1103/PhysRevD.92.072014.

- [29] Belle Collaboration. Measurement of the τ lepton polarization and $R(D^*)$ in the decay $\bar{B} \rightarrow D^* \tau^- \bar{\nu}_\tau$. *Phys. Rev. Lett.*, 118(21):211801, 2017. doi: 10.1103/PhysRevLett.118.211801.
- [30] Belle Collaboration. Measurement of the τ lepton polarization and $R(D^*)$ in the decay $\bar{B} \rightarrow D^* \tau^- \bar{\nu}_\tau$ with one-prong hadronic τ decays at Belle. *Phys. Rev. D*, 97(1):012004, 2018. doi: 10.1103/PhysRevD.97.012004.
- [31] Belle Collaboration. Measurement of $R(D)$ and $R(D^*)$ with a semileptonic tagging method. *Phys. Rev. Lett.*, 124(16):161803, 2020. doi: 10.1103/PhysRevLett.124.161803.
- [32] Belle-II Collaboration. Test of lepton flavor universality with a measurement of $R(D^*)$ using hadronic B tagging at the Belle II experiment. *Phys. Rev. D*, 110(7):072020, 2024. doi: 10.1103/PhysRevD.110.072020.
- [33] Belle-II Collaboration. Test of lepton flavor universality with measurements of $R(D^+)$ and $R(D^{*+})$ using semileptonic B tagging at the Belle II experiment. *Phys. Rev. D*, 112(3):032010, 2025. doi: 10.1103/fmn3-h8fy.
- [34] LHCb Collaboration. Measurement of the ratios of branching fractions $\mathcal{R}(D^*)$ and $\mathcal{R}(D^0)$. *Phys. Rev. Lett.*, 131:111802, 2023. doi: 10.1103/PhysRevLett.131.111802.
- [35] LHCb Collaboration. Test of lepton flavor universality using $B^0 \rightarrow D^{*-} \tau^+ \nu_\tau$ decays with hadronic τ channels. *Phys. Rev. D*, 108(1):012018, 2023. doi: 10.1103/PhysRevD.108.012018. [Erratum: *Phys.Rev.D* 109, 119902 (2024)].
- [36] LHCb Collaboration. Measurement of the Branching Fraction Ratios $R(D^+)$ and $R(D^{*+})$ Using Muonic τ Decays. *Phys. Rev. Lett.*, 134(6):061801, 2025. doi: 10.1103/PhysRevLett.134.061801.
- [37] J. Wess and B. Zumino. Supergauge transformations in four dimensions. *Nucl. Phys. B*, 70:39–50, 1974. doi: 10.1016/0550-3213(74)90355-1.
- [38] Stephen P. Martin. A supersymmetry primer. 1997.
- [39] Savas Dimopoulos and Howard Georgi. Softly broken supersymmetry and $su(5)$. *Nucl. Phys. B*, 193:150–162, 1981. doi: 10.1016/0550-3213(81)90522-8.
- [40] Glennys R. Farrar and Pierre Fayet. Phenomenology of the production, decay, and detection of new hadronic states associated with supersymmetry. *Phys. Lett. B*, 76: 575–579, 1978. doi: 10.1016/0370-2693(78)90858-4.
- [41] H. Goldberg. Constraint on the photino mass from cosmology. *Phys. Rev. Lett.*, 50: 1419–1422, 1983. doi: 10.1103/PhysRevLett.50.1419.
- [42] John R. Ellis, J. S. Hagelin, D. V. Nanopoulos, Keith A. Olive, and M. Srednicki. Supersymmetric relics from the big bang. *Nucl. Phys. B*, 238:453–476, 1984. doi: 10.1016/0550-3213(84)90461-9.
- [43] Savas Dimopoulos, Stuart Raby, and Frank Wilczek. Supersymmetry and the scale of unification. *Phys. Rev. D*, 24:1681–1683, 1981. doi: 10.1103/PhysRevD.24.1681.

- [44] Ugo Amaldi, Wim de Boer, and Hermann Fürstenau. Comparison of grand unified theories with electroweak and strong coupling constants measured at lep. *Phys. Lett. B*, 260:447–455, 1991. doi: 10.1016/0370-2693(91)91641-8.
- [45] ATLAS Collaboration. The quest to discover supersymmetry at the atlas experiment. *Phys. Rep.*, 1116:1–76, 2025. doi: 10.1016/j.physrep.2025.01.004.
- [46] R. D. Peccei and Helen R. Quinn. Cp conservation in the presence of pseudoparticles. *Phys. Rev. Lett.*, 38:1440–1443, 1977. doi: 10.1103/PhysRevLett.38.1440.
- [47] R. D. Peccei and Helen R. Quinn. Constraints imposed by cp conservation in the presence of pseudoparticles. *Phys. Rev. D*, 16:1791–1797, 1977. doi: 10.1103/PhysRevD.16.1791.
- [48] Steven Weinberg. A new light boson? *Phys. Rev. Lett.*, 40:223–226, 1978. doi: 10.1103/PhysRevLett.40.223.
- [49] Frank Wilczek. Problem of strong p and t invariance in the presence of instantons. *Phys. Rev. Lett.*, 40:279–282, 1978. doi: 10.1103/PhysRevLett.40.279.
- [50] Jihn E. Kim and Gianpaolo Carosi. Axions and the strong cp problem. *Rev. Mod. Phys.*, 82:557–601, 2010. doi: 10.1103/RevModPhys.82.557.
- [51] Jogesh C. Pati and Abdus Salam. Lepton number as the fourth color. *Phys. Rev. D*, 10:275–289, 1974. doi: 10.1103/PhysRevD.10.275.
- [52] Howard Georgi and Sheldon L. Glashow. Unity of all elementary-particle forces. *Phys. Rev. Lett.*, 32:438–441, 1974. doi: 10.1103/PhysRevLett.32.438.
- [53] I. Dorsner, S. Fajfer, A. Greljo, J. F. Kamenik, and N. Kosnik. Physics of leptoquarks in precision experiments and at particle colliders. *Phys. Rept.*, 641:1–68, 2016. doi: 10.1016/j.physrep.2016.06.001.
- [54] Claudia Cornella, Javier Fuentes-Martin, and Gino Isidori. Revisiting the vector leptoquark explanation of the b -physics anomalies. *JHEP*, 07:168, 2019. doi: 10.1007/JHEP07(2019)168.
- [55] Michael J. Baker, Javier Fuentes-Martin, Gino Isidori, and Matthias König. High- p_t signatures in vector-leptoquark models. *Eur. Phys. J. C*, 79:334, 2019. doi: 10.1140/epjc/s10052-019-6853-x.
- [56] Jason Aebischer, Gino Isidori, Marko Pesut, Ben A. Stefanek, and Felix Wilsch. Confronting the vector leptoquark hypothesis with new low- and high-energy data. *Eur. Phys. J. C*, 83:153, 2023. doi: 10.1140/epjc/s10052-023-11304-5.
- [57] ATLAS Collaboration. Search for leptoquarks decaying into the $b\tau$ final state in pp collisions at $\sqrt{s} = 13$ tev with the atlas detector. *JHEP*, 10:001, 2023. doi: 10.1007/JHEP10(2023)001.
- [58] CMS Collaboration. Search for singly and pair-produced leptoquarks coupling to third-generation fermions in proton-proton collisions at $\sqrt{s} = 13$ tev. *Phys. Lett. B*, 819:136446, 2021. doi: 10.1016/j.physletb.2021.136446.

- [59] CMS Collaboration. Search for a third-generation leptoquark coupled to a τ lepton and a b quark through single, pair, and nonresonant production in proton-proton collisions at $\sqrt{s} = 13$ tev. *JHEP*, 05:311, 2024. doi: 10.1007/JHEP05(2024)311.
- [60] ATLAS Collaboration. A measurement of the high-mass $\tau\bar{\tau}$ production cross-section at $\sqrt{s} = 13$ tev with the atlas detector and constraints on new particles and couplings. *JHEP*, 10:054, 2025. doi: 10.1007/JHEP10(2025)054.
- [61] CMS Collaboration. Searches for additional higgs bosons and for vector leptoquarks in $\tau\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ tev. *JHEP*, 07:073, 2023. doi: 10.1007/JHEP07(2023)073.
- [62] ATLAS Collaboration. Leptoquark summary plots for scalar or vector models. Technical Report ATL-PHYS-PUB-2025-013, CERN, 2025.
- [63] CMS Collaboration. Search for new physics in the τ lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ tev. *JHEP*, 09:051, 2023. doi: 10.1007/JHEP09(2023)051.
- [64] Motoi Endo, Syuhei Iguro, Teppei Kitahara, Michihisa Takeuchi, and Ryoutaro Watanabe. Non-resonant new physics search at the lhc for the $b \rightarrow c\tau\nu$ anomalies. *JHEP*, 02:106, 2022. doi: 10.1007/JHEP02(2022)106.
- [65] CERN. About CERN. <https://home.cern/about/>, . Accessed: 2026-06-11.
- [66] Lyndon Evans and Philip Bryant. LHC Machine. *JINST*, 3:S08001, 2008. doi: 10.1088/1748-0221/3/08/S08001.
- [67] CERN. The Large Hadron Collider. <https://home.cern/science/accelerators/large-hadron-collider/>, . Accessed: 2026-06-11.
- [68] Esma Mobs. The CERN accelerator complex - August 2018. Complexe des accélérateurs du CERN - Août 2018. CERN Document Server, 2018. URL <https://cds.cern.ch/record/2636343>. General Photo, CERN-GRAPHICS-2018-005.
- [69] Werner Herr and B. Muratori. Concept of luminosity. 2006. doi: 10.5170/CERN-2006-002.361. URL <https://cds.cern.ch/record/941318>.
- [70] ATLAS Collaboration. Luminosity determination in pp collisions at $\sqrt{s} = 13$ tev using the atlas detector at the lhc. *Eur. Phys. J. C*, 83:982, 2023. doi: 10.1140/epjc/s10052-023-11747-w.
- [71] ATLAS Collaboration. LHC Run 3: physics at record energy starts tomorrow. 2022. URL <https://atlas.cern/Updates/Press-Statement/LHC-Run3-Starts>. Accessed: 2026-06-11.
- [72] ATLAS Collaboration. ATLAS data quality operations and performance for 2015–2018 data-taking. *JINST*, 15:P04003, 2020. doi: 10.1088/1748-0221/15/04/P04003.
- [73] ATLAS Collaboration. The atlas experiment at the cern large hadron collider. *JINST*, 3:S08003, 2008. doi: 10.1088/1748-0221/3/08/S08003.

- [74] ATLAS Collaboration. The atlas experiment at the cern large hadron collider: A description of the detector configuration for run 3. *JINST*, 19:P05063, 2024. doi: 10.1088/1748-0221/19/05/P05063.
- [75] Fabian Kuger. Signal formation processes in micromegas detectors and quality control for large size detector construction for the atlas new small wheel, 2017.
- [76] B. Abbott et al. Production and integration of the atlas insertable b-layer. *JINST*, 13:T05008, 2018. doi: 10.1088/1748-0221/13/05/T05008.
- [77] ATLAS Collaboration. The Phase-I Trigger Readout Electronics Upgrade of the ATLAS Liquid Argon Calorimeters. *JINST*, 17(05):P05024, 2022. doi: 10.1088/1748-0221/17/05/P05024.
- [78] ATLAS Collaboration. Operation of the atlas trigger system in run 2. *JINST*, 15:P10004, 2020. doi: 10.1088/1748-0221/15/10/P10004.
- [79] Johan Alwall et al. The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations. *JHEP*, 07:079, 2014. doi: 10.1007/JHEP07(2014)079.
- [80] Christian Bierlich et al. A comprehensive guide to the physics and usage of PYTHIA 8.3. *SciPost Phys. Codebases*, page 8, 2022. doi: 10.21468/SciPostPhysCodeb.8.
- [81] Richard D. Ball et al. Parton distributions for the LHC run II. *JHEP*, 04:040, 2015. doi: 10.1007/JHEP04(2015)040.
- [82] ATLAS Collaboration. ATLAS Pythia 8 tunes to 7 tev data. Technical Report ATL-PHYS-PUB-2014-021, CERN, 2014. URL <https://cds.cern.ch/record/1966419>.
- [83] Richard D. Ball et al. Parton distributions with LHC data. *Nucl. Phys. B*, 867:244–289, 2013. doi: 10.1016/j.nuclphysb.2012.10.003.
- [84] Leif Lönnblad and Stefan Prestel. Matching tree-level matrix elements with interleaved showers. *JHEP*, 03:019, 2012. doi: 10.1007/JHEP03(2012)019.
- [85] David J. Lange. The EvtGen particle decay simulation package. *Nucl. Instrum. Meth. A*, 462:152–155, 2001. doi: 10.1016/S0168-9002(01)00089-4.
- [86] ATLAS Collaboration. AtlFast3: The next generation of fast simulation in ATLAS. *Comput. Softw. Big Sci.*, 6:7, 2022. doi: 10.1007/s41781-021-00079-7.
- [87] Enrico Bothmann et al. Event generation with Sherpa 2.2. *SciPost Phys.*, 7:034, 2019. doi: 10.21468/SciPostPhys.7.3.034.
- [88] Steffen Schumann and Frank Krauss. A parton shower algorithm based on Catani–Seymour dipole factorisation. *JHEP*, 03:038, 2008. doi: 10.1088/1126-6708/2008/03/038.
- [89] Stefan Höche, Frank Krauss, Marek Schönherr, and Frank Siegert. A critical appraisal of NLO+PS matching methods. *JHEP*, 09:049, 2012. doi: 10.1007/JHEP09(2012)049.

- [90] Stefan Höche, Frank Krauss, Marek Schönherr, and Frank Siegert. QCD matrix elements + parton showers. the NLO case. *JHEP*, 04:027, 2013. doi: 10.1007/JHEP04(2013)027.
- [91] Stefano Catani, Frank Krauss, Bryan R. Webber, and Ralf Kuhn. QCD matrix elements + parton showers. *JHEP*, 11:063, 2001. doi: 10.1088/1126-6708/2001/11/063.
- [92] Stefan Höche, Frank Krauss, Steffen Schumann, and Frank Siegert. QCD matrix elements and truncated showers. *JHEP*, 05:053, 2009. doi: 10.1088/1126-6708/2009/05/053.
- [93] Stefano Frixione, Paolo Nason, and Giovanni Ridolfi. A positive-weight next-to-leading-order monte carlo for heavy flavour hadroproduction. *JHEP*, 09:126, 2007. doi: 10.1088/1126-6708/2007/09/126.
- [94] Paolo Nason. A new method for combining NLO QCD with shower monte carlo algorithms. *JHEP*, 11:040, 2004. doi: 10.1088/1126-6708/2004/11/040.
- [95] Stefano Frixione, Paolo Nason, and Carlo Oleari. Matching NLO QCD computations with parton shower simulations: the POWHEG method. *JHEP*, 11:070, 2007. doi: 10.1088/1126-6708/2007/11/070.
- [96] Simone Alioli, Paolo Nason, Carlo Oleari, and Emanuele Re. A general framework for implementing NLO calculations in shower monte carlo programs: the POWHEG BOX. *JHEP*, 06:043, 2010. doi: 10.1007/JHEP06(2010)043.
- [97] Torbjörn Sjöstrand et al. An introduction to PYTHIA 8.2. *Comput. Phys. Commun.*, 191:159–177, 2015. doi: 10.1016/j.cpc.2015.01.024.
- [98] Michał Czakon and Alexander Mitov. Top++: A program for the calculation of the top-pair cross-section at hadron colliders. *Comput. Phys. Commun.*, 185:2930–2938, 2014. doi: 10.1016/j.cpc.2014.06.021.
- [99] ATLAS Collaboration. The ATLAS simulation infrastructure. *Eur. Phys. J. C*, 70: 823–874, 2010. doi: 10.1140/epjc/s10052-010-1429-9.
- [100] S. Agostinelli et al. Geant4—a simulation toolkit. *Nucl. Instrum. Meth. A*, 506: 250–303, 2003. doi: 10.1016/S0168-9002(03)01368-8.
- [101] ATLAS Collaboration. Performance of electron and photon triggers in ATLAS during LHC run 2. *Eur. Phys. J. C*, 80:47, 2020. doi: 10.1140/epjc/s10052-019-7500-2.
- [102] ATLAS Collaboration. Performance of the ATLAS muon triggers in run 2. *JINST*, 15: P09015, 2020. doi: 10.1088/1748-0221/15/09/P09015.
- [103] ATLAS Collaboration. Performance of the missing transverse momentum triggers for the ATLAS detector during run-2 data taking. *JHEP*, 08:080, 2020. doi: 10.1007/JHEP08(2020)080.
- [104] ATLAS Collaboration. Performance of the ATLAS track reconstruction algorithms in dense environments in LHC run 2. *Eur. Phys. J. C*, 77:673, 2017. doi: 10.1140/epjc/s10052-017-5225-7.

- [105] ATLAS Collaboration. Muon reconstruction and identification efficiency in ATLAS using the full run 2 pp collision data set at $\sqrt{s} = 13$ TeV. *Eur. Phys. J. C*, 81:578, 2021. doi: 10.1140/epjc/s10052-021-09233-2.
- [106] ATLAS Collaboration. Software and computing for run 3 of the ATLAS experiment at the LHC. *Eur. Phys. J. C*, 85:234, 2025. doi: 10.1140/epjc/s10052-024-13701-w.
- [107] ATLAS Collaboration. Vertex reconstruction performance of the ATLAS detector at $\sqrt{s} = 13$ TeV. Technical Report ATL-PHYS-PUB-2015-026, CERN, 2015. URL <https://cds.cern.ch/record/2037717>.
- [108] ATLAS Collaboration. Electron and photon performance measurements with the ATLAS detector using the 2015–2017 LHC proton–proton collision data. *JINST*, 14: P12006, 2019. doi: 10.1088/1748-0221/14/12/P12006.
- [109] ATLAS Collaboration. Electron reconstruction and identification in the ATLAS experiment using the 2015 and 2016 LHC proton–proton collision data at $\sqrt{s} = 13$ TeV. *Eur. Phys. J. C*, 79:639, 2019. doi: 10.1140/epjc/s10052-019-7140-6.
- [110] ATLAS Collaboration. Electron and photon efficiencies in LHC run 2 with the ATLAS experiment. *JHEP*, 05:162, 2024. doi: 10.1007/JHEP05(2024)162.
- [111] ATLAS Collaboration. Muon isolation with machine learning algorithms with the ATLAS detector. Public performance plots, MUON-2024-01, 2024. URL <https://atlas.web.cern.ch/Atlas/GROUPS/PHYSICS/PLOTS/MUON-2024-01/index.php>.
- [112] ATLAS Collaboration. Studies of the muon momentum calibration and performance of the ATLAS detector with pp collisions at $\sqrt{s} = 13$ TeV. *Eur. Phys. J. C*, 83:686, 2023. doi: 10.1140/epjc/s10052-023-11584-x.
- [113] ATLAS Collaboration. Jet reconstruction and performance using particle flow with the ATLAS detector. *Eur. Phys. J. C*, 77:466, 2017. doi: 10.1140/epjc/s10052-017-5031-2.
- [114] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez. The anti- k_t jet clustering algorithm. *JHEP*, 04:063, 2008. doi: 10.1088/1126-6708/2008/04/063.
- [115] ATLAS Collaboration. Selection of jets produced in 13 TeV proton–proton collisions with the ATLAS detector. Technical Report ATLAS-CONF-2015-029, CERN, 2015. URL <https://cds.cern.ch/record/2037702>.
- [116] ATLAS Collaboration. Tagging and suppression of pileup jets with the ATLAS detector. Technical Report ATLAS-CONF-2014-018, CERN, 2014. URL <https://cds.cern.ch/record/1700870>.
- [117] ATLAS Collaboration. JetVertexNNTagger: neural-network jet-vertex tagging. Athena software, release 25.2.51. URL <https://gitlab.cern.ch/atlas/athena/-/blob/release/25.2.51/Reconstruction/Jet/JetMomentTools/Root/JetVertexNNTagger.cxx>. Accessed 11 September 2026.

- [118] ATLAS Collaboration. NNJVT FixedEffPt working-point configuration. ATLAS calibration data, JetPileupTag/NNJvt/2022-03-22, 2022. URL https://atlas-groupdata.web.cern.ch/atlas-groupdata/JetPileupTag/NNJvt/2022-03-22/NNJVT.Cuts.FixedEffPt.Offline.Nonprompt_All_MaxW.json. Accessed 11 September 2026.
- [119] ATLAS Collaboration. Jet energy scale and resolution measured in proton–proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector. *Eur. Phys. J. C*, 81:689, 2021. doi: 10.1140/epjc/s10052-021-09402-3.
- [120] ATLAS Collaboration. Transforming jet flavour tagging at ATLAS. *Nature Communications*, 17:541, 2026. doi: 10.1038/s41467-025-65059-6.
- [121] ATLAS Collaboration. Reconstruction, identification, and calibration of hadronically decaying tau leptons with the ATLAS detector for the LHC run 3 and reprocessed run 2 data. Technical Report ATL-PHYS-PUB-2022-044, CERN, 2022. URL <https://cds.cern.ch/record/2827111>.
- [122] ATLAS Collaboration. Identification of hadronic tau lepton decays using neural networks in the ATLAS experiment. Technical Report ATL-PHYS-PUB-2019-033, CERN, 2019. URL <https://cds.cern.ch/record/2688062>.
- [123] ATLAS Collaboration. The performance of missing transverse momentum reconstruction and its significance with the ATLAS detector using 140 fb^{-1} of $\sqrt{s} = 13$ TeV pp collisions. *Eur. Phys. J. C*, 85:606, 2025. doi: 10.1140/epjc/s10052-025-14062-8.
- [124] ATLAS Collaboration. Formulae for estimating significance. ATL-PHYS-PUB-2020-025, 2020. URL <https://cds.cern.ch/record/2736148>.
- [125] ATLAS Collaboration. Modelling and computational improvements to the simulation of single vector-boson plus jet processes for the ATLAS experiment. *JHEP*, 08:089, 2022. doi: 10.1007/JHEP08(2022)089.
- [126] ATLAS Collaboration. Measurements of the production cross-section for a Z boson in association with b - or c -jets in proton–proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector. *Eur. Phys. J. C*, 84:984, 2024. doi: 10.1140/epjc/s10052-024-13159-w.
- [127] ATLAS Collaboration. Measurements of WH and ZH production with Higgs boson decays into bottom quarks and direct constraints on the charm Yukawa coupling in 13 TeV pp collisions with the ATLAS detector. *JHEP*, 04:075, 2025. doi: 10.1007/JHEP04(2025)075.

2524 **Appendix A**

2525 **Correction of the signal internal event** 2526 **weights**

2527 For the right-handed interpretation, the dependence of the signal prediction on g_U and
2528 β_L^{23} is obtained by reweighting the baseline sample generated at each M_U value. The
2529 MADGRAPH5_AMC@NLO internal event weights encode the change in the matrix element
2530 when these coupling parameters are varied. Their performance is evaluated by comparing the
2531 reweighted predictions with samples generated independently at selected parameter points.

2532 The internal event weights alone do not fully reproduce the reference predictions in
2533 regions where the relative contributions of resonant and non-resonant production change. The
2534 overall normalisation is therefore adjusted to that of the corresponding reference prediction.
2535 A bin-by-bin two-dimensional correction in the leading-jet p_T and $m_{b\tau}$ is then applied to
2536 account for the remaining shape differences. These observables provide sensitivity to the
2537 resonant and non-resonant components and allow both contributions to be corrected within
2538 the same sample.

2539 The corrected predictions are validated using truth-level selections that reproduce the main
2540 signal-region requirements. At each stage of the selection, the corrected and independently
2541 generated event yields agree within 10% for both the resonant and non-resonant signal
2542 regions. This difference is smaller than the theoretical uncertainties assigned to the signal
2543 prediction.

2544 Appendix B

2545 Multijet background study

2546 The multijet suppression used in Section 7.1.2 is studied with simulated dijet and leptoquark
 2547 events. The motivation and discriminating variables are introduced there; this appendix
 2548 presents the study selection, threshold scan and residual-background checks.

2549 B.1 Study selection and distributions

2550 The study selection is kept loose to retain enough simulated multijet events to examine the
 2551 relevant distributions. It requires the E_T^{miss} trigger, exactly one hadronic τ candidate, no
 2552 selected electron or muon, and at least one jet, as listed in Table B.1. The final SR thresholds
 2553 on E_T^{miss} , p_T^τ , m_T and the visible invariant mass are not imposed at this stage. No b -tag
 2554 multiplicity is required.

Table B.1 Loose selection used to study the multijet suppression. This selection defines the denominator of the rejection fractions shown below.

Requirement	Selection
Trigger	Lowest-threshold unprescaled E_T^{miss} trigger
Hadronic τ multiplicity	$N_\tau = 1$
Electron and muon veto	$N_\ell = 0$
Jet multiplicity	$N_j \geq 1$

2555 Figure B.1 shows the two-dimensional distributions after this selection. The multijet
 2556 sample is concentrated at small $\Delta\phi_{\text{min}}$ and relatively small $R_{\text{miss}/j}$. The signal occupies a
 2557 broader angular range and extends to larger momentum ratios. Approximately 90% of the
 2558 multijet events have $R_{\text{miss}/j} < 6$, compared with approximately 55% of the signal in the
 2559 study sample. The ratio requirement alone would therefore reject a substantial fraction of the

signal. It is instead combined with the angular requirement so that either a large angle or a large ratio is sufficient to retain an event.

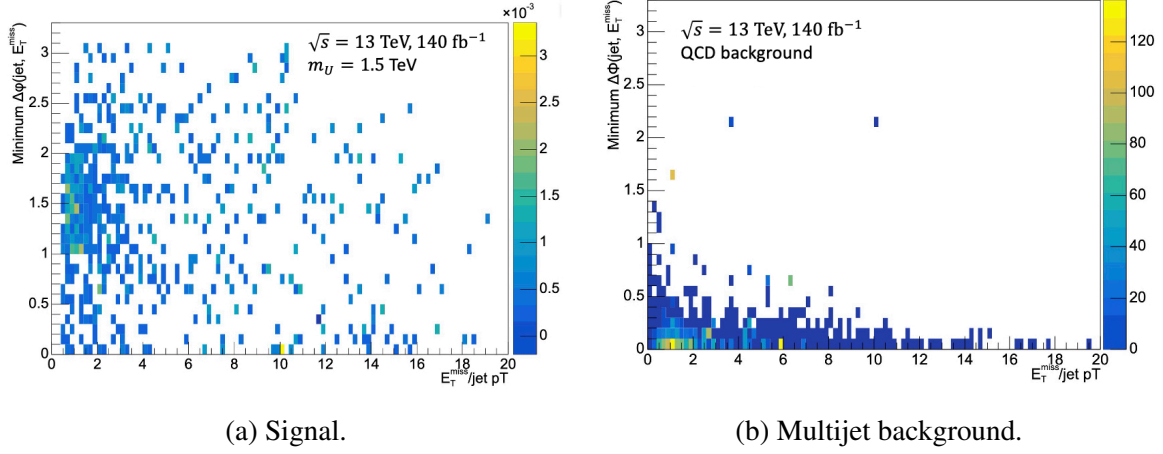


Fig. B.1 Correlation between the minimum jet- $\vec{p}_T^{\text{miss}}$ azimuthal separation and the ratio of E_T^{miss} to the p_T of the closest jet in azimuth, for (a) simulated signal with $M_U = 1.5$ TeV and (b) simulated multijet background after the loose study selection. The concentration of multijet events at small values of both variables motivates a combined requirement.

B.2 Threshold scan and signal retention

The angular threshold is scanned while the momentum-ratio threshold is fixed at six. For an angular threshold X , the rejected fraction is defined as

$$\text{Rejection ratio} = \frac{N(\Delta\phi_{\min} < X \text{ and } R_{\text{miss}/j} < 6)}{N(\text{all events after preselection})}. \quad (\text{B.1})$$

The numerator requires both small angular separation and a small momentum ratio. Its complement is the accepted region, defined by the logical OR in Eq. 7.4. The rejected fraction is evaluated separately for signal and multijet events, using each sample's yield after the loose selection as its denominator.

Figure B.2 shows the resulting rejection fractions. At $X = 0.4$, approximately 86% of the multijet events are rejected, while the signal loss is 3.6%. The corresponding signal efficiency for the cleaning cut is therefore $1 - 0.036 = 96.4\%$. Increasing the angular threshold rejects more multijet events but also removes more signal. The selected threshold provides substantial multijet rejection with a small signal loss in the studied sample.

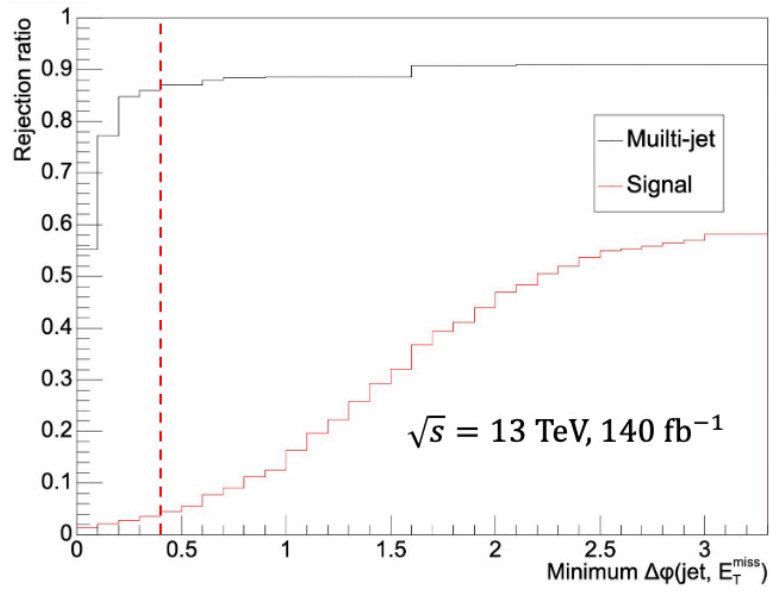


Fig. B.2 Rejected fractions of simulated signal and multijet events as a function of the minimum jet- $\vec{p}_T^{\text{miss}}$ angular threshold X , with $R_{\text{miss}/j} < 6$ also required in the rejected sample. The red and black curves show signal loss and multijet rejection, respectively. The vertical dotted line marks the adopted threshold $X = 0.4$. The denominator is the loose selection in Table B.1.

B.3 Residual multijet contribution

The residual multijet contribution is examined after applying the cleaning cut and the full SR requirements. No simulated multijet events survive those selections in the samples used for the study. This supports neglecting the multijet contribution in the nominal SR prediction, but zero surviving simulated events does not establish an exactly zero physical rate or a quantitative upper bound without the effective sample size.

Jets misidentified as τ candidates can still occur in other background processes. In particular, $Z \rightarrow \nu\nu$ production with jets contains genuine missing momentum and can pass the multijet cleaning even when the selected τ candidate is a jet. This contribution is included in the simulated Z+jets background. A supplementary selection requiring Medium τ identification while failing Tight also gives a negligible multijet contribution in the simulated study, supporting the effectiveness of the cleaning beyond the nominal Tight selection.

Appendix C

Zero- b -jet signal-region optimisation

The $E_{\text{T}}^{\text{miss}}$ threshold in SR0b is studied to assess whether tighter selections can improve the sensitivity at large β_L^{23} . A higher threshold can suppress both the SM background and the destructive interference, but also reduces the retained signal. The significance is therefore evaluated as the threshold is varied, including only the leading systematic uncertainties.

The threshold is scanned from 200 to 1500 GeV in 100 GeV steps. The signal parameter points are $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 0.5, 1.8)$ for SR0b-Res and $(3.0 \text{ TeV}, 2.0, 1.8)$ for SR0b-NonRes. The large value of β_L^{23} makes these benchmarks relevant to the zero- b -jet categories discussed in Section 7.3. The scan outcomes and adopted thresholds are summarised in Table C.1.

Table C.1 Signal benchmarks and preferred thresholds in the supplementary SR0b $E_{\text{T}}^{\text{miss}}$ scan. The last column gives the thresholds used in the final analysis.

Region	$(M_U [\text{TeV}], g_U, \beta_L^{23})$	Scan maximum	Adopted $E_{\text{T}}^{\text{miss}}$ cut
SR0b-Res	(1.5, 0.5, 1.8)	600 GeV	$> 200 \text{ GeV}$
SR0b-NonRes	(3.0, 2.0, 1.8)	400 GeV	$> 400 \text{ GeV}$

For SR0b-NonRes, the significance is maximised at $E_{\text{T}}^{\text{miss}} > 400 \text{ GeV}$. The scan therefore supports retaining the threshold used in the final region definition. For SR0b-Res, the maximum occurs at $E_{\text{T}}^{\text{miss}} > 600 \text{ GeV}$, giving an improvement of approximately 20% in the significance at the chosen benchmark. The final resonant selection nevertheless retains $E_{\text{T}}^{\text{miss}} > 200 \text{ GeV}$ to keep its kinematic requirements aligned with SR1b-Res. The higher threshold also reduces the magnitude of the destructive interference in SR0b-Res. This demonstrates the role of $E_{\text{T}}^{\text{miss}}$ in suppressing both the SM background and the interference contribution, although the gain must be balanced against the loss of signal events.

